4 NOISE IMPACTS

4.1 Introduction

4.1.1 This section presents an assessment of the potential noise impacts associated with the construction and operation of the Project. The noise impact assessment is conducted in accordance with the requirements of Annex 5 and Annex 13 of the Technical Memorandum on Environmental Impact Assessment Process (EIAO-TM) as well as the requirements set out under Clause 3.4.5 of the EIA Study Brief (ESB-291/2015).

4.1.2 An application for an Environmental Permit (EP) would be submitted for the following Schedule 2 Designated Projects (DPs) (refer to **Figure 1.3**) and these DPs during construction and operational phases are included in this noise impact assessment:

- Construction of new primary distributor road (Road P1) (DP1)
- Construction of eight new distributor roads (Road D1 to D8) (DP2)
- Construction of slip roads between: Road D8 Junction and existing Castle Peak Road; Junction of D8/P1 and Junction of D7/P1; and Kong Sham Western Highway (KSWH) connection to Road D3 (DP5)
- Construction of partly depressed and partly decked-over roads located at Road D2, Road D4 and Road D6 (DP6)
- Construction of four new SPSs at Sites 2-34, 3-41, 3-48 and 4-35 (DP9)
- Construction of Road P1 and a slip-road from KSWH to Road D3 partly located within the "CA" of Yuen Tau Shan (DP12)

4.1.3 The following DPs would apply the EP through separated EIA studies but its potential impacts during construction phase and operational phase are also included in this noise impact assessment:

- Construction of new West Rail Hung Shui Kiu (HSK) Station (Site 4-34) (DP3)
- Construction of environmentally friendly transport services (EFTS) (DP4) – Potential DP subject to further review
- Construction of a new container back-up and storage area (Sites 3-1, 3-4, 3-5, 3-13 and 3-14) (DP7) – Potential DP subject to further review
- Construction of new HSK sewage treatment works (STW) (Site 3-26 and part of existing San Wai STW) (DP8)
- Construction of one refuse transfer station (RTS) (Site 3-12) (DP11)
- Construction of flushing water service reservoirs for reuse of reclaimed water at Tan Kwai Tsuen and Fung Kong Tsuen (Sites 3-3 and 5-40) (DP10)

4.2 Environmental Legislation, Policies, Plans, Standards and Criteria

General

4.2.1 Noise impacts are assessed in accordance with the criteria and methodology given in the Technical Memoranda (TMs) under the Noise Control Ordinance (NCO), and the EIAO.

4.2.2 The NCO and EIAO provide the statutory framework for noise control. Assessment procedures and standards are set out in the following TMs:

- EIAO-TM;
- TM on Noise from Construction Work other than Percussive Piling (GW-TM);
- TM on Noise from Percussive Piling (PP-TM);
- TM on Noise from Construction Work in Designated Areas (DA-TM); and
- TM on Noise from Places other than Domestic Premises, Public Places or Construction Sites (IND-TM).

4.2.3 With regard to the assessments of the construction noise impact during restricted hours and operational fixed plant noise impact, the NCO designates acceptable noise levels for Noise Sensitive Receivers (NSRs) on the basis of an Area Sensitivity Rating (ASR), based on the characteristics of the area within which they are located such as rural, village, low-density residential, or urban (see **Table 4.1**). Within these areas, the presence of "influencing factors" (such as the presence of industrial activities or major roads) can further affect the ASR and hence the acceptable noise levels.

Table 4.1 Area Sensitivity Ratings

Type of Area Containing NSR	Degree to which NSR is Affected by Influencing Factor		
	Not Affected	Indirectly Affected	Directly Affected
Rural area	A	B	B
Urban area	B	C	C
Low density residential area consisting of low-rise or isolated high-rise developments	A	B	C
Area other than those above	B	B	C

Construction Noise

4.2.4 The NCO provides the statutory framework for noise control of construction works, other than percussive piling, using powered mechanical equipment (PME) between the hours of 1900 and 0700 hours or at any time on Sundays and general holiday (that is, restricted hours). Noise control on construction activities taking place at other times is subject to the Criteria for Evaluating Noise Impact stated in Table 1B of Annex 5 in the EIAO-TM. The criteria are summarised in **Table 4.2**.

Table 4.2 Noise Criteria for Daytime Construction Noise

Uses	Daytime Noise Criteria in Leq(30min), dB(A))
Domestic premises	75
Educational Institution	70
Educational Institution (during examination)	65

Note: The above standards apply to uses which rely on opened windows for ventilation.

4.2.5 Between 1900 and 0700 hours and all day on Sundays and public holidays, activities involving the use of PME for the purpose of carrying out construction works is prohibited unless a construction noise permit (CNP) has been obtained. A CNP may be granted provided that the Acceptable Noise Level (ANL) for the NSRs can be complied with. ANLs are assigned depending upon the ASR. The corresponding basic noise levels (BNLs) for evening and night-time periods are given in **Table 4.3**.

Table 4.3 Basic Noise Levels (BNLs)

Time Period	Basic Noise Level (BNLs), dB(A)		
	ASR A	ASR B	ASR C
Evening (1900 to 2300 hours) (1)	60	65	70
Night (2300 to 0700 hours)	45	50	55

Note:(1) Includes Sundays and Public Holidays during daytime and evening.

4.2.6 The Noise Control Authority will consider a well-justified CNP application, for construction works within restricted hours as guided by the relevant Technical Memoranda issued under the NCO. The Noise Control Authority will take into account of contemporary conditions / situations of adjoining land uses and any previous complaints against construction activities at the site before making his decision in granting a CNP. Nothing in this EIA shall bind the Noise Control Authority in making his decision. If a CNP is to be issued, the Noise Control Authority shall include in it any condition he thinks fit. Failure to comply with any such conditions will lead to cancellation of the CNP and prosecution action under the NCO.

4.2.7 Under the DA-TM, the use of five types of Specified Powered Mechanical Equipment (SPME) and three types of Prescribed Construction Work (PCW) within a designated area during restricted hours would require a valid CNP. The SPME includes hand-held breaker, bulldozer, concrete lorry mixer, dump truck and hand-held vibratory poker. The PCW are:

- Erecting or dismantling of formwork or scaffolding;
- Loading, unloading or handling of rubble, wooden boards, steel bars, wood or scaffolding material; and
- Hammering.

4.2.8 In general, it should not be presumed that a CNP would be granted for carrying out PCW within a designated area during restricted hours. The CNP may be granted for the execution of construction works during restricted hours involving the use of PME and/ or SPME if the relevant Acceptable Noise Levels and criteria stipulated in the GW-TM and DA-TM can be met.

4.2.9 Percussive piling is prohibited between 1900 and 0700 hours on any weekday not being a general holiday and at any time on Sunday or general holiday. A CNP is required for the carrying out of percussive piling between 0700 and 1900 hours on any day not being a general holiday. PP-TM sets out the requirements for working and determination of the permitted hours of operations. Referring to the construction method for the Project, percussive piling is not required.

Road Traffic Noise

4.2.10 For road traffic noise, the following $L_{10(1\ hour)}$ criteria stipulated in Annex 5, Table 1A of EIAO-TM are adopted for different types of noise sensitive receivers (NSRs):

- 70 dB(A) at 1 m from the external façades of residential dwellings, hotels, offices;
- 65 dB(A) at 1 m from the external façades of schools, places of public worship, courts of law, places where unaided voice communication is required; and
- 55 dB(A) at 1 m from the external façades of hospital or clinics.

Rail Noise

4.2.11 The EIAO-TM and IND-TM stipulate the appropriate ANL for rail noise. The ANLs are dependent on ASRs of the NSRs and are shown in **Table 4.4**. The EIAO-TM specifies the noise limit of L_{max} to be 85 dB(A) during night-time. In accordance with the Hong Kong Planning Standards and Guidelines (HKPSG), a 24-hour averaged noise level of 65dB(A) $L_{eq,24hr}$ has been specified.

Table 4.4 Rail Noise Criteria

NSR Description	ANL on Different Area Sensitivity Rating for Rail Noise, (Leq, 30min, dB(A))					
	Daytime and Evening time (0700 to 2300 hrs)			Night-time (2300 to 0700 hrs)		
	A	B	C	A	B	C
Domestic premises, hotel, hostel and hospitals	60	65	70	50	55	60
Churches, school, medical clinics, and performing arts centre [b]	60	65	70	[a]		

Note:
[a] No sensitive use/activities during this period.
[b] Churches, school, performing arts centre and clinics are considered to be noise sensitive during daytime and evening time only.

Fixed Plant Noise

4.2.12 Fixed plant noise sources are controlled by Section 13 of the NCO and IND-TM. Noise criteria for fixed plant noise impact with a noise criteria of 5dB(A) below the appropriate ANL shown in Table 2 of the IND-TM or the prevailing background noise levels (for quiet areas with level 5 dB(A) below the ANL) recommended in the Annex 5 of EIAO-TM for planning purposes are applied to this study. The ANLs and criteria for different ASRs are summarised in **Table 4.5** below.

Table 4.5 Noise Criteria for Fixed Noise Sources

Time Period	Fixed Plant Noise for Different Area Sensitivity Rating ($L_{eq\ 30\ min}$, dB(A))					
	ANL, dB(A)			ANL-5, dB(A) for Fixed Plant Noise Criteria		
	ASR A	ASR B	ASR C	ASR A	ASR B	ASR C
Day (0700 to 1900 hrs)	60	65	70	55	60	65
Evening (1900 to 2300 hrs)	60	65	70	55	60	65
Night (2300 to 0700 hrs)	50	55	60	45	50	55

4.2.13 The Project areas are located in the proximity to developing areas, mixed with busy heavy vehicles road traffic and rail traffic, but also with scattered village developments. Thus, the prevailing background noise measurements at selected representative NSRs have been conducted. The ASR of the type of area within the NSR located is determined for assessment. When determining the ASR for a NSR, the following are considered:

- The type of area within the concerned NSR is located;
- The Influencing Factor (IF) identified;
- The Degree to which the NSR is affected by IF; and
- If applicable, nearby OZP "Industrial" or "Industrial Estates" zone.

4.2.14 A summary of the prevailing background noise level measured at the representative existing NSRs within the Project is given in **Table 4.6** and the locations of background noise measurement are shown in **Appendix 4.2.1 and Appendix 4.2.2**.

Table 4.6 Summary of Measured Prevailing Background Noise Level

ID	Measurement Location	Minimum Measured Noise Level, Leq (30 min) dB(A)		
		Daytime (0700-1900)	Evening (1900-2300)	Night-time (2300-0700)
M01	San Lee Uk Tsuen (Along Tin Ha Road, outside the Zion Church of U.B. in Christ)	68	68	58
M02	Tin Sum Tsuen (Outside Tin Sum Playground)	49	49	39
M03	San Sang Tsuen (Along Tin Ha Road, opposite to San Sang Tsuen Sitting-out Area)	53	50	40
M04	Sha Chau Lei Tsuen (Opposite to Ching Chung Care & Attention Home for the Aged)	52	49	42
M05	Sik Kong Tsuen (Near Long View Garden Block 9 & 10)	52	50	45
M06	Tseung Kong Wai (Near Tseung Kong Wai Public Toilet)	55	55	46
M07	Fung Kong Tsuen	51	46	39

Helicopter Noise

4.2.15 There is an existing helipad near Ha Tsuen Interchange in the vicinity of the Project. According to Table 1A of Annex 5 "A Summary of Noise Criteria" in the EIAO-TM, the criteria for helicopter noise is L_{max} 90 dB(A) for offices and L_{max} 85 dB(A) for domestic premises, hotels, hostels, education institutes, places of worship, hospital and clinics during 0700-1900 hours. The standards apply to uses that rely on openable windows for ventilation and the standards should be viewed as the maximum permissible noise levels assessed at 1 m from the external façade.

4.2.16 In accordance with the EIAO-TM, there are no specified evening and night-time assessment criteria for helicopter noise yet. Based on the requirements under Annex 5(d) and section 4.4.2(c) of EIAO-TM, criteria for evaluating such helicopter noise impacts shall be determined on a case-by-case basis so as to assess whether the proposed methodology are consistent with the methodologies adopted for Hong Kong Projects having similar issues or with methodologies accepted by recognised national / international organisation. By referring to the approved EIA Reports on Helipad at Yung Shue Wan, Lamma Island (AEIAR-094/2006, hereinafter YSW EIA) and Peng Chau Helipad (AEIAR-087/2005), those helipads are intended for emergency operation only. In view of emergency nature and without scheduled flight, noise criteria were not proposed for emergency helicopter operation during evening and night-time period.

4.2.17 As review the helicopter noise assessment criteria adopted in several similar cities/countries including London (United Kingdom), California (United States), Singapore, Hamilton (New Zealand) and Victoria (Australia). The findings reveal that there is no evening and night-time noise criteria or guidelines adopted for emergency helicopter operation, such as fire-fighting, medical emergencies and law enforcement, etc. As stipulated in the approved EIA Report of "Deep Bay Link" (EIA-078/2002), this existing helipad is for emergency use. The use of this helipad during evening and night-time period has similar situations to those reviewed cities/countries.

4.3 Baseline Conditions

4.3.1 The HSK NDA (herein referred to as the "Project") is located at the north of Castle Peak Road – HSK Section, where bounded by Tin Ying Road, Kiu Hung Road on the east, and KSWH on the west. There is a small hill extending from Yuen Tau Shan to its immediately northwest and further north is Deep Bay. The locality of area within Project area is predominantly in rural. There are some scattered existing village developments (i.e. Ha Tsuen, Kau Lee Uk Tsuen, San Lee Uk Tsuen and Tin Sam) in the north and west of the Project. While some existing mid to high-rise residential developments are located in the south along Castle Peak Road and southeast along Hung Tin Road of the Project. Industrial premises/factories are also scattered throughout the area. Other facilities such as Tin Hau Temple, ancestral hall, shrine and study hall are also located in the area. Currently, the Project area mainly comprises zones of villages, agriculture, open storage and industrial uses.

4.3.2 Even though the area within Project is mainly rural area, its surrounding areas are developed sub-urban area. To the south of the Project area along Castle Peak Road – HSK is a mixture of village houses and mid-rise residential developments. Developed Tin Shui Wai New Town with high-rise residential development is in its east.

4.3.3 The existing noise climate is dominated by the road traffic noise from Castle Peak Road, KSWH, Yuen Long Highway and Hung Tin Road as well as the railway noise from the West Rail Line (WRL) and Light Rail Transit (LRT) of MTR.

4.3.4 Castle Peak Road and Yuen Long Highway are heavily trafficked trunk road and expressway, respectively. The background noise level in the area along Castle Peak Road and Yuen Long Highway is high.

4.3.5 Areas around Tin Sam, San Lee Uk Tsuen, San Sang Tsuen, Kau Lee Uk Tsuen and Shek Po Tsuen are clustered rural villages, the WRL, which is an above-ground railway passing through the area would be the dominant source of the background noise.

4.3.6 Isolated industrial operations scattered throughout the Project area also contribute to the overall ambient noise levels. These operations include open storage facilities, canning factory, container back-up areas and car stripping/repair workshops. Noise emitted from the fixed plant equipment in the industrial premises is the major noise sources contributing to the existing background.

4.3.7 The northern area of the Project, including the area around Fung Kong Tsuen, Sha Kong Wai and San Hing Tsuen is mainly a mixture of clustered villages and open storage areas. The major noise source in the area is from the movement of heavy vehicles on local roads and the loading/unloading of containers at the storage site. Areas around Ngau Hom Shek are natural slopes, with no significant noise source in the area.

4.4 Noise Sensitive Receivers

4.4.1 In accordance with Annex 13 of the EIAO-TM, any domestic premises including temporary housing, school, educational institution, hospitals, medical clinics, homes for the aged, convalescent homes, places of worship, libraries, courts of law, performing arts centres, auditoria, amphitheatres are identified as NSRs. The assessment area is the areas within 300 m from the Project area.

4.4.2 All existing NSRs, committed and planned noise developments and uses are identified based on the Revised RODP and the below latest Outline Zoning Plans. The identified NSRs are listed in **Table 4.7**. The locations of these NSRs are shown in **Appendix 4.4.1**.

- Ha Tsuen OZP No. S/YL-HT/10
- Lam Tei and Yick Yuen OZP No. S/TM-LTYY/8
- Tong Yan San Tsuen OZP No. S/YL-TYST/12
- Ping Shan OZP No. S/YL-PS/16
- Lau Fau Shan and Tsim Bei Tsui OZP No. S/YL-LFS/7
- Tin Shui Wai OZP No. S/TSW/12

Table 4.7 Identified Existing, Committed and Planned Noise Sensitive Receivers

NSR ID	NSR Description	Existing / Planned Use
Existing NSRs		
E1-OA	Tin Wah Estate	Existing Residential
E1-OB_R	Tin Shing Court	Existing Residential
E1-OB_E0	Q.E.S. Old Students' Association Branch Primary School	Existing Educational
E1-OB_E1	YLPMSAA Tang Siu Tong Secondary School	Existing Educational
E1-OB_E2	Lions Clubs International Ho Tak Sum Primary School	Existing Educational
E2-OA	Village House at Lau Fau Shan	Existing Village
E2-OB	San Hing Tsuen	Existing Village

NSR ID	NSR Description	Existing / Planned Use
E2-OC	Sha Kong Wai	Existing Village
E2-OD	Tin Yan Estate	Existing Residential
E4-OA	Tsing Cheun Wai	Existing Village
E4-OB	Nai Wai	Existing Village
E4-OC	Sun Fung Wai	Existing Village
E4-OD	Chung Uk Tsuen	Existing Village
E5-OA	Tong Fong Tsuen	Existing Village
E5-OB	Hang Mei Tsuen	Existing Village
E5-OC	Tai Tao Tsuen	Existing Village
E5-OD	Uptown	Existing Residential
E5-OE	Shung Tak Catholic English College	Existing Educational
E5-OF	Treasure Court	Existing Residential
E5-OG	Lai Hung Garden	Existing Residential
E5-OH	Bauhinia Garden	Existing Residential
E1-IA	Tseung Kong Wai	Existing Village
E1-IB	Hong Mei Tsuen	Existing Village
E1-IC	Tung Tau Tsuen	Existing Village
E1-ID	Lo Uk Tsuen	Existing Village
E1-IE	Sik Kong Wai	Existing Village
E1-IF	Ha Tsuen Shi	Existing Village
E1-IG	San Uk Tsuen	Existing Village
E1-IH	Sha Chau Lei	Existing Village
E1-II	Kau Lee Uk Tsuen	Existing Village
E1-IJ	San Sang Tsuen	Existing Village
E1-IK	Shek Po Tsuen	Existing Village
E2-IA	Fung Kong Tsuen	Existing Village
E4-IA	Tin Sum Tsuen	Existing Village
E4-IB	Tsing Chuen Wai	Existing Village
E5-IA	Kiu Tau Wai	Existing Village
E5-IB_R	Hung Uk Tsuen	Existing Village
E5-IB_C	Chinese Mission Seminary	Existing Church
E5-IC	San Lee Uk Tsuen	Existing Village
E1-26	Existing Homes for the Elderly (Yeung Chun Pui and Ching Chung)	IC
E4-30	Ling Liang Worldwide Evangelistic Mission HSK Ling Liang Church (located at the junction of Castle Peak Road and Yick Yuen Road)	IC
E5-23	Sheffield Villas	R3
E5-25	Hung Fuk Estate	RSc
E5-31	Existing Residential Development	R2
E5-35	Bellevue Court	R4
E5-39	Existing Residential Development	R3
E5-44	The Church of Jesus Christ of Latter-Day Saints (Hung Shui Kiu) (located at the junction of Castle Peak Road and Tin Ha Road)	R3
Committed NSRs by Other Projects		
O5-06	Planned Subsidised Sales Flat	R2(SSF)c
O5-33	Planned Residential Development	R2

NSR ID	NSR Description	Existing/Planned Use
Planned NSRs under HSK Project		
P1-02	Site 1-02	R2(SSF)c
P1-03	Site 1-03	E
P1-04	Site 1-04	R2c
P1-05	Site 1-05	R2c
P1-06	Site 1-06	R2c
P1-08	Site 1-08	R2c
P1-09	Site 1-09	R2c
P1-12	Site 1-12	R2
P1-15	Site 1-15	E
P1-17	Site 1-17	R2
P1-18	Site 1-18	E
P1-19	Site 1-19	R2
P1-21	Site 1-21	R2c
P1-22	Site 1-22	E
P2-02	Site 2-02	R3
P2-03	Site 2-03	R3
P2-07	Site 2-07	R3
P2-09	Site 2-09	E
P2-10	Site 2-10	R3
P2-11	Site 2-11	R3c
P2-14	Site 2-14	R3
P2-15	Site 2-15	E
P2-16	Site 2-16	E
P2-18	Site 2-18	RR4
P2-19	Site 2-19	RR4
P2-24	Site 2-24	E
P2-25	Site 2-25	G
P2-26	Site 2-26	E
P2-28	Site 2-28	E
P2-30	Site 2-30	RSc
P2-31	Site 2-31	R2(SSF)c
P4-01	Site 4-01	G
P4-03	Site 4-03	R2(SSF)c
P4-04	Site 4-04	RSc
P4-05	Site 4-05	RSc
P4-06	Site 4-06	R2(SSF)c
P4-08	Site 4-08	E
P4-09	Site 4-09	R1c
P4-17	Site 4-17	OU
P4-20	Site 4-20	RR4
P4-22	Site 4-22	RSc
P4-24	Site 4-24	R2(SSF)c
P4-25a	Site 4-25a	OU(C&R)
P4-25b	Site 4-25b	OU(C&R)
P4-25c	Site 4-25c	OU(C&R)
P4-26	Site 4-26	R1c

NSR ID	NSR Description	Existing/Planned Use
P4-28	Site 4-28	RSc
P4-29	Site 4-29	OU(C&R)
P4-32	Site 4-32	G
P4-33	Site 4-33	E
P4-36	Site 4-36	E
P5-07a	Site 5-07a	OU(C&R)
P5-07b	Site 5-07b	OU(C&R)
P5-08	Site 5-08	OU(C&R)
P5-09	Site 5-09	OU(C&R)
P5-14	Site 5-14	E
P5-16	Site 5-16	R2(SSF)c
P5-17	Site 5-17	RSc
P5-21	Site 5-21	E
P5-22	Site 5-22	R3
P5-24	Site 5-24	R2(SSF)c
P5-26	Site 5-26	R2(SSF)
P5-28	Site 5-28	G
P5-32	Site 5-32	OU(C&R)
P5-34	Site 5-34	E
P5-37	Site 5-37	E

Note: Full names of the abbreviations of uses should be referred to the List of Abbreviations.

4.4.3 In order to evaluate the construction and operation noise impacts from the Project, representative existing/committed/planned NSRs located within the assessment area are identified for assessment. Only the first layer of NSRs has been identified for assessment because it would provide acoustic shielding to those receivers at further distance behind. For planned hospital, clinic and tertiary institution, these developments are anticipated to be provided with central air-conditionings and therefore would not considered as representative points for noise impact assessment. Noise assessment points (NAPs) to represent the identified NSRs are selected for carrying out construction and operation impact assessment and presented in following sections.

Construction Noise

4.4.4 To evaluate the construction noise impacts from the Project, the first layer of NSRs located closest to the Project work sites and with direct line of sight to the Project work sites, have been selected as representative NSRs for noise prediction and evaluation.

4.4.5 Representative NAPs considered to be potentially worst affected NSRs are selected for the construction noise assessment. The details of the APs are presented in **Table 4.8** and **Table 4.9**, and their locations are indicated in **Figure 4.4.1** – **Figure 4.4.15**.

Table 4.8 Representative Noise Sensitive Receivers and Noise Assessment Points for Construction Noise Impact Assessment (Existing NSRs outside Project Area)

NSR ID	NSR Description	Land Use	No. of Storeys	NAP ID
E4-OA	Tsing Cheun Wai	Residential	3	ETCW02
E4-OA	Tsing Cheun Wai	Residential	3	ETCW03
E4-OB	Nai Wai	Residential	3	ENW02
E4-OD	Chung Uk Tsuen	Residential	3	ESFW01
E4-OC	Sun Fung Wai	Residential	3	ESFW02
E4-OC	Sun Fung Wai	Residential	3	ESFW03
E4-OC	Sun Fung Wai	Residential	3	ESFW04
E4-OD	Chung Uk Tsuen	Residential	3	ECUT01
E4-OD	Chung Uk Tsuen	Residential	3	ECUT02
E5-OE	Shung Tak Catholic English College	Educational Institution	7	ESTC51
E5-OD	Uptown	Residential	17	EUPT02
E2-OA	Village House at Lau Fau Shan	Residential	3	ELFS01
E2-OB	San Hing Tsuen	Residential	3	ELFS02
E2-OC	Sha Kong Wai	Residential	3	ELFS03
E2-OD	Wing Jan School/Wing Jan Lutheran Church	Educational Institution	1	ELFS04
E2-OB	San Hing Tsuen	Residential	3	ELFS05
E1-OA	Tin Wah Estate	Residential	21	ETSW01
E1-OB_E1	YLPMSAA Tang Siu Tong Secondary School	Educational Institution	7	ETSW02
E1-OC	Tin Shui Estate	Residential	39	ETSW03
E1-OD	The Yuen Institute MFBM Nei Ming Chan Lui Chung Tak Memorial College	Educational Institution	7	ETSW04
E1-OC	Tin Shui Estate	Residential	37	ETSW05
E1-OE	Tin Oi Court	Residential	39	ETSW06
E1-OF	Locwood Court	Residential	27	ETSW07
E1-OG	VTC Youth College	Residential	6	ETSW08
E1-OI	Tin Shing Court	Residential	40	ETSW09
E1-OB_R	Tin Shing Court	Residential	40	ETSW10
E1-OB_E1	YLPMSAA Tang Siu Tong Secondary School	Educational Institution	7	ETSW11

Table 4.9 Representative Noise Sensitive Receivers and Noise Assessment Points for Construction Noise Impact Assessment (Existing and Planned NSRs within Project Area)

NSR ID	NSR Description	Use	No. of Domestic Storey	NAP ID
Planned NSRs within Project area				
P5-26	Site 5-26RSc	R2(SSF)	21	52602,52604
P5-24	Site 5-24R2(SSF)c	R2(SSF)c	29	52401,52407,52408
P2-18	Site 2-18RR4	RR4	-	21801,21802,21803
P2-19	Site 2-19RR4	RR4	-	21901
R4-20	Site 4-20RR4	RR4	-	42001,42005,42006

NSR ID	NSR Description	Use	No. of Domestic Storey	NAP ID
P4-22	Site 4-22RSc	RSc	33	42203,42216,42251
P4-28	Site 4-28RSc	RSc	38	42806,42853,42859
P5-32	Site 5-32OU(C&R)	O(C&R)	28	53208
P5-16	Site 5-16R2(SSF)c	R2(SSF)c	28	51601,51602
P5-17	Site 5-17RSc	RSc	29	51702,51703
P5-21	Site 5-21E	E	-	52151
P4-03	Site 4-3R2(SSF)c	R2(SSF)c	26	40305
P4-06	Site 4-6R2(SSF)c	R2(SSF)c	38	40602
P1-03	Site 1-3E	E	-	10353,10354
P2-31	Site 2-31R2(SSF)c	R2(SSF)c	32	23101
P1-12	Site 1-12R2	R2	30	11203
P1-15	Site 1-15E	E	-	11556
P1-17	Site 1-17R2	R2	31	11702
Existing NSRs within Project area				
E4-IB	Tsing Chuen Wai	Existing Village	3	ETCW01
E5-39	Existing Residential Development	Existing Residential	3	E53902
E5-39	Existing Residential Development	Existing Residential	3	E53903
E4-IA	Tin Sum Tsuen	Existing Village	3	ETST02
E4-IA	Tin Sum Tsuen	Existing Village	3	ETST04
E4-IA	Tin Sum Tsuen	Existing Village	3	ETST05a
E4-IA	Tin Sum Tsuen	Existing Village	3	ETST05b
E4-IA	Tin Sum Tsuen	Existing Village	3	ETST06
E5-IC	San Lee Uk Tsuen	Existing Village	3	ESLUT01
E5-IC	San Lee Uk Tsuen	Existing Village	3	ESLUT03
E5-IC	San Lee Uk Tsuen	Existing Village	3	ESLUT04
E5-IC	San Lee Uk Tsuen	Existing Village	3	ESLUT05
E5-35	Bellevue Court	Existing Village	3	EBC01
E5-39	Existing Residential Development	Existing Residential	3	E53905
E5-39	Existing Residential Development	Existing Residential	3	E53906
E5-31	Existing Residential Development	Existing Residential	10	E53104
E5-31	Existing Residential Development	Existing Residential	10	E53106
E5-25	Hung Fuk Estate	Existing Residential	41	E52503
E5-25	Hung Fuk Estate	Existing Residential	41	E52504
E5-25	Hung Fuk Estate	Existing Residential	41	E52505
E5-25	Hung Fuk Estate	Existing Residential	41	E52506
E5-IB_R	Hung Uk Tsuen	Existing Village	3	EHUT01
E5-IB_R	Hung Uk Tsuen	Existing Village	3	EHUT02
E5-IA	Kiu Tau Wai	Existing Village	3	EHUT03
E5-IA	Kiu Tau Wai	Existing Village	3	EHUT04
E1-IK	Shek Po Tsuen	Existing Village	3	ESPT01
E1-IK	Shek Po Tsuen	Existing Village	3	ESPT04
E1-IK	Shek Po Tsuen	Existing Village	3	ESPT05
E1-IK	Shek Po Tsuen	Existing Village	3	ESPT06
E1-IJ	San Sang Tsuen	Existing Village	3	ESST01
E1-IJ	San Sang Tsuen	Existing Village	3	ESST06
E1-IJ	San Sang Tsuen	Existing Village	3	ESST07
E1-II	Kau Lee Uk Tsuen	Existing Village	3	EKLUT01
E1-IH	Sha Chau Lei	Existing Village	3	ESCL01
E1-IH	Sha Chau Lei	Existing Village	3	ESCL03
E1-IH	Sha Chau Lei	Existing Village	3	ESCL04
E1-IG	San Uk Tsuen	Existing Village	3	E12501
E1-IG	San Uk Tsuen	Existing Village	3	E12502

NSR ID	NSR Description	Use	No. of Domestic Storey	NAP ID
E1-IG	San Uk Tsuen	Existing Village	3	E12503
E1-IF	Ha Tsuen Shi	Existing Village	3	EHTS01
E1-IE	Sik Kong Wai	Existing Village	3	ELUT01
E1-IE	Sik Kong Wai	Existing Village	3	ESKT02
E1-IA	Tseung Kong Wai	Existing Village	3	ETKW01
E1-IB	Hong Mei Tsuen	Existing Village	3	EHMT01
E1-IC	Tung Tau Tsuen	Existing Village	3	ETTT01
E1-IC	Tung Tau Tsuen	Existing Village	3	ETTT02
E2-IA	Fung Kong Tsuen	Existing Village	3	EFKT01
E2-IA	Fung Kong Tsuen	Existing Village	3	EFKT03
E2-IA	Fung Kong Tsuen	Existing Village	3	EFKT04

Note: Full names of the abbreviations of uses should be referred to the List of Abbreviations.

Road Traffic Noise

4.4.6 To evaluate the operational traffic noise impact from the Project, the first layer of existing/planned NSRs located in the vicinity of proposed road networks and the planned NSRs under the Project located in the vicinity of existing roads, have been selected as representative NSRs for road traffic noise prediction and evaluation because it would provide acoustic shielding to those receivers at further distance behind.

4.4.7 Representative NAPs considered to be worst affected NSRs are selected for traffic noise impact assessment. The details of NAPs are presented in **Table 4.10** and **Table 4.11**, and their locations are indicated in **Figure 4.4.16 – Figure 4.4.36**.

Table 4.10 Representative Noise Sensitive Receivers and Noise Assessment Points for Traffic Noise Impact Assessment (Existing NSRs outside Project Area)

NSR ID	Description	Existing / Planned Use	No. of Storey	NAP ID
E1-OA	Tin Wah Estate	Existing Residential	35-40	E1-OA_R01
E1-OB_R	Tin Shing Court	Existing Residential	35-40	E1-OB_R01
E1-OB_E2	Lions Clubs International Ho Tak Sum Primary School	Existing Educational	8	E1-OB_E21
E1-OB_E1	YLPMSAA Tang Siu Tong Secondary School	Existing Educational	8	E1-OB_E11 to E1-OB_E12
E1-OB_E0	Q.E.S. Old Students' Association Branch Primary School	Existing Educational	8	E1-OB_E01 to E1-OB_E02
E2-OA	Village House at Lau Fau Shan	Existing Village	1-3	E2-OA_R01
E2-OB	San Hing Tsuen	Existing Village	1-3	E2-OB_R01 to E2-OB_R03
E2-OC	Sha Kong Wai	Existing Village	1-3	E2-OC_R01 to E2-OC_R04
E5-OD	Uptown	Existing Residential	1-19	E5-OD_R01
E5-OE	Shung Tak Catholic English College	Existing Educational	8	E5-OE_E01
E5-OF	Treasure Court	Existing Residential	4-6	E5-OF_R01
E5-OG	Lai Hung Garden	Existing Residential	10	E5-OG_R01
E5-OH	Bauhinia Garden	Existing Residential	6	E5-OH_R01
E4-OA	Tsing Cheun Wai	Existing Village	3	E4-OA_R01 to E4-OA_R04
E4-OB	Nai Wai	Existing Village	3	E4-OB_R01

NSR ID	Description	Existing / Planned Use	No. of Storey	NAP ID
E4-OC	Sun Fung Wai	Existing Village	3	E4-OC_R01 to E4-OC_R03
E4-OD	Chung Uk Tsuen	Existing Village	3	E4-OD_R01 to E4-OD_R04

Table 4.11 Representative Noise Sensitive Receivers and Noise Assessment Points for Traffic Noise Impact Assessment (Existing and Planned NSRs within Project area)

NSR ID	Description	Use	No. of Storey	NAP ID
Existing NSRs within Project Area				
E1-IA	Tseung Kong Wai	Existing Village	3	E1-IA_R01
E1-IB	Hong Mei Tsuen	Existing Village	3	E1-IB_R01
E1-IC	Tung Tau Tsuen	Existing Village	3	E1-IC_R01 to E1-IC_R03
E1-ID	Lo Uk Tsuen	Existing Village	3	E1-ID_R01
E1-IE	Sik Kong Wai	Existing Village	3	E1-IE_R01
E1-IF	Ha Tsuen Shi	Existing Village	3	E1-IF_R01 to E1-IF_R02
E1-IG	San Uk Tsuen	Existing Village	3	E1-IG_R01 to E1-IG_R03
E1-IH	Sha Chau Lei	Existing Village	3	E1-IH_R01 to E1-IH_R02
E1-IJ	San Sang Tsuen	Existing Village	3	E1-IJ_R01 to E1-IJ_R05
E1-IK	Shek Po Tsuen	Existing Village	3	E1-IK_R01 to E1-IK_R03
E2-IA	Fung Kong Tsuen	Existing Village	3	E2-IA_R01 to E2-IA_R02
E4-IA	Tin Sum Tsuen	Existing Village	3	E4-IA_R01 to E4-IA_R03
E4-IB	Tsing Chuen Wai	Existing Village	3	E4-IB_R01
E5-IA	Kiu Tau Wai	Existing Village	3	E5-IA_R01
E5-IB_R	Hung Uk Tsuen	Existing Village	3	E5-IB_R01 to E5-IB_R02
E5-IB_C	Chinese Mission Seminary	Existing Church	1	E5-IB_C01
E5-IC	San Lee Uk Tsuen	Existing Village	3	E5-IC_R01
E1-26	Existing Homes for the Elderly (Yeung Chun Pui and Ching Chung)	IC	3	E1-26_H01 to E1-26_H02
E4-30	Ling Liang Worldwide Evangelistic Mission Hung Shui Kiu Ling Liang Church	IC	1	E4-30_C01
E5-23	Sheffield Villas	R3	6	E5-23_R01 to E5-23_R12
E5-25	Hung Fuk Estate	RSc	24	E5-25_R01 to E5-25_R41
E5-31	Coronet Court	R2	11	E5-31_R01
	Park Nara		11	E5-31_R05
	Beauty Court		9	E5-31_R11
	Aster Court		10	E5-31_R21 to E5-31_R22
E5-35	Bellevue Court	R4	3	E5-35_R01
E5-39	Parkview Garden	R3	3	E5-39_R01
	Li Fong Yuen		2	E5-39_R02
	Casa De Oro		6	E5-39_R03

NSR ID	Description	Use	No. of Storey	NAP ID
	Tin Sum Villa		7	E5-39_R04
	Tak Cheung Building		6	E5-39_R05
E5-44	The Church of Jesus Christ of Latter-Day Saints (Hung Shui Kiu)	R3	1	E5-44_C01 to E5-44_C02
O5-06	Planned Subsidised Sales Flat	R2(SSF)c	33	O5-06_R01 to O5-06_R24
O5-33	Planned Residential Development	R2	13	O5-33_R01 to O5-33_R31
Planned NSRs within Project Area by Project				
P1-02	Site 1-02	R2(SSF)c	27	P1-02_R01 to P1-02_R31
P1-03	Site 1-03	E	8	P1-03_E01 to P1-03_E11
P1-04	Site 1-04	R2c	30	P1-04_R01
P1-05	Site 1-05	R2c	31	P1-05_R01 to P1-05_R11
P1-06	Site 1-06	R2c	34	P1-06_R01 to P1-06_R11
P1-08	Site 1-08	R2c	34	P1-08_R01 to P1-08_R22
P1-09	Site 1-09	R2c	32	P1-09_R01 to P1-09_R12
P1-12	Site 1-12	R2	31	P1-12_R01
P1-15	Site 1-15	E	8	P1-15_E01 to P1-15_E05
P1-17	Site 1-17	R2	27	P1-17_R01
P1-18	Site 1-18	E	8	P1-18_E01 to P1-18_E03
P1-19	Site 1-19	R2	23	P1-19_R01 to P1-19_R21
P1-21	Site 1-21	R2c	31	P1-21_R01 to P1-21_R21
P1-22	Site 1-22	E	8	P1-22_E01 to P1-22_E11
P2-02	Site 2-02	R3	6	P2-02_R01 to P2-02_R32
P2-03	Site 2-03	R3	6	P2-03_R01 to P2-03_R21
P2-07	Site 2-07	R3	6	P2-07_R01 to P2-07_R21
P2-09	Site 2-09	E	8	P2-09_E01 to P2-09_E02
P2-10	Site 2-10	R3	6	P2-10_R01 to P2-10_R31
P2-11	Site 2-11	R3c	6	P2-11_R01 to P2-11_R31
P2-14	Site 2-14	R3	6	P2-14_R01 to P2-14_R12
P2-15	Site 2-15	E	8	P2-15_R01
P2-16	Site 2-16	E	8	P2-16_E01 to P2-16_R15
P2-18	Site 2-18	RR4	6	P2-18_R01 to P2-18_R05
P2-19	Site 2-19	RR4	6	P2-19_R01
P2-24	Site 2-24	E	8	P2-24_E01 to P2-24_E11
P2-25	Site 2-25	G	8	P2-25_H01
P2-26	Site 2-26	E	8	P2-26_E01 to P2-26_E11
P2-28	Site 2-28	E	8	P2-28_E01 to P2-28_E02
P2-30	Site 2-30	RSc	8	P2-30_R01 to P2-30_R21
P2-31	Site 2-31	R2(SSF)c	8	P2-31_R01 to P2-31_R21
P4-01	Site 4-01	G	25	P4-01_R01 to P4-01_R13
P4-03	Site 4-03	R2(SSF)c	26	P4-03_R01 to P4-03_R11
P4-04	Site 4-04	RSc	35	P4-04_R01 to P4-04_R24
P4-05	Site 4-05	RSc	42	P4-05_R01 to P4-05_R14
P4-06	Site 4-06	R2(SSF)c	39	P4-06_R01 to P4-06_R12
P4-08	Site 4-08	E	8	P4-08_E01 to P4-08_E11
P4-09	Site 4-09	R1c	40	P4-09_E06
P4-20	Site 4-20	RR4	3	P4-20_R01 to P4-20_R04
P4-22	Site 4-22	RSc	33	P4-22_R01 to P4-22_R42
P4-24	Site 4-24	R2(SSF)c	29	P4-24_R01 to P4-24_R46
P4-25a	Site 4-25a	OU(C&R)	38	P4-25a_R01
P4-25b	Site 4-25b	OU(C&R)	38	P4-25b_R01 to P4-25b_R11
P4-25c	Site 4-25c	OU(C&R)	39	P4-25c_R01 to P4-25c_R11

NSR ID	Description	Use	No. of Storey	NAP ID
P4-26	Site 4-26	R1c	36	P4-26_R01 to P4-26_R21
P4-28	Site 4-28	RSc	38	P4-28_R01 to P4-28_R44
P4-29	Site 4-29	OU(C&R)	38	P4-29_R01 to P4-29_R31
P4-32	Site 4-32	G	32	P4-32_R01 to P4-32_R14
P4-33	Site 4-33	E	8	P4-33_E01 to P4-33_E03
P4-36	Site 4-36	E	8	P4-36_E02 to P4-36_R01
P5-07a	Site 5-07a	OU(C&R)	31	P5-07a_R01 - P5-07a_R21
P5-07b	Site 5-07b	OU(C&R)	31	P5-07b_R01 to P5-07b_R21
P5-08	Site 5-08	OU(C&R)	22	P5-08_R01 to P5-08_R11
P5-09	Site 5-09	OU(C&R)	33	P5-09_R01 to P5-09_R31
P5-14	Site 5-14	E	8	P5-14_E01 to P5-14_E11
P5-16	Site 5-16	R2(SSF)c	28	P5-16_R01 to P5-16_R11
P5-17	Site 5-17	RSc	29	P5-17_E04 to P5-17_R12
P5-21	Site 5-21	E	8	P5-21_E01 to P5-21_E02
P5-22	Planned Residential Development	R3	40	P5-22_R01 to P5-22_R03
P5-24	Site 5-24	R2(SSF)c	35	P5-24_R01 to P5-24_R23
P5-26	Site 5-26	R2(SSF)	22	P5-26_R01 to P5-26_R04
P5-28	Site 5-28	G	8	P5-32_R01 to P5-32_R82
P5-32	Site 5-32	OU(C&R)	29	P5-32_R01 to P5-32_R82
P5-34	Site 5-34	E	8	P5-34_E01 to P5-34_E11
P5-37	Site 5-37	E	8	P5-37_E01 to P5-37_E02

Note: Full names of the abbreviations of uses should be referred to the List of Abbreviations.

Rail Noise

4.4.8 To evaluate the rail noise impact from the Project, the first layer of existing/planned NSRs located in the vicinity of WRL and LRT, have been selected as representative NSRs for rail noise prediction and evaluation because it would provide acoustic shielding to those receivers at further distance behind.

4.4.9 Representative NAPs are considered to be worst within the affected NSRs are selected for rail noise impact assessment. The details of NAPs are presented in **Table 4.12** and their locations are indicated in **Figure 4.4.31** – **Figure 4.4.36**.

Table 4.12 Summary of Existing and Planned Noise Sensitive Receivers for Rail Noise Assessment

NSR ID	Description	Land Use	Max mPD	Type of Area Containing NSR[1]	Degree of Influence by IF[2]	Area Sensitive Rating	Criteria, dB(A) (0700 to 2300 hrs)	Criteria, dB(A) (2300 to 0700 hrs)	NAP ID
Existing WRL NSRs									
E4-IB	Premises to the southwest of Yick Yuen Tsuen	Existing Village	8.7	[i]	Not Affected	A	60	50	WR-E1
E4-IB	Premises to the east of Oakland Court	Existing Village	8.5	[i]	Not Affected	A	60	50	WR-E2
E4-IA	Premises to the southwest of Tin Sum Tsuen	Existing Village	7.7	[i]	Not Affected	A	60	50	WR-E3
E1-IH	Premises to the West of San Lee Uk Tsuen	Existing Village	6.5	[i]	Not Affected	A	60	50	WR-E4
E1-IH	Premises to the West of San Lee Uk Tsuen	Existing Village	9.9	[i]	Not Affected	A	60	50	WR-E5
E1-IJ	Block 3,Galore Garden	Existing Village	12.4	[i]	Not Affected	A	60	50	WR-E6
E1-IK	Shek Po Tsuen Village House	Existing Village	8.2	[i]	Not Affected	A	60	50	WR-E7
Project Planned WRL NSRs									
P4-06	Residential Development in Site 4-6	Residential	141	[iii]	Not Affected	B	65	55	WR-P1
P4-08	School in Site 4-8	Education	47	[iii]	Not Affected	B	65	NA	WR-P2
P4-10	Residential Development in Site 4-10	Residential	166	[iii]	Not Affected	B	65	55	WR-P3
P4-20	Residential Development in Site 4-20	Residential	19	[iv]	Not Affected	B	65	55	WR-P4
P4-22	Residential Development in Site 4-22	Residential	127	[iii]	Not Affected	B	65	55	WR-P5a
P4-22	Residential Development in Site 4-22	Residential	127	[iii]	Not Affected	B	65	55	WR-P5b
P4-22	Residential Development in Site 4-22	Residential	127	[iii]	Not Affected	B	65	55	WR-P5c
P4-22	Residential Development in Site 4-22	Residential	127	[iii]	Not Affected	B	65	55	WR-P5d
P4-22	Residential Development in Site 4-22	Residential	127	[iii]	Not Affected	B	65	55	WR-P5e
P4-22	Residential Development in Site 4-22	Residential	127	[iii]	Not Affected	B	65	55	WR-P5f

NSR ID	Description	Land Use	Max mPD	Type of Area Containing NSR[1]	Degree of Influence by IF[2]	Area Sensitive Rating	Criteria, dB(A) (0700 to 2300 hrs)	Criteria, dB(A) (2300 to 0700 hrs)	NAP ID
P4-25	Residential Development in Site 4-25	Commercial & Residential	178	[iii]	Not Affected	B	65	55	WR-P6a
P4-25	Residential Development in Site 4-25	Commercial & Residential	178	[iii]	Not Affected	B	65	55	WR-P6b
P4-25	Residential Development in Site 4-25	Commercial & Residential	178	[iii]	Not Affected	B	65	55	WR-P6c
P4-29	Residential Development in Site 4-29	Commercial & Residential	171	[iii]	Not Affected	B	65	55	WR-P7a
P4-29	Residential Development in Site 4-29	Commercial & Residential	171	[iii]	Not Affected	B	65	55	WR-P7b
P4-29	Residential Development in Site 4-29	Commercial & Residential	171	[iii]	Not Affected	B	65	55	WR-P7c
P4-36	School in Site 4-36	Education	48	[iii]	Not Affected	B	65	NA	WR-P8
P5-17	Residential Development in Site 5-17	Residential	111	[iii]	Not Affected	B	65	55	WR-P9a
P5-17	Residential Development in Site 5-17	Residential	111	[iii]	Not Affected	B	65	55	WR-P9b
P5-21	School in Site 5-21	Education	43	[iii]	Not Affected	B	65	NA	WR-P10
P5-34	School in Site 5-34	Education	43	[iii]	Not Affected	B	65	NA	WR-P11
Existing LRT NSRs									
E5-39	Existing Residential Development	R3	21.5	[iii]	Directly Affected by [a]	C	70	60	LR-E1
E5-39	Existing Residential Development	R3	29.6	[iii]	Directly Affected by [a]	C	70	60	LR-E2
E5-31	Existing Residential Development	R2	46.4	[iii]	Directly Affected by [a]	C	70	60	LR-E3
E5-31	Existing Residential Development	R2	46.1	[iii]	Directly Affected by [a]	C	70	60	LR-E4
E5-31	Existing Residential Development	R2	46.6	[iii]	Directly Affected by [a]	C	70	60	LR-E5
E5-IB_R	Village House in Hung Uk Tsuen facing Kiu Hung Road	Existing Village	14.5	[iii]	Directly Affected by [a]	C	70	60	LR-E6

NSR ID	Description	Land Use	Max mPD	Type of Area Containing NSR[1]	Degree of Influence by IF[2]	Area Sensitive Rating	Criteria, dB(A) (0700 to 2300 hrs)	Criteria, dB(A) (2300 to 0700 hrs)	NAP ID
E4-30	Ling Liang Worldwide Evangelistic Mission Hung Shui Kiu Ling Liang Church	IC	14.5	[iii]	Directly Affected by [a]	C	70	NA	LR-E7
Project Planned LRT NSRs									
P4-22	Residential Development in Site 4-22	Residential	127	[iii]	Directly Affected by [a]	C	70	60	LR-P1
P4-26	Residential Development in Site 4-26	Residential	152	[iii]	Directly Affected by [a]	C	70	60	LR-P2
P4-28	Residential Development in Site 4-28	Residential	144	[iii]	Directly Affected by [a]	C	70	60	LR-P3a
P4-28	Residential Development in Site 4-28	Residential	144	[iii]	Directly Affected by [a]	C	70	60	LR-P3b
P4-33	School in Site 4-33	Education	48	[iii]	Directly Affected by [a]	C	70	NA	LR-P4
P5-07	Residential Development in Site 5-7b	Commercial & Residential	142	[iii]	Not Affected	B	65	55	LR-P5a
P5-07	Residential Development in Site 5-7b	Commercial & Residential	142	[iii]	Not Affected	B	65	55	LR-P5b
P5-07	Residential Development in Site 5-7b	Commercial & Residential	142	[iii]	Not Affected	B	65	55	LR-P5c
P5-09	Residential Development in Site 5-9	Commercial & Residential	152	[iii]	Not Affected	B	65	55	LR-P6

Notes:
[1] Type of Area Containing NSR: [i] – Rural area, including country parks or village type developments
[ii] – Low density residential area consisting of low-rise or isolated high-rise developments
[iii] – Urban area (High-rise buildings within mixture of residential, commercial and governmental institution, etc.)
[iv] – Area other than [i], [ii] and [iii]
[2] Influencing Factor (IF): [a] – Castle Peak Road – Hung Shui Kui Section (AADT > 30,000)

Fixed Plant Noise

4.4.10 The planned NSRs in the vicinity of existing fixed plant noise sources within the Project, and the existing and planned NSRs in the vicinity of the planned fixed plant noise sources are assessed to evaluate the potential fixed plant noise impact. Representative NAPs considered to be potentially worst affected NSRs are selected for the fixed plant noise assessment. The details of NAPs are presented in **Table 4.13**, and their locations are indicated in **Figure 4.4.37 – Figure 4.4.51**.

Table 4.13 Representative Noise Sensitive Receivers and Assessment Points for Fixed Plant Noise Impact Assessment

NSR ID / Site Ref. No.	Use	Description	No. of Domestic Storey	Height, mAG	Representative NAP ID
E1-1A	V	Tseung Kong Wai	3	8.2	E1-IA_R01 to E1-IA_R02
E1-IJ	V	San Sang Tsuen	3	8.2	E1-IJ_R05
E1-IK	V	Shek Po Tsuen	3	8.2	E1-IK_R04 to E1-IK_R06
E2-IA	V	Fung Kong Tsuen	3	8.2	E2-IA_R03
E1-OB_E1	E	YLPMSAA Tang Siu Tong Secondary School	8	36	E1-OB_E11
E1-OB_R	R	Tin Shing Court	35	100	E1-OB_R02 to E1-OB_R03
Site 1-12	R	Private Residential Development	31	109	P1-12_R01
Site 1-19	R	Private Residential Development	24	89	P1-19_R11
Site 1-21	R	Private Residential Development			P1-21_R22
Site 2-24	E	1 Primary School & 1 Secondary School	8	36	P2-24_E03
Site 2-26	E	1 Primary School & 1 Secondary School	8	36	P2-26_E11
Site 2-28	E	1 Primary School	8	36	P2-28_R02
Site 4-1	R	Fire Station Staff Quarters	30	134.8	P4-01_R02
Site 4-3	R	Subsidised Sales Flat (with Commercial)	26	92	P4-03_R01 to P4-03_R02
Site 4-4	R	Public Rental Housing (with Commercial)	42	143	P4-04_R13
Site 4-5	R	Public Rental Housing (with Commercial)	42	143	P4-05_R03 to P4-05_R13
Site 4-6	R	Subsidised Sales Flat (with Commercial)	39	131	P4-06_R11, P4-06_R12
Site 4-10	R	Private Residential (with Commercial)	43	155	P4-10_R01 to P4-10_R11
Site 4-20	V	Village Resite	3	8.2	P4-20_R01, P4-20_R05
Site 4-22	R	Public Rental Housing (with Commercial)	34	116	P4-22_R01
Site 4-25c	C/R	Commercial & Residential Development	45	165.7	P4-25c_R01 to P4-25c_R11
Site 4-28	R	Public Rental Housing (with Commercial)	39	131	P4-28_R41
Site 4-29	C/R	Commercial & Residential Development	43	158	P4-29_R21
Site 5-6	R	Planned Subsidised Sales Flat (with Commercial)	33	103	O5-06_R13, O5-06_R25
Site 5-8	C/R	Commercial & Residential Development	39	151.9	P5-08_R11 to P5-08_R31

NSR ID / Site Ref. No.	Use	Description	No. of Domestic Storey	Height, mAG	Representative NAP ID
Site 5-9	C/R	Commercial & Residential Development	40	147	P5-09_R32
Site 5-14	E	1 Primary School & 1 Secondary School	8	36	P5-14_E01
Site 5-16	R	Subsidised Sales Flat (with Commercial)	29	101	P5-16_R05
Site 5-21	E	1 Secondary School	8	36	P5-21_E01
Site 5-34	E	1 Primary School & 1 Secondary School	8	36	P5-34_E01

Note: Full names of the abbreviations of uses should be referred to the List of Abbreviations.

Helicopter Noise

4.4.11 There is no planned helipad to be proposed within the Project area. In the northwestern Project area, there is an existing helipad near Ha Tsuen Interchange for emergency uses of KSWH. Referring to the Revised RODP, the developments in the vicinity of this existing helipad would be port-back up area, logistic facilities, RTS, STWs which are not considered as NSRs. The nearest planned NSRs under Project are located at least 1 km from this existing helipad. **Table 4.14** presents the identified representative planned and existing NSRs nearest to the helipad. The locations of the helipad and the nearest NSRs are presented in **Figure 4.4.52**.

Table 4.14 Representative Noise Sensitive Receivers and Assessment Points for Helicopter Noise Impact Assessment

ID / Site Ref. No.	Use	Description	No. of Domestic Storey	Height, mAG	Representative NAP ID
Site 2-28	E	Planned Primary School	8	36	P2-28_E01
E1-IA	V	Existing Village - Tseung Kong Wai	3	8.2	E1-IA_R01

Note: Full names of the abbreviations of uses should be referred to the List of Abbreviations.

4.5 Identification of Pollutant Sources

Construction Phase

4.5.1 The Project has divided into five stages for development including Advance Works, and Stage 1 to Stage 4. Based on the construction programme, potential construction noise impacts of the Project would be arise from the use of PME during the following major construction works:

- Site formation;
- Construction of Road P1 (DP1);
- Construction of Road D1, D2, D3, D4, D5, D6, D7, D8 (DP2), partly depressed and partly decked-over road located at Road D2, Road D4 and Road D6 (DP6) and local roads;
- Construction of slip roads: between Road D8 Junction and existing Castle Peak Road; between Junction of D8/P1 and Junction of D7/P1; and from KSWH connection to Road D3 (DP5);
- Construction of Road P1 and slip-road from KSWH to Road D3 partly located within the “Conservation Area” of Yuen Tau Shan (DP12);

- Site development for residential, "G/IC", commercial, industrial development and open space areas;
- Site development for a new district cooling system (DCS), port back-up and storage use sites, RTS, logistics and enterprise & technology park sites; and
- Construction of four SPSs (DP9) and associated rising mains.

4.5.2 The plant inventory as provided by the Project Engineer and details of PME adopted for the assessment are presented in **Appendix 4.5.1**. **Appendix 4.5.2** presents the work programme, locations of the work sites and distances between the notional sources and the NSRs. The plant inventory and associated plant utilisation rates adopted for the assessment have been confirmed by the Project Engineer. **Appendix 4.5.3** presents the sound power levels (SWLs) of various PME and relevant source of reference.

4.5.3 The currently envisaged construction programme would not require any works using PME during restricted hours (i.e. the hours of 1900 to 0700 hours or at any time on a general holiday) nor percussive piling works. Therefore, only the impacts from daytime general construction activities are assessed in this EIA study.

4.5.4 If the Contractors in future use percussive piling for some construction works or the use of PMEs within the restricted hours, CNP should be obtained from the Noise Control Authority prior to commencement of the relevant construction works.

Operation Phase

4.5.5 Operational phase impacts will arise from the following operations:

- Road traffic noise within 300 m area from the Project boundary;
- Rail noise (WRL, LRT and proposed EFTS within the Project);
- Fixed plant noise from existing fixed plant noise sources within 300 m assessment area and planned fixed plant noise sources of the Project.

Road Traffic Noise

4.5.6 The proposed road network within the Project comprises of primary distributor (Road P1), district distributors (Road D1 to D8) and local roads. Together with the existing heavily trafficked roads surrounding the development site such as Castle Peak Road, KSWH, Hung Tin Road and Tin Ha Road, potential impact by road traffic noise would affect the planned NSRs within the development site.

Rail Noise

4.5.7 The operation of the existing WRL and LRT would pose potential noise impact on the some planned NSRs in the vicinity of its alignments and stations.

4.5.8 The EFTS is proposed to be introduced to operate as a major internal mode of transport within the Project. It is proposed to provide feeder services between TSW LRT station, Nai Wai LRT Station and proposed HSK WRL Station so as to connect the Project with Tuen Mun, TSW and Yuen Long New Towns and the urban area. The mode of EFTS may be in the form of rail based or road based mode of transport subject to further study. For the purpose of the EIA study, the assessment presented has assumed the EFTS to be designed as rail based. As discussed in **Section 1.5** of this report, the proposed EFTS is a potential Schedule 2 DP under the EIAO if the selected system is rail type. The proposed EFTS will be designed, constructed and operated by the others. The associated environmental impacts will be adequately addressed in a further detailed EIA study to be prepared and submitted under the EIAO by the respective project proponent.

Fixed Plant Noise

4.5.9 Existing and planned fixed noise sources as identified within the Project area are summarised in **Table 4.15**.

Table 4.15 List of Existing and Planned Fixed Plant Noise Sources

Types	Noise Sources	Reference Location[1]
Existing Sources	Pumping Stations	Sik Kong Tsuen, Sik Kong Wai, Lo Uk Tsuen, Kiu Tau Wai, Shung Cheung Wai, Site 1-23
	San Wai STW	Site 3-17
Planned Sources	SPSs	Sites 2-34, 3-41, 3-48, 4-35
	New HSK STW	Site 3-26
	Port Back-up, Storage and Workshop Uses	Sites 3-1, 3-4, 3-5, 3-6, 3-7, 3-8, 3-11, 3-13 and 3-14
	District Cooling System (DCS)	Sites 1-24, 4-18
	Electricity Substation (ESS)	Sites 4-23, 4-37, 5-19
	Public Transport Interchange (PTI) and Vehicle Depot	Sites 2-23, 4-13a, 4-21, 4-29, 5-5
	Sports Ground	Site 1-29
	Fire Station	Site 4-1
	Hospital	Site 5-13
	Flood Retention Facilities	Sites 1-21, 1-32, 2-13, 2-32, 3-31, 4-11, 4-19, 5-11 and 5-43
	EFTS Depot	Site 3-18
	Parking and Operational Facilities for EFTS	Site 2-29
	Proposed HSK Station	Site 4-34
	Industrial Areas	Sites 3-43, 3-44, 3-45, 3-46, 3-47, 3-50, 3-51 and 3-52

Notes: [1] Reference source no. is shown in Figures 4.4.37 – 4.4.51.

4.5.10 All flood retention facilities will be installed underground with a small concrete building structure for maintenance use. The ventilation fans would be operated during maintenance period only. A preliminary environmental review for the flood retention facilities would be conducted in the design stage under ETWB Technical Circular (Works) No. 13/2003 to address the potential impact. The nature and operation of the other fixed plant noise sources are described in **Section 4.9**.

4.5.11 There are some existing electricity substation (ESS) located within the 300 m study area. The major noise sources come from the transformer and ventilation system. Based on the site observation, their equipment which generate noise are enclosed in a building structure and the exhaust of the ventilation system is located away from any NSRs. Acoustic louvers are generally to be installed at the exhaust which effectively attenuate the noise generated during operation of the ESS. The locations of the exhaust of the ventilation system or chillers should be located away from the NSRs as far as possible. Noise impact to the NSRs from the operation of the ESS is therefore insignificant.

4.5.12 There are existing industrial developments/factories at Kiu Tau Wai which pose potential fixed plant/industrial noise impacts on the nearby NSRs. In accordance with the programme, the building construction works at planned residential developments at Site 5-16 and 5-17 which are located in the vicinity of Kiu Tau Wai would be completed in Year 2030 and may be occupied in the same year. The site clearance at Kiu Tau Wai would be commenced in Year 2031. It may have one year industrial/residential interface

issue. However, the building construction works at commercial developments at Site 5-3a, 5-3b, 5-18a and 5-18b would be completed in Year 2030. These developments are located between the concerned residential sites and Kiu Tau Wai which can provide the screening effect for fixed plant/industrial noise impacts and the potential noise impacts on these two residential sites would be minimised. In addition, the planned industrial sites located next to KSWH would be ready by Year 2030. It is possible to arrange the industries at Kiu Tau Wai to move to these planned sites early so as to reduce the extent of the noise impact on these two residential sites.

Helicopter Noise

4.5.13 The existing helipad near Ha Tsuen Interchange is for emergency uses of KSWH. There is no routine flight. Referring to YSW EIA, a setback of 246 m is required for the approach/departure of the Super Puma AS332 L2 Type Helicopter which is the loudest helicopter used by Government Flight Services. As the nearest planned NSR located at more than 1km from this helipad, the operation of this helipad would not pose helicopter noise impact to the planned NSRs within the Project.

4.6 Construction Noise Impact Assessment

Assessment Approach and Methodology

4.6.1 The construction activities of the Project taking place concurrently within 300 m of a given NSR are considered to contribute to the cumulative impact at the NSR. Any construction activities of the work areas taking place within 300 m of a given NSR are included in the assessment. Based on the available information of the concurrent projects as presented in **Section 2.12** of the EIA Report, the work sites of the concurrent projects with confirmed programme would not be located within 500 m assessment area during construction phase of the Project, therefore, no cumulative noise impact would be anticipated.

4.6.2 The assessment of construction noise impact is undertaken based on standard acoustic principles. SWLs of PME are taken from Table 3 of the Technical Memorandum on Noise from Construction Work other than Percussive Piling (GW-TM) issued under the NCO. Where no SWL is provided in the GW-TM, reference is made to British Standard BS5228 “Code of practice for noise and vibration control on construction and open sites”, Part 1 or other previous similar studies or from measurements taken at other sites in Hong Kong. The availability in Hong Kong of the PME adopted for this assessment is confirmed by the Project Engineer who is responsible for the design of construction works of the Project. A positive 3 dB façade correction is applied to the predicted noise levels to account for the façade effect at each noise assessment point.

Prediction and Evaluation of Environmental Impacts

4.6.3 For normal daytime working hours, exceedances of the construction noise criteria (Leq (30 min) 75 dB(A) for residential uses and 70 dB(A) for educational institutions (65 dB(A) during examinations)) are predicted at representative NSRs in the absence of mitigation measures. Details of the construction noise calculations and results for the unmitigated scenario are presented in **Appendix 4.6.1**.

4.6.4 A summary of the unmitigated construction noise levels at the representative NSRs during normal daytime working hours within the construction period of the Project is listed in **Table 4.16 - Table 4.18**. Noise mitigation measures would therefore be required to reduce noise levels at the affected NSRs for compliance with the noise standard.

Table 4.16 Summary of Unmitigated Construction Noise Assessment Results (Existing NSRs outside the Project Area) during Normal Daytime Working Hours

NAP ID	NAP Description	Criterion, dB(A) [(a)]	Predicted Mitigated Construction Noise Levels, $L_{eq\ (30min)}$ dB(A)	Duration of Exceedance (Months)
ETCW02	No. 739 Oaklands Court	75	68 - 85	38
ETCW03	No. 703A, Chung Long Terrace	75	67 - 79	35
ENW02	No. 202, Shun Fung Wai	75	68 - 81	23
ESFW01	No. 332, Chung Uk Tsuen	75	73 - 79	47
ESFW02	Village house, Nai Wai	75	64 - 82	32
ESFW03	No. 368, Sun Fung Wai	75	67 - 82	35
ESFW04	No. 716, Chung Uk Tsuen	75	69 - 81	24
ECUT01	No. 16, Chung Uk Tsuen	75	68 - 79	32
ECUT02	No. 203, Chung Uk Tsuen	75	71 - 80	21
ESTC51	Shung Tak Catholic English College	70 (65)	64 - 73	22
EUPT02	Flat C, Tower 1, Uptown	75	74 - 74	0
ELFS01	No. 30 Lau Fau Shan	75	60 - 82	7
ELFS02	No. 3H, San Hing Tsuen	75	65 - 83	7
ELFS03	No. 310 Sha Kong Wai	75	61 - 82	25
ELFS04	Wing Jan School/Wing Jan Lutheran Church	70 (65)	63 - 81	7
ELFS05	No. 1, Lau Fau Shan Road	75	65 - 81	29
ETSW01	Wah Yat House, Tin Wah Estate	75	72 - 78	15
ETSW02	Caritas Lok Kan School	75	67 - 79	18
ETSW03	Shui Lung House, Tin Shui Estate	75	71 - 78	34
ETSW04	The Yuen Institute MFBM Nei Ming Chan Lui Chung Tak Memorial College	70 (65)	69 - 78	23
ETSW05	Shui Fung House, Tin Shui Estate	75	63 - 80	10
ETSW06	Oi Tao House, Tin Oi Court	75	73 - 80	36
ETSW07	Block 7, Locwood Court	75	66 - 82	26
ETSW08	VTC Youth College (Tin Shui Wai)	70 (65)	71 - 84	93
ETSW09	Shing Ting House, Tin Shing Court	75	64 - 85	48
ETSW10	Shing Kan House, Tin Shing Court	75	66 - 80	28
ETSW11	YLPMSAA Tang Siu Tong Secondary School	70 (65)	66 - 81	73

Notes:

(a) Construction noise assessment criterion is 75 dB(A) for domestic premises and 70dB(A) for education institutions (65dB(A) during examinations).

Table 4.17 Summary of Unmitigated Construction Noise Assessment Results (Existing NSRs within the Project Area) during Normal Daytime Working Hours

NAP ID	NAP Description	Criterion, dB(A) [(a)]	Predicted Unmitigated Construction Noise Levels, $L_{eq\ (30min)}$ dB(A)	Duration of Exceedance (Months)
ETCW01	No. 430, village house	75	67 - 82	23
E53902	No. 125, Lee Fong Yuen	75	68 - 84	42
E53903	Village house, Tin Ha Road	75	69 - 84	42
ETST02	No. 13, Tin Sum	75	65 - 78	11
ETST04	No.211, Tin Sum	75	69 - 71	0
ETST05a	No. 143 ,Tin Sum (South Façade)	75	68 - 86	36
ETST05b	No. 143, Tin Sum, (West Façade)	75	68 - 87	39
ETST06	No. 290, Tin Sum	75	68 - 82	50
ESLUT01	No. 46A, San Lee Uk Tsuen	75	67 - 88	26
ESLUT03	No. 62, San Lee Uk Tsuen	75	67 - 90	108
ESLUT04	No. 254, San Lee Uk Tsuen	75	65 - 92	56
ESLUT05	No. 120, San Lee Uk Tsuen	75	68 - 93	65
EBC01	House 15, Bellevue Court	75	64 - 89	80
E53905	Village House, Hung Shui Kiu Tin Sam Road	75	64 - 81	26
E53906	Flat C, Tin Sam Villa	75	66 - 86	12
E53104	Flat C, Block 1, Aster Court	75	69 - 83	12
E53106	Flat D, Block 1, Aster Court	75	68 - 82	74
E52503	Hung Long House, Hung Fuk Estate	75	68 - 82	31
E52504	Hung Yan House, Hung Fuk Estate	75	66 - 82	26
E52505	Hung Yan House, Hung Fuk Estate	75	67 - 85	82
E52506	Hung Shing House, Hung Fuk Estate	75	63 - 81	95
EHUT01	No. 141, Hung Uk Tsuen	75	72 - 86	31
EHUT02	No. 81A, Hung Uk Tsuen	75	80 - 85	54
EHUT03	No. 112, Kiu Tau Wai	75	76 - 86	54
EHUT04	No. 85A, Kiu Tau Wai	75	76 - 87	23
ESPT01	No. 181, Shek Po Tsuen	75	66 - 80	31
ESPT04	Village house, Shek Po Tsuen	75	62 - 84	71
ESPT05	No. 215, Shek Po Tsuen	75	62 - 85	36
ESPT06	No. 201, Shek Po Tsuen	75	62 - 87	36
ESST01	Block L, Bun Shing Garden	75	67 - 86	48
ESST06	Block H, Fu Lam Garden	75	62 - 86	68
ESST07	No, 60, San Sang Tsuen	75	64 - 94	38
EKLUT01	Village house, Kau Lee Uk Tsuen	75	75 - 84	52
ESCL01	No. 06, Sha Chau Lei Tsuen	75	65 - 89	55

NAP ID	NAP Description	Criterion, dB(A) [(a)]	Predicted Unmitigated Construction Noise Levels, $L_{eq\ (30min)}$ dB(A)	Duration of Exceedance (Months)
ESCL03	No. 45, Sha Chau Lei Tsuen	75	68 - 89	39
ESCL04	No. 18, Sha Chau Lei Tsuen	75	65 - 85	55
E12501	No. 108, Ha Tsuen Shi	75	75 - 84	55
E12502	No. 115, Ha Tsuen Shi	75	76 - 83	57
E12503	No. 216, Ha Tsuen Shi	75	61 - 82	36
EHTS01	No. 1B, San Uk Tsuen	75	69 - 91	36
ELUT01	Block 11, Yan Wu Garden	75	82 - 87	37
ESKT02	No. 151, Sik Kong Wai	75	76 - 87	24
ETKW01	No.108C, Tseung Kong Wai	75	65 - 82	19
EHMT01	No. 411, San Wai	75	61 - 88	21
ETTT01	Block A Luxor Garden, Tung Tau Tsuen	75	69 - 85	43
ETTT02	No. 111, Lo Uk Tsuen	75	68 - 84	38
EFKT01	No.61, Fung Kong Tsuen	75	64 - 85	42
EFKT03	No.80, Fung Kong Tsuen	75	64 - 82	54
EFKT04	No.23F, Fung Kong Tsuen	75	70 - 80	41

Notes:
(a) Construction noise assessment criterion is 75 dB(A) for domestic premises and 70dB(A) for education institutions (65dB(A) during examinations).

Table 4.18 Summary of Unmitigated Construction Noise Assessment Results (Planned NSRs) during Normal Daytime Working Hours

NAP ID	NAP Description	Criterion, dB(A) [(a)]	Predicted Unmitigated Construction Noise Levels, $L_{eq\ (30min)}$ dB(A)	Duration of Exceedance (Months)
10353	Planned School in Site 1-3	70 (65)	73 - 85	12
10354	Planned School in Site 1-3	70 (65)	72 - 83	10
11203	Planned Residential Development in Site 1-12	75	71 - 82	8
11556	Planned School in Site 1-15	70 (65)	70 - 84	25
11702	Planned Residential Development in Site 1-17	75	68 - 83	8
21801	Planned Residential Development in Site 2-18	75	75 - 84	59
21802	Planned Residential Development in Site 2-18	75	73 - 82	59
21803	Planned Residential Development in Site 2-18	75	69 - 85	29
21901	Planned Residential Development in Site 2-19	75	64 - 75	0
23101	Planned Residential Development in Site 2-31	75	68 - 83	10
40305	Planned Residential Development in Site 4-3	75	80 - 84	16

NAP ID	NAP Description	Criterion, dB(A) [(a)]	Predicted Unmitigated Construction Noise Levels, $L_{eq\ (30min)}$ dB(A)	Duration of Exceedance (Months)
40602	Planned Residential Development in Site 4-6	75	74 - 82	10
42001	Planned Residential Development in Site 4-20	75	75 - 90	77
42005	Planned Residential Development in Site 4-20	75	72 - 91	77
42006	Planned Residential Development in Site 4-20	75	74 - 93	80
42203	Planned Residential Development in Site 4-22	75	68 - 84	27
42216	Planned Residential Development in Site 4-22	75	65 - 87	21
42251	Planned Residential Development in Site 4-22	75	66 - 96	36
42806	Planned Residential Development in Site 4-28	75	73 - 87	42
42853	Planned Residential Development in Site 4-28	75	71 - 83	42
42859	Planned Residential Development in Site 4-28	75	78 - 85	54
51601	Planned Residential Development in Site 5-16	75	66 - 90	51
51602	Planned Residential Development in Site 5-16	75	74 - 80	37
51702	Planned Residential Development in Site 5-17	75	74 - 86	60
51703	Planned Residential Development in Site 5-17	75	64 - 76	3
52151	Planned School in Site 5-21	70 (65)	62 - 84	49
52401	Planned Residential Development in Site 5-24	75	62 - 85	36
52407	Planned Residential Development in Site 5-24	75	61 - 86	62
52408	Planned Residential Development in Site 5-24	75	70 - 87	61
52602	Planned Residential Development in Site 5-26	75	68 - 85	60
52604	Planned Residential Development in Site 5-26	75	68 - 87	60
53208	Planned Residential Development in Site 5-32	75	77 - 80	31

Notes:

(a) Construction noise assessment criterion is 75 dB(A) for domestic premises and 70dB(A) for education institutions (65dB(A) during examinations).

Mitigation Measures

4.6.5 The construction noise assessment results indicate that, in the absence of any mitigation measures, there would be exceedance of the construction noise criteria at some of the NSRs. Various mitigation options listed below have thus been considered:

- Good site practices to limit noise emissions at the sources;
- Use of quiet PME;
- Use of construction noise barriers or enclosures to screen noise from construction plant.

4.6.6 The above mitigation measures would be implemented in work sites as good practices where appropriate. Detailed descriptions of these mitigation measures are given in the following sections.

Good Site Practice

4.6.7 Although the noise mitigation effects are not easily quantifiable and the benefits may vary with site conditions and operating conditions, good site practices are easy to implement and do not impact upon the works schedule. The site practices listed below should be followed during construction works:

- Only well-maintained plant should be operated on-site and plant should be serviced regularly during the construction program;
- Silencers or mufflers on construction equipment should be utilised and should be properly maintained during the construction program;
- Mobile plant, is any, should be sited as far from NSRs as possible;
- Machines and plant (such as trucks) that may be in intermittent use should be shut down between work periods or should be throttled down to a minimum;
- Plant known to emit noise strongly in one direction should, wherever possible, be orientated so that the noise is directed away from the nearby NSRs;
- Material stockpiles and other structures should be effectively utilised, wherever practicable, in screening noise from on-site construction activities.

4.6.8 To reduce the noise impacts at the affected NSRs during normal daytime working hours, mitigation measure such as adopting quiet PME is recommended. The type of quiet PME adopted in this assessment is for reference only. The contractors may adopt alternative quiet PME as long as it can be demonstrated that they would not result in construction noise impacts worse than those predicted in this assessment.

4.6.9 For the use of quiet plant associated with the construction works, reference has been made to quiet PME list available on the Environmental Protection Department (EPD) website, which contains the SWLs for specific quiet PME. The SWLs for quiet PME adopted for the assessment are detailed in **Appendix 4.6.2**.

Management of locations of Workfronts

4.6.10 For some large works areas, road construction works and channel revitalisation works, construction activities will be conducted by multi-workfront. Considering the large size of works area, the locations of the workfronts can be well separated (i.e. 300 m) such that NSRs will be limited to expose to only one workfront at any one time. The works areas that require special workfront management are identified and listed in **Table 4.19** below.

Table 4.19 Details of Workfront Management

Works area	Construction activity	Workfront Management
#0201-3	Construction of Road D7 and Road D8 - part 1	Only 1 workfront should be allowed to operate within 300 m from the planned NSRs within Site 4-22. Other workfront should be located more than 300 m or have no direct line of sight from the planned NSR within Site 4-22.
#0201-4	Construction of Road D8 - part 2	Maximum 2 workfronts should be allowed to operate within 300 m from the planned NSR within Site 4-22 at any one time. Other workfront should be located more than 300 m or have no direct line of sight from the planned NSR within Site 4-22.
#0201-5	Construction of Road D8 - part 3	Only 1 workfront should be allowed to operate within 300 m from the planned NSR within Site 4-28 at any one time. Other workfront should be located more than 300 m or have no direct line of sight from the planned NSR within Site 4-28.
#0209-S5	Site formation for sites 4-19 and associated area	Only 1 workfront should be allowed to operate within 300 m from the planned NSR within Site 4-22 at any one time. Other workfront should be located more than 300 m or have no direct line of sight from the planned NSR within Site 4-22.
#0301-2	Construction of Road D2 - part 2	Only 1 workfront should be allowed to operate within 300 m from the existing NSRs within Ha Tsuen Shi and Yan Wu Garden (i.e. NSR E12051, E12052 and ELUT01) at any one time. Other workfront should be located more than 300 m or have no direct line of sight from the these NSRs.
#0301-2	Construction of Road D2 - part 3	Only 1 workfront should be allowed to operate within 300 m from the existing YLPMSAA Tang Siu Tong Secondary School and existing village houses in Shek Po Tsuen (i.e. NSR ETSW11 and ESPT01) at any one time. Other workfront should be located more than 300 m or have no direct line of sight from the these NSRs.
#0302-1	Site development for Site No. 2-28, 2-33,1-15,1-19,1-20,1-22	Maximum 2 workfronts should be allowed to operate within 300 m from the existing NSRs within Yan Wu Garden (i.e. ELUT01) at any one time. Other workfront should be located more than 300 m or have no direct line of sight from the NSRs.
#0303-2	Construction of Road D5	Only 1 workfront should be allowed to operate within 300 m from the existing NSRs within San Uk Tsuen (i.e. NSR EHTS01) at any one time. Other workfront should be located more than 300 m or have no direct line of sight from the NSRs.
#0315-S1	Site Development for Sites 3-24, 3-31 and 3-34	Only 1 workfront should be allowed to operate within 300 m from the existing NSRs within San Sang Tsuen (i.e. NSR ESST07) at any one time. Other workfront should be located more than 300 m or have no direct line of sight from the NSRs.
#0307-2	Revitalisation of existing HSK Main Channel	Only 1 workfront should be allowed to operate within 300 m from the existing NSRs within Bellevue Court, Tin Sam Villa, Aster Court and Shek Po Tsuen (i.e. NSR EBC01, E53906, E53104, E53106, ESPT05 and ESPT06) at any one time. Other workfront should be located more than 300 m or have no direct line of sight from the NSRs.
#0317-S1	Site Formation for Sites 1-28, 5-34 and 2-32 (portion 2)	Maximum 2 workfronts should be allowed to operate within 300 m from the existing NSRs within San Lee Uk Tsuen (i.e. ESLUT04 and ESLUT05) at any one time. Other workfront should be located more than 300 m or have no direct line of sight from the NSRs.

Works area	Construction activity	Workfront Management
#0317-S2	Site Formation for Sites 1-27 (portion 2), 1-29, 1-31,1-30	Maximum 2 workfronts should be allowed to operate within 300 m from the existing NSRs within Sha Chau Lei Tsuen (i.e. ESCL01, ESCL03 and ESCL04) at any one time. Other workfront should be located more than 300 m or have no direct line of sight from the NSRs.
#0308-1	Site Development for Sites 1-28, 1-32, 5-34 and 2-32 (portion 2)	Only 1 workfront should be allowed to operate within 300 m from the existing NSRs within San Lee Uk Tsuen (i.e. ESLUT04) at any one time. Other workfront should be located more than 300 m or have no direct line of sight from the NSRs.
#0309-S6	Site formation for Sites 5-11, 5-13, 5-14 and 5-42	Only 1 workfront should be allowed to operate within 300 m from the planned school at Site 5-21 (i.e. 52151) at any one time. Other workfront should be located more than 300 m or have no direct line of sight from the NSR.
#0401-3	Removal of existing Tin Ying Road - part 3	Only 1 workfront should be allowed to operate within 300 m from the existing VTC Youth College (Tin Shui Wai) (i.e. ETSW08) at any one time. Other workfront should be located more than 300 m or have no direct line of sight from the NSR.
#0402-1	Drainage Channel Revitalisation - part 1	Only 1 workfront should be allowed to operate within 300 m from the existing VTC Youth College (Tin Shui Wai) (i.e. ETSW08) at any one time. Other workfront should be located more than 300 m or have no direct line of sight from the NSR.

Construction Noise Barriers

4.6.11 To alleviate the construction noise impact on the affected NSRs, construction noise barriers or enclosures would be erected to provide screening from the construction plant.

4.6.12 Noise barriers will become more effective when located immediately adjacent to the working area, and can reduce the noise level by up to 5 dB(A) and 10 dB(A) for mobile and stationary plants, respectively. The Contractor shall be responsible for design of the movable noise barrier with due consideration given to the size of the PME and the requirement of intercepting the line of sight between the NSRs and PME. The minimum surface density of the movable noise barrier is 10 kg/m^2.

4.6.13 The use of full enclosure has been considered in this assessment to shelter relatively static plant such as air compressor and generator. These enclosures can provide about 15 dB noise attenuation. The typical configurations of noise barrier and enclosure are shown in **Appendix 4.6.3**.

4.6.14 The mitigated plant inventory is presented in **Appendix 4.6.4**.

4.6.15 With the implementation of the above proposed mitigation measures, the mitigated noise levels at the representative NSRs would be in the range of 50 – 77 dB(A). The predicted construction noise levels at the representative NSRs are summarised in **Table 4.20** to **Table 4.22** and the detailed mitigated noise assessment results are presented in **Appendix 4.6.5**.

Table 4.20 Summary of Mitigated Construction Noise Assessment Results (Existing NSRs outside the Project area) during Normal Daytime Working Hours

NAP ID	NAP Description	Criterion, dB(A) [(a)]	Predicted Mitigated Construction Noise Levels, $L_{eq\ (30min)}$ dB(A)	Duration of Exceedance (Months) [(b)]
ETCW02	No. 739 Oaklands Court	75	58 - 74	-
ETCW03	No. 703A, Chung Long Terrace	75	53 - 72	-
ENW02	No. 202, Shun Fung Wai	75	58 - 67	-
ESFW01	No. 332, Chung Uk Tsuen	75	61 - 67	-
ESFW02	Village house, Nai Wai	75	57 - 68	-
ESFW03	No. 368, Sun Fung Wai	75	58 - 69	-
ESFW04	No. 716, Chung Uk Tsuen	75	57 - 68	-
ECUT01	No. 16, Chung Uk Tsuen	75	59 - 67	-
ECUT02	No. 203, Chung Uk Tsuen	75	62 - 67	-
ESTC51	Shung Tak Catholic English College	70 (65)	54 - 60	-
EUPT02	Flat C, Tower 1, Uptown	75	61 - 61	-
ELFS01	No. 30, Lau Fau Shan	75	56 - 68	-
ELFS02	No. 3H, San Hing Tsuen	75	57 - 69	-
ELFS03	No. 310, Sha Kong Wai	75	57 - 72	-
ELFS04	Wing Jan School/Wing Jan Lutheran Church	70 (65)	58 - 67	- (1)
ELFS05	No. 1, Lau Fau Shan Road	75	58 - 68	-
ETSW01	Wah Yat House, Tin Wah Estate	75	60 - 65	-
ETSW02	Caritas Lok Kan School	75	57 - 64	-
ETSW03	Shui Lung House, Tin Shui Estate	75	60 - 65	-
ETSW04	The Yuen Yuen Institute MFBM Nei Ming Chan Lui Chung Tak Memorial College	70 (65)	59 - 65	-
ETSW05	Shui Fung House, Tin Shui Estate	75	59 - 70	-
ETSW06	Oi Tao House, Tin Oi Court	75	56 - 70	-
ETSW07	Block 7, Locwood Court	75	57 - 72	-
ETSW08	VTC Youth College (Tin Shui Wai)	70 (65)	58 - 70	- (9)
ETSW09	Shing Ting House, Tin Shing Court	75	54 - 69	-
ETSW10	Shing Kan House, Tin Shing Court	75	56 - 65	-
ETSW11	YLPMSAA Tang Siu Tong Secondary School	70 (65)	56 - 67	- (2)

Notes:

(a) Construction noise assessment criterion is 75 dB(A) for domestic premises and 70 dB(A) for education institutions (65 dB(A) during examinations).

(b) For educational institutes, number in the round bracket denotes the number of months of noise exceedance during examination period. Examination period assumed to be March – June and December of each year.

Table 4.21 Summary of Mitigated Construction Noise Assessment Results (Existing NSRs within the Project area) during Normal Daytime Working Hours

NAP ID	NAP Description	Criterion, dB(A)	Predicted Mitigated Construction Noise Levels, $L_{eq\ (30min)}$ dB(A) (a)&(b)	Duration of Exceedance (Months)
ETCW01	No. 430, village house	75	57 - 73	-
E53902	No. 125, Lee Fong Yuen	75	57 - 71	-
E53903	Village house, Tin Ha Road	75	58 - 71	-
ETST02	No. 13, Tin Sum	75	50 - 65	-
ETST04	No.211, Tin Sum	75	57 - 61	-
ETST05a	No. 143 ,Tin Sum (South Façade)	75	58 - 73	-
ETST05b	No. 143, Tin Sum, (West Façade)	75	58 - 74	-
ETST06	No. 290, Tin Sum	75	57 - 70	-
ESLUT01	No. 46A, San Lee Uk Tsuen	75	59 - 75	-
ESLUT03	No. 62, San Lee Uk Tsuen	75	56 - **77**	11
ESLUT04	No. 254, San Lee Uk Tsuen	75	55 - 75	-
ESLUT05	No. 120, San Lee Uk Tsuen	75	57 - 75	-
EBC01	House 15, Bellevue Court	75	53 - 73	-
E53905	Village House, Hung Shui Kiu Tin Sam Road	75	54 - 70	-
E53906	Flat C, Tin Sam Villa	75	56 - 72	-
E53104	Flat C, Block 1, Aster Court	75	59 - 69	-
E53106	Flat D, Block 1, Aster Court	75	59 - 74	-
E52503	Hung Long House, Hung Fuk Estate	75	60 - 69	-
E52504	Hung Yan House, Hung Fuk Estate	75	52 - 69	-
E52505	Hung Yan House, Hung Fuk Estate	75	53 - 74	-
E52506	Hung Shing House, Hung Fuk Estate	75	52 - 73	-
EHUT01	No. 141, Hung Uk Tsuen	75	63 - 73	-
EHUT02	No. 81A, Hung Uk Tsuen	75	67 - 73	-
EHUT03	No. 112, Kiu Tau Wai	75	67 - 74	-
EHUT04	No. 85A, Kiu Tau Wai	75	67 - 75	-
ESPT01	No. 181, Shek Po Tsuen	75	54 - 68	-
ESPT04	Village house, Shek Po Tsuen	75	51 - 71	-
ESPT05	No. 215, Shek Po Tsuen	75	51 - 71	-
ESPT06	No. 201, Shek Po Tsuen	75	51 - 73	-
ESST01	Block L, Bun Shing Garden	75	51 - 72	-
ESST06	Block H, Fu Lam Garden	75	51 - 73	-
ESST07	No, 60, San Sang Tsuen	75	53 - 77	4
EKLUT01	Village house, Kau Lee Uk Tsuen	75	59 - 70	-
ESCL01	No. 06, Sha Chau Lei Tsuen	75	54 - 74	-
ESCL03	No. 45, Sha Chau Lei Tsuen	75	55 - 75	-
ESCL04	No. 18, Sha Chau Lei Tsuen	75	55 - 71	-

NAP ID	NAP Description	Criterion, dB(A)	Predicted Mitigated Construction Noise Levels, $L_{eq\ (30min)}$ dB(A) (a)&(b)	Duration of Exceedance (Months)
E12501	No. 108, Ha Tsuen Shi	75	59 - 71	-
E12502	No. 115, Ha Tsuen Shi	75	63 - 69	-
E12503	No. 216, Ha Tsuen Shi	75	50 - 68	-
EHTS01	No. 1B, San Uk Tsuen	75	58 - **76**	10
ELUT01	Block 11, Yan Wu Garden	75	68 - 74	-
ESKT02	No. 151, Sik Kong Wai	75	64 - 73	-
ETKW01	No.108C, Tseung Kong Wai	75	60 - 70	-
EHMT01	No. 411, San Wai	75	52 - 71	-
ETTT01	Block A Luxor Garden, Tung Tau Tsuen	75	53 - 74	-
ETTT02	No. 111, Lo Uk Tsuen	75	52 - 75	-
EFKT01	No.61, Fung Kong Tsuen	75	58 - 73	-
EFKT03	No.80, Fung Kong Tsuen	75	58 - 71	-
EFKT04	No.23F, Fung Kong Tsuen	75	63 - 74	-

Notes:
(a) Construction noise assessment criterion is 75 dB(A) for domestic premises and 70 dB(A) for education institutions (65 dB(A) during examinations).
(b) Bold faced value indicates the predicted noise level exceeds the relevant noise limit.

Table 4.22 Summary of Mitigated Construction Noise Assessment Results (Planned NSRs) during Normal Daytime Working Hours

NAP ID	NSR	Criterion, dB(A)	Predicted Mitigated Construction Noise Levels, $L_{eq\ (30min)}$ dB(A) [(a)]	Duration of Exceedance (Months) [(c)]
10353	Planned School in Site 1-3	70 (65)	57 - 70	- (3)
10354	Planned School in Site 1-3	70 (65)	56 - 69	- (3)
11203	Planned Residential Development in Site 1-12	75	55 - 68	-
11556	Planned School in Site 1-15	70 (65)	54 - 69	- (10)
11702	Planned Residential Development in Site 1-17	75	56 - 68	-
21801	Planned Residential Development in Site 2-18	75	66 - 74	-
21802	Planned Residential Development in Site 2-18	75	66 - 72	-
21803	Planned Residential Development in Site 2-18	75	62 - 72	-
21901	Planned Residential Development in Site 2-19	75	57 - 63	-
23101	Planned Residential Development in Site 2-31	75	58 - 68	-
40305	Planned Residential Development in Site 4-3	75	69 - 72	-
40602	Planned Residential Development in Site 4-6	75	64 - 69	-
42001	Planned Residential Development in Site 4-20	75	64 - 75	-
42005	Planned Residential Development in Site 4-20	75	62 - 74	-
42006	Planned Residential Development in Site 4-20	75	63 - **77**	8
42203	Planned Residential Development in Site 4-22	75	57 - 69	-
42216	Planned Residential Development in Site 4-22	75	54 - 69	-
42251	Planned Residential Development in Site 4-22	75	59 - **77**	12
42806	Planned Residential Development in Site 4-28	75	64 - 72	-
42853	Planned Residential Development in Site 4-28	75	61 - 70	-
42859	Planned Residential Development in Site 4-28	75	68 - 73	-
51601	Planned Residential Development in Site 5-16	75	53 - 75	-
51602	Planned Residential Development in Site 5-16	75	59 - 66	-
51702	Planned Residential Development in Site 5-17	75	56 - 74	-
51703	Planned Residential Development in Site 5-17	75	53 - 62	-
52151	Planned School in Site 5-21	70 (65)	51 - 70	- (8)
52401	Planned Residential Development in Site 5-24	75	51 - 70	-
52407	Planned Residential Development in Site 5-24	75	50 - 74	-
52408	Planned Residential Development in Site 5-24	75	57 - 75	-
52602	Planned Residential Development in Site 5-26	75	55 - 72	-
52604	Planned Residential Development in Site 5-26	75	55 - 74	-
53208	Planned Residential Development in Site 5-32	75	59 - 67	-

Notes:

(a) Construction noise assessment criterion is 75 dB(A) for domestic premises and 70 dB(A) for education institutions (65 dB(A) during examinations).

(b) Bold faced value indicates the predicted noise level exceeds the relevant noise limit.

(c) For educational institutes, number in the round bracket denotes the number of months of noise exceedance during examination period. Examination period assumed to be March – June and December of each year.

(d) Further Noise Mitigation Measures.

4.6.16 With implementation of practical noise mitigation measures, including the use of quiet PME, movable noise barriers, fixed type noise barrier, noise enclosure and proper workfront management, noise exceedance is still anticipated at some of the NSRs due to the close proximity to the proposed works areas. Details of noise exceedance are listed in **Table 4.23** below.

Table 4.23 Detail of Noise Exceedance

NAP	NAP ID	Range of exceedance, dB(A)	Duration of Noise Exceedance	Critical Construction Activities (Work area) lead to the Exceedance
Wing Jan School/Wing Jan Lutheran Church	ELFS04	2 (during examination)	Dec 2031	Site formation for Sites 2-1, 2-2, 2-3, 2-4, 2-5, 2-7 and 2-35 (#0407-1)
VTC Youth College (Tin Shui Wai)	ETSW08	1-5 (during examination)	Dec 2033, Mar – Jun 2034 Dec 2035 Mar – May 2036	Tim Sam Channel (#0307-1) Tin Ying Road - part 3 (#0401-3) Channel revitalisation - part 1 (#0402-1) Site formation for a new DCS at site 1-24 (#0405) Site formation for Sites 1-8, 1-9, 1-10, 1-11, 1-13, 1-14 and 1-16 (#0407-5)
YLPMSAA Tang Siu Tong Secondary School	ETSW11	2 (during examination)	Dec 2031 and Mar 2032	Construction of Road D2 - part 3 (#0301-3)
No. 62, San Lee Uk Tsuen	ESLUT03	2	Mar 2025 – Jan 2026	Site formation for Site 5-36, 5-37 and 5-38 (#0016-2)
No, 60, San Sang Tsuen	ESST07	2	June – Sept 2033	Site development for Sites 3-24, 3-31 and 3-34 (#0315-S1)
No. 1B, San Uk Tsuen	EHTS01	1	Aug 2032 - May 2033	Construction of Road D5 (#0303-2)
Planned School in Site 1-3	10353	5 (during examination)	Apr –June 2033	Site formation for Sites 1-2, 1-4, 1-5, 1-6 and 1-7 (#0407-4)
Planned School in Site 1-3	10354	4 (during examination)	Apr –June 2033	Site formation for Sites 1-2, 1-4, 1-5, 1-6 and 1-7 (#0407-4)
Planned School in Site 1-15	11556	2 -4 (during examination)	Dec 2033 Mar – Jun 2034 Dec 2037 Mar – Jun 2038	Site formation for Sites 1-8, 1-9, 1-10, 1-11, 1-13, 1-14 and 1-16 (#0407-5) Site development for Sites 1-10, 1-11 and 1-14 (#0408-4)
Planned Residential Development in Site 4-20	42006	2	Nov 2028 - Jun 2029	Construction of Road D7 and Road D8 - part 1 (#0201-3)
Planned Residential Development in Site 4-22	42251	2	Jan – Dec 2028	Site formation for sites 4-19 and associated area (#0209-S5)

NAP	NAP ID	Range of exceedance, dB(A)	Duration of Noise Exceedance	Critical Construction Activities (Work area) lead to the Exceedance
Planned School in Site 5-21	52151	3-5 (during examination)	Dec 2031 Mar 2032 Dec 2034 Mar – June, Dec 2035	Site formation for Sites 5-11, 5-13, 5-14 and 5-42 (#0309-S6) Site development for S3 (Site No. 1-1,1-3,2-25,5-11,5-14,5-13,2-31) (#0302-3) Construction of Road D5 (#0303-2)

4.6.17 The noise exceedance at these existing or planned residential NSRs are predicted to be no more than 2 dB(A). The noise exceedance can be further reduced by grouping of PME during conducting the construction activities at the critical work areas identified in **Table 4.23**. For site formation works, dump trucks should not be used simultaneously with excavation and breaking works (i.e. use of excavators, backhoes and breakers, handheld). For site development works and road construction works, concreting activities (i.e. use of concrete lorry mixers, vibratory pokers and concrete pumps) should not be used simultaneously with other works such as bored piling, excavation, road paving. The works areas and works activities that requires grouping of PME are shown in **Appendix 4.6.6**. The predicted construction noise levels after grouping of PME adopted are presented in **Appendix 4.6.7**. With proper grouping of PME at the criterial works area, the maximum predicted construction noise level would reduce to 75 dB(A) at all existing and planned residential NSRs shown in **Table 4.23**.

4.6.18 Referring to the results listed in **Table 4.23** above, it can be noted that, for education institutions, the noise exceedance would only occur during the examination period. The Contractor shall liaise with the school representative(s) to obtain the examination schedule. Considering the large size of the critical works areas, the noise exceedance can be further mitigated by maintaining certain separation between the critical works area and the school during the examination period. **Appendix 4.6.8** presents the required minimum separations between the schools and the critical works areas during the examination period. For critical works area #0408-4 near the planned school in Site 1-15 (NAP ID11556), the construction activities have no room to maintain a sufficient separation to minimum the noise exceedance due to limited space of this works area. In this case, grouping of PME during the examination period, as shown in **Appendix 4.6.9,** is recommended. The predicted construction noise levels with implementation of the above-mentioned arrangements during the examination period are presented in **Appendix 4.6.10**. The maximum predicted construction noise levels reduced to 62 - 65 dB(A) at the existing and planned schools shown in **Table 4.23**.

4.6.19 With implementation of practical noise mitigation measures, grouping of PME at critical works areas and maintaining recommended separations between the schools and the critical works areas during examination period, the construction noise levels are predicted to comply with the standard stipulated in the EIAO-TM and no residual impact is anticipated. Notwithstanding this, due to more stringent noise criterion of 65 dB(A), it is recommended that the Contractor should liaise with the school representative(s) to obtain the examination schedule so as to avoid noisy construction activities during school examination period.

Evaluation of Residual Environmental Impacts

4.6.20 With the implementation of the above proposed mitigation measures, residual construction noise impact would not be anticipated. An Environmental Monitoring and Audit (EM&A) programme is recommended to ensure the proper implementation of the proposed mitigation measures and take remedial measures when non-compliance is recorded. As the construction of the Project may involve different parties, it is proposed to set up a liaison group among relevant government departments, contractors of the Works contracts, etc. during construction phase of the Project so as to ensure the proper implementation of the proposed noise mitigation measures.

4.7 Traffic Noise Impact Assessment

Assessment Approach and Methodology

4.7.1 Road traffic noise impact is predicted using the methodology provided in the UK Department of Transport Calculation of Road Traffic Noise (CRTN) 1988. The computer model RoadNoise 2000 which complies with the CRTN is used for the prediction. The noise levels are presented in terms of noise levels exceeded for 10% of the one-hour period during the peak traffic flow, i.e. $L_{10(1hour)}$ dB(A).

4.7.2 Traffic noise is predicted based on the worst-case year traffic forecast within 15 years upon commencement of operation of the proposed roadwork according to Clause 3.4.5.2(vi)(a) of the EIA Study Brief. Based on the latest project programme, the proposed road network is expected to be in operation in operation in Year 2033. The assessment year for road traffic noise is taken as Year 2048 which is the maximum traffic projection within 15 years after full operation of the proposed roadwork. The peak traffic flow with the Project in Year 2048 is shown in **Appendix 4.7.1**. Transport Department (TD) has no comment on the traffic forecast adopted in this EIA study.

4.7.3 The following scenarios are studied in the assessment:

- Unmitigated scenario in Year 2048;
- Mitigated scenario in Year 2048; and
- Prevailing scenario in Year 2018 for indirect technical remedies eligibility assessment.

4.7.4 For the purpose of the traffic noise assessment in the EIA study, the roads within 300 m from the proposed Project boundary are included in the assessment. Road-plots of the traffic noise model with the extent of Project roads are depicted in **Appendix 4.7.2**. The friction course on Hung Tin Road (excluding slip roads) and Yuen Long Highway, existing noise barriers at Tin Ha Road, Hung Tin Road (in the vicinity of Hung Fuk Estate), Hung Chi Road, Hung Shui Kiu Tin Sum Road and all traffic noise mitigation measures and friction course as stated in the approved EIA Report of “Deep Bay Link” (EIA-078/2002) have been taken into account in the assessment. In this study, the planned Tuen Mun Western Bypass (TMWB) which is located next to planned Site 4-4 is also considered in the assessment with the assumption of the semi-enclosures on its slip-roads under unmitigated and mitigated scenarios.

4.7.5 The traffic speeds adopted for the roads in the noise models are summarised as follows and the traffic speed adopted for each road is presented in **Appendix 4.7.2**:

(a) KSWH and Yuen Long Highway: 80 km per hour;

(b) Tin Ying Road (for prevailing scenario only), Hung Tin Road and certain sections of Castle Peak Road: 70 km per hour; and

(c) Concerned roads other than those listed in (a) and (b) above: 50 km per hour.

4.7.6 Direct mitigation measures would be proposed in all situations where the predicted traffic noise level exceeds the criteria set in Table 1A of Annex 5 to the EIAO-TM by 1 dB(A) or more.

4.7.7 If any façades of NSRs are still exposed to predicted noise levels exceeding the relevant noise criteria after the implementation of all direct mitigation measures, provision of indirect technical remedies in the form of acoustic insulation and air conditioning should be considered under the EIAO-TM and EIAO Guidance Note No. 12/2010. The eligibility for indirect technical remedies would be tested against the following three criteria:

- The predicted overall noise level from the Project roads, together with other traffic noise in the vicinity must be above a specified noise level (for example, 70 dB(A) for domestic premises and 65 dB(A) for educational institutions. All are in $L_{10\ (1hour)}$);
- The predicted overall noise level is at least 1.0 dB(A) more than the prevailing traffic noise level, i.e. the total traffic noise level existing before the works to construct Project roads were commenced; and
- The contribution to the increase in the predicted overall noise level from the Project roads must be at least 1.0 dB(A).

Prediction and Evaluation of Environmental Impacts

4.7.8 The Revised RODP has incorporated environmentally friendly designs to avoid and/or reduce traffic noise impact to sensitive receivers. These designs include deletion of Tin Ying Road, provision of depressed roads with or without decking-over, allocation of non-sensitive uses at the noisy roads, traffic controls for heavy goods vehicles which centralising the cargo truck travelling along KSWH, Planned Road P1 and D1 would also be implemented to reduce the road traffic noise impact at those NSRs within the Project area. In addition, provision of EFTS would potentially reduce the road-based traffic volume, thereby reducing the potential traffic noise impact arising from the Project. In the Stage 2 Community Engagement which consulted the public on the PODP and facilitated formulation of the RODP, there were comments expressing concerns over the noise impact brought by Castle Peak Road. Taking into account the public concern, low noise road surfacing (LNRS) would be implemented, subject to design review, on the road section of Castle Peak Road in the vicinity of Project area. Different materials for LNRS are being reviewed and tested by the Government for application in Hong Kong. Environmental review will be conducted in due course to study how the use of new road surfacing material could help reduce traffic noise impacts.

4.7.9 Road traffic noise assessment has been conducted for the representative NSRs in the worst-case year 2048. The predicted traffic noise levels at the representative NSRs under the unmitigated scenario are presented in **Table 4.24** to **Table 4.26**. Whereas **Appendix 4.7.3** shows the details of the noise assessment results of the each assessment points of the NSRs at different assessment levels under unmitigated scenario.

Table 4.24 Summary of Unmitigated Road Traffic Noise Assessment Results (Existing NSRs outside the Project Area)

NSR ID	Description	Existing / Planned Use	NAP ID	Criteria, dB(A)	Predicted Unmitigated Overall Noise Level, dB(A)	Compliance of Noise Criteria
E1-OA	Tin Wah Estate	Existing Residential	E1-OA_R01	70	66-67	Yes
E1-OB_R	Tin Shing Court	Existing Residential	E1-OB_R01	70	67-68	Yes
E1-OB_E0	Q.E.S. Old Students' Association Branch Primary School	Existing Educational	E1-OB_E01 to E1-OB_E02	65	56-74	No
E1-OB_E1	YLPMSAA Tang Siu Tong Secondary School	Existing Educational	E1-OB_E11 to E1-OB_E12	65	61-74	No
E1-OB_E2	Lions Clubs International Ho Tak Sum Primary School	Existing Educational	E1-OB_E21	65	57-69	No
E2-OA	Village House at Lau Fau Shan	Existing Village	E2-OA_R01	70	66-67	Yes
E2-OB	San Hing Tsuen	Existing Village	E2-OB_R01 to E2-OB_R03	70	71-72	No
E2-OC	Sha Kong Wai	Existing Village	E2-OC_R01 to E2-OC_R04	70	71-76	No
E2-OD	Tin Yan Estate	Existing Residential	E2-OD_R01	70	66-67	Yes
E4-OA	Tsing Cheun Wai	Existing Village	E4-OA_R01 to E4-OA_R04	70	71-73	No
E4-OB	Nai Wai	Existing Village	E4-OB_R01	70	77	No
E4-OC	Sun Fung Wai	Existing Village	E4-OC_R01 to E4-OC_R03	70	72-80	No
E4-OD	Chung Uk Tsuen	Existing Village	E4-OD_R01 to E4-OD_R04	70	74-78	No
E5-OA	Tong Fong Tsuen	Existing Village	E5-OA-R01 & E5-OA-R02	70	67-79	No
E5-OB	Hang Mei Tsuen	Existing Village	E5-OB-R01	70	75	No
E5-OC	Tai Tao Tsuen	Existing Village	E5-OC-R01	70	71	No
E5-OD	Uptown	Existing Residential	E5-OD_R01	70	71-76	No
E5-OE	Shung Tak Catholic English College	Existing Educational	E5-OE_E01	65	73-74	No
E5-OF	Treasure Court	Existing Residential	E5-OF_R01	70	73	No
E5-OG	Lai Hung Garden	Existing Residential	E5-OG_R01	70	74	No
E5-OH	Bauhinia Garden	Existing Residential	E5-OH_R01	70	74-75	No

Note: Full names of the abbreviations of uses should be referred to the List of Abbreviations.

Table 4.25 Summary of Unmitigated Road Traffic Noise Assessment Results (Existing NSRs within the Project Area)

NSR ID	Description	Use	NAP ID	Criteria, dB(A)	Predicted Unmitigated Overall Noise Level, dB(A)	Compliance of Noise Criteria
E1-IA	Tseung Kong Wai	Existing Village	E1-IA_R01	70	76	No
E1-IB	Hong Mei Tsuen	Existing Village	E1-IB_R01	70	54-55	Yes
E1-IC	Tung Tau Tsuen	Existing Village	E1-IC_R01 to E1-IC_R03	70	64-69	Yes
E1-ID	Lo Uk Tsuen	Existing Village	E1-ID_R01	70	59	Yes
E1-IE	Sik Kong Wai	Existing Village	E1-IE_R01	70	69-70	Yes
E1-IF	Ha Tsuen Shi	Existing Village	E1-IF_R01 to E1-IF_R02	70	64-76	No
E1-IG	San Uk Tsuen	Existing Village	E1-IG_R01 to E1-IG_R03	70	65-69	Yes
E1-IH	Sha Chau Lei	Existing Village	E1-IH_R01 to E1-IH_R02	70	66-70	Yes
E1-IJ	San Sang Tsuen	Existing Village	E1-IJ_R01 to E1-IJ_R05	70	68-73	No
E1-IK	Shek Po Tsuen	Existing Village	E1-IK_R01 to E1-IK_R03	70	64-73	No
E2-IA	Fung Kong Tsuen	Existing Village	E2-IA_R01 to E2-IA_R02	70	72-76	No
E4-IA	Tin Sum Tsuen	Existing Village	E4-IA_R01 to E4-IA_R03	70	72-73	No
E4-IB	Tsing Chuen Wai	Existing Village	E4-IB_R01	70	69-72	No
E5-IA	Kiu Tau Wai	Existing Village	E5-IA_R01	70	64-69	Yes
E5-IB_R	Hung Uk Tsuen	Existing Village	E5-IB_R01 to E5-IB_R02	70	63-74	No
E5-IB_C	Chinese Mission Seminary	Existing Church	E5-IB_C01	65	73-74	No
E5-IC	San Lee Uk Tsuen	Existing Village	E5-IC_R01	70	73	No
E1-26	Existing Homes for the Elderly (Yeung Chun Pui and Ching Chung)	IC	E1-26_H01 to E1-26_H02	70	59-63	Yes
E4-30	Ling Liang Worldwide Evangelistic Mission Hung Shui Kiu Ling Liang Church	IC	E4-30_C01	65	72	No
E5-23	Sheffield Villas	R3	E5-23_R01 to E5-23_R12	70	64-66	Yes
E5-25	Hung Fuk Estate	RSc	E5-25_R01 to E5-25_R41	70	59-73	No
E5-31	Existing Residential Development	R2	E5-31_R01 to E5-31_R22	70	61-73	No
E5-35	Bellevue Court	R4	E5-35_R01	70	64-73	No
E5-39	Existing Residential Development	R3	E5-39_R01 to E5-39_R05	70	68-77	No
E5-44	The Church of Jesus Christ of Latter-Day Saints (Hung Shui Kiu)	R3	E5-44_C01 to E5-44_C02	65	68-71	No

NSR ID	Description	Use	NAP ID	Criteria, dB(A)	Predicted Unmitigated Overall Noise Level, dB(A)	Compliance of Noise Criteria
O5-06	Committed Subsidised Sales Flat	R2(SSF)c	O5-06_R01 to O5-06_R24	70	59-75	No
O5-33	Committed Residential Development	R2	O5-33_R01 to O5-33_R31	70	72-73	No

Note: Full names of the abbreviations of uses should be referred to the List of Abbreviations.

4.7.10 Referring to **Table 4.24** and **Table 4.25**, the predicted noise levels at the representative existing NSRs outside and within the Project area would be in the range of 56 to 79 dB(A) and 54 to 77 dB(A), respectively. Most of representative existing NSRs would expose to the noise levels exceeding the relevant noise criteria. Mitigation measures are required to alleviate the potential adverse traffic noise impact.

Table 4.26 Summary of Unmitigated Road Traffic Noise Assessment Results (Planned NSRs within the Project Area)

NSR ID	Use	Description	NAP ID	Criteria, dB(A)	Predicted Unmitigated Overall Noise Level, dB(A)	Compliance of Noise Criteria
P1-02	R2(SSF)c	Site 1-02	P1-02_R01 to P1-02_R31	70	57-75	No
P1-03	E	Site 1-03	P1-03_E01 to P1-03_E11	65	53-64	Yes
P1-04	R2c	Site 1-04	P1-04_R01	70	47-60	Yes
P1-05	R2c	Site 1-05	P1-05_R01 to P1-05_R11	70	57-67	Yes
P1-06	R2c	Site 1-06	P1-06_R01 to P1-06_R11	70	64-70	Yes
P1-08	R2c	Site 1-08	P1-08_R01 to P1-08_R22	70	66-74	No
P1-09	R2c	Site 1-09	P1-09_R01 to P1-09_R12	70	68-75	No
P1-12	R2	Site 1-12	P1-12_R01	70	68-73	No
P1-15	E	Site 1-15	P1-15_E01 to P1-15_E05	65	65-73	No
P1-17	R2	Site 1-17	P1-17_R01	70	57-63	Yes
P1-18	E	Site 1-18	P1-18_E01 to P1-18_E03	65	57-68	No
P1-19	R2	Site 1-19	P1-19_R01 to P1-19_R21	70	69-76	No
P1-21	R2c	Site 1-21	P1-21_R01 to P1-21_R21	70	54-69	Yes
P1-22	E	Site 1-22	P1-22_E01 to P1-22_E11	65	63-79	No
P2-02	R3	Site 2-02	P2-02_R01 to P2-02_R32	70	61-67	Yes
P2-03	R3	Site 2-03	P2-03_R01 to P2-03_R21	70	52-65	Yes
P2-07	R3	Site 2-07	P2-07_R01 to P2-07_R21	70	62-66	Yes
P2-09	E	Site 2-09	P2-09_E01 to P2-09_E02	65	57-66	No

NSR ID	Use	Description	NAP ID	Criteria, dB(A)	Predicted Unmitigated Overall Noise Level, dB(A)	Compliance of Noise Criteria
P2-10	R3	Site 2-10	P2-10_R01 to P2-10_R31	70	62-70	Yes
P2-11	R3c	Site 2-11	P2-11_R01 to P2-11_R31	70	66-74	No
P2-14	R3	Site 2-14	P2-14_R01 to P2-14_R12	70	62-68	Yes
P2-15	E	Site 2-15	P2-15_R01	70	62-64	Yes
P2-16	E	Site 2-16	P2-16_E01 to P2-16_R15	65	63-69	No
P2-18	RR4	Site 2-18	P2-18_R01 to P2-18_R05	70	69-71	No
P2-19	RR4	Site 2-19	P2-19_R01	70	66-67	Yes
P2-24	E	Site 2-24	P2-24_E01 to P2-24_E11	65	59-70	Yes
P2-25	G	Site 2-25	P2-25_H01	70	69-71	No
P2-26	E	Site 2-26	P2-26_E01 to P2-26_E11	65	63-70	No
P2-28	E	Site 2-28	P2-28_E01 to P2-28_E02	65	60-68	No
P2-30	RSc	Site 2-30	P2-30_R01 to P2-30_R21	70	48-70	Yes
P2-31	R2(SSF)c	Site 2-31	P2-31_R01 to P2-31_R21	70	61-68	Yes
P4-01	G	Site 4-01	P4-01_R01 to P4-01_R13	70	64-75	No
P4-03	R2(SSF)c	Site 4-03	P4-03_R01 to P4-03_R11	70	61-76	No
P4-04	RSc	Site 4-04	P4-04_R01 to P4-04_R24	70	56-71	No
P4-05	RSc	Site 4-05	P4-05_R01 to P4-05_R14	70	65-72	No
P4-06	R2(SSF)c	Site 4-06	P4-06_R01 to P4-06_R12	70	66-74	No
P4-08	E	Site 4-08	P4-08_E01 to P4-08_E11	65	61-65	Yes
P4-09	R1c	Site 4-09	P4-09_E06	70	62-68	Yes
P4-17	OU	Site 4-17	P4-17_E01 to P4-17_E06	65	57-73	No
P4-20	RR4	Site 4-20	P4-20_R01 to P4-20_R04	70	64-72	No
P4-22	RSc	Site 4-22	P4-22_R01 to P4-22_R42	70	52-71	No
P4-24	R2(SSF)c	Site 4-24	P4-24_R01 to P4-24_R46	70	66-75	No
P4-25a	OU(C&R)	Site 4-25a	P4-25a_R01	70	65-69	Yes
P4-25b	OU(C&R)	Site 4-25b	P4-25b_R01 to P4-25b_R11	70	61-71	No
P4-25c	OU(C&R)	Site 4-25c	P4-25c_R01 to P4-25c_R11	70	52-60	Yes
P4-26	R1c	Site 4-26	P4-26_R01 to P4-26_R21	70	64-71	No

NSR ID	Use	Description	NAP ID	Criteria, dB(A)	Predicted Unmitigated Overall Noise Level, dB(A)	Compliance of Noise Criteria
P4-28	RSc	Site 4-28	P4-28_R01 to P4-28_R44	70	54-73	No
P4-29	OU(C&R)	Site 4-29	P4-29_R01 to P4-29_R31	70	50-71	No
P4-32	G	Site 4-32	P4-32_R01 to P4-32_R14	70	63-73	No
P4-33	E	Site 4-33	P4-33_E01 to P4-33_E03	65	67-75	No
P4-36	E	Site 4-36	P4-36_E02 to P4-36_R01	65	66-70	No
P5-07a	OU(C&R)	Site 5-07a	P5-07a_R01 – P5-07a_R21	70	55-71	No
P5-07b	OU(C&R)	Site 5-07b	P5-07b_R01 to P5-07b_R21	70	60-73	No
P5-08	OU(C&R)	Site 5-08	P5-08_R01 to P5-08_R11	70	69-73	No
P5-09	OU(C&R)	Site 5-09	P5-09_R01 to P5-09_R31	70	60-71	No
P5-14	E	Site 5-14	P5-14_E01 to P5-14_E11	65	67-74	No
P5-16	R2(SSF)c	Site 5-16	P5-16_R01 to P5-16_R11	70	58-79	No
P5-17	RSc	Site 5-17	P5-17_E04 to P5-17_R12	65	54-75	No
P5-21	E	Site 5-21	P5-21_E01 to P5-21_E02	65	61-66	No
P5-22	Site 5-22	R3	P5-22_R01 to P5-22_R03	70	65-77	No
P5-24	R2(SSF)c	Site 5-24	P5-24_R01 to P5-24_R23	70	43-68	Yes
P5-26	R2(SSF)	Site 5-26	P5-26_R01 to P5-26_R04	70	63-74	No
P5-28	G	Site 5-28	P5-28_H01 to P5-32_R01 to P5-32_R82	70	64-72	No
P5-32	OU(C&R)	Site 5-32	P5-32_R01 to P5-32_R82	70	62-80	No
P5-34	E	Site 5-34	P5-34_E01 to P5-34_E11	65	60-73	No
P5-37	E	Site 5-37	P5-37_E01 to P5-37_E02	65	70-74	No

Note: Full names of the abbreviations of uses should be referred to the List of Abbreviations.

4.7.11 For the planned NSRs within the Project area (see **Table 4.26**), the predicted road traffic noise levels would be in the range of 43 to 80 dB(A). Despite the environmentally friendly designs are considered in the Revised RODP, results indicate that most planned residential and educational NSRs would still exceed the noise criteria by up to 10 dB(A) and 14 dB(A) respectively. Hence, direct mitigation measures for these affected NSRs are required. For those non-compliance planned educational institutes, mitigation measures as discussed in **Section 4.7.15** are considered under the Class Assessment Document following ETWB Technical Circular (Works) No. 13/2003.

4.7.12 The total number of dwellings, classrooms and other NSRs that will be exposed to noise impact exceeding the criteria set in Annex 5 in the EIAO-TM have been calculated. It is estimated that a total of 786 of existing dwellings, 4352 planned dwellings, 120 existing classrooms, 445 planned classroom, 2 existing churches, 1 existing and 3 planned buildings of home for the elderly would be exposed to traffic noise impact under unmitigated scenario.

Direct Mitigation Measures

4.7.13 High volume of traffic flows of some major roads within or surrounding the Project, namely KSWH, Yuen Long Highway, Ping Ha Road, Tin Wah Road, Hung Tin Road, Castle Peak Road, Planned Road P1 and Planned Road D1, is expected in Year 2048. With the implementation of environmentally friendly designs in the Revised RODP, the noise barriers on KSWH according to its approved EIA Report and the LNRS on Castle Peak Road, exceedance of the noise criteria at some existing and planned NSRs are still predicted in the unmitigated scenario. Noise mitigation measures would therefore be required to alleviate the potential road traffic noise impact.

4.7.14 Noise mitigation measures in terms of: (i) vertical barriers and cantilever noise barriers on the Project roads; (ii)LNRS on Project roads; and (iii) alternative residential development layout (for Sites 5-8, 5-26 and 5-32) are proposed to alleviate adverse traffic noise impact on the affected NSRs. The details of proposed LNRS and noise barriers are summarised in below **Table 4.27** and **Table 4.28** with total length of mitigation measures rounded off to the nearest 10 m. The locations of the vertical barriers, cantilever noise barriers and extent of LNRS are shown in **Figures 4.7.1 – Figure 4.7.16**. The alternative residential development layouts are presented in **Figure 4.7.17 – Figure 4.7.27**. Different materials for LNRS are being reviewed and tested by the Government for application in Hong Kong. Environmental reviews will be conducted at the later design stage to review the proposed noise mitigation measures taking into account the latest design standard at that time for the application of the LNRS materials.

Table 4.27 Locations of Proposed Low Noise Road Surfacing

ID	Road	Length, m
LNS1	Not used	-
LNS2	Tin Wah Road	350
LNS3	Proposed Roundabout at Junction J2	210
LNS4	Proposed Road D1	500
LNS5	Proposed Road near Site2-26	280
LNS6	Proposed Road D1	400
LNS7	Proposed Road D1	640
LNS8	Proposed Road D1 (Depressed Section)	120
LNS9	Proposed Road L1	670
LNS10	Proposed Road P1	560
LNS11	Proposed Road L5	440
LNS12	Proposed Road D2	280
LNS13	Proposed Road D2 (Depressed Section)	180
LNS14	Proposed Road D2	1260
LNS15	Proposed Road L3	390
LNS16	Proposed Road D2	370
LNS17	Proposed Roundabout at Junction J8	190
LNS18	Ping Ha Road	110
LNS19	Not used	-

ID	Road	Length, m
LNS20	Kiu Fat Street	280
LNS22	Proposed Road near Site 5-13	730
LNS23	Hung Chi Road / Proposed Road L5	340
LNS24	Not used	-
LNS25	Note used	-
LNS26	Not used	-
LNS27	Proposed Road near Site 4-24	140
LNS28	Not used	-
LNS29	Proposed Road L2	670
LNS30	Proposed Road near Site 4-29	270
LNS31	Not used	-
LNS32	Proposed Road P1	790
LNS33	Proposed Road P1	380
LNS34	Proposed Road P1	610
LNS35	Proposed Road D8	480
LNS36	Proposed Road near Site 4-33	210

Table 4.28 List of Proposed Traffic Noise Mitigation Measures

Noise Barrier ID	Location	Barrier Type	Height, mAG	Length, m
CB1	Proposed Road D2	Cantilever	6+4.2 m at 45 degree	50
CB2	Not Used	-	-	-
CB3a	Proposed Road P1	Cantilever	5.5+2.5 m at 45 degree	80
CB3b	Proposed Road P1	Cantilever	5.5+2.5 m at 45 degree	130
CB4	Not Used	-	-	-
CB5	Proposed Road D2, near Site 1-22	Cantilever	6+4.2 m at 45 degree	130
VB2	Tin Wah Road / Lau Fau Shan Road	Vertical	5	140
VB3	Proposed Road D1	Vertical	3	120
VB4	Proposed Road D2	Vertical	5	140
VB5	Proposed Road D1	Vertical	5	90
VB6	Proposed Road D1	Vertical	5	150
VB7	Proposed Road D3	Vertical	4	60
VB8	Proposed Road D3	Vertical	4	50
VB9	Proposed Road D3	Vertical	4	50
VB10	Proposed Road D2	Vertical	4	180
VB11	Not used	-	-	-
VB12	Proposed Road D2	Vertical	6	110
VB13	Not used	-	-	-
VB14	Not used	-	-	-
VB15	Proposed Road D2	Vertical	5	30
VB16	Proposed Road D2	Vertical	5	60
VB17	Proposed Road D2	Vertical	5	20

Noise Barrier ID	Location	Barrier Type	Height, mAG	Length, m
VB17	Proposed Road D2	Vertical	5	20
VB18	Proposed Road D2	Vertical	4	60
VB19	Not used	-	-	-
VB20	Proposed Road L3	Vertical	7	110
VB21	Proposed Road D2/ Ping Ha Road	Vertical	7	50
VB22	Not used	-	-	-
VB23	Proposed Road D2/ Ping Ha Road	Vertical	7	130
VB24	Ping Ha Road	Vertical	7	160
VB25	Not used	-	-	-
VB26	Proposed Road L1	Vertical	5	70
VB27	Not used	-	-	-
VB28	Hung Tin Road	Vertical	5	90
VB29	Hung Tin Road	Vertical	7	80
VB30	Not used	-	-	-
VB31	Hung Tin Road	Vertical	7	60
VB32	Not used	-	-	-
VB33	Not used	-	-	-
VB34	Not used	Vertical	7	30
VB35	Proposed Road D5	Vertical	5	80
VB36	Proposed Road D5	Vertical	5	70
VB37	Proposed Road D5	Vertical	3	70
VB38	Proposed Road D5	Vertical	3	80
VB39	Proposed Road P1	Vertical	3	100
VB40	Proposed Road P1	Vertical	3	130
VB41	Proposed Road D2	Vertical	5	130
VB42	Not used	-	-	-

4.7.15 Regarding the noise exceedance on the planned education institutes at Sites 1-18, 1-22, 2-16, 4-33, 4-36 and 5-34, alternative layouts and 3 m high boundary wall (for Site 2-9, 4-33, 4-36 and 5-34 only) are also considered under the Class Assessment Document following ETWB Technical Circular (Works) No. 13/2003. The details of the proposed boundary wall are presented in **Table 4.29** and the alternative layouts are presented in **Figure 4.7.17 – Figure 4.7.27**.

Table 4.29 Details of Proposed Boundary Wall for Planned Education Institutes

Boundary Wall ID	Location	Barrier Type	Height, mAG	Length, m
VB1	Boundary Wall at Site 2-9	Vertical	3	65
VB43	Boundary wall at Site 4-36	Vertical	3	60
VB44	Boundary wall at Site 4-36	Vertical	3	120
VB45	Boundary wall at Site 4-33	Vertical	3	200
VB46	Boundary wall at Site 5-34	Vertical	3	150
VB47	Boundary wall at Site 5-34	Vertical	3	160

4.7.16 With the implementation of the above recommended noise mitigation measures, the noise levels at some of the representative NSRs would comply with the traffic noise criteria. However, exceedance of the traffic noise criteria would still be predicted at some of the representative existing NSRs due to existing roads and some planned NSRs due to existing roads including KSWH and planned local roads.

4.7.17 The predicted mitigated traffic noise levels at the representative existing NSRs outside and within the Project area, and the major noise source causing exceedance on the affected existing NSRs are presented in below **Table 4.30** and **Table 4.31**. **Table 4.32** summarises the predicted mitigated traffic noise levels at the planned representative NSRs.

Table 4.30 Summary of Mitigated Road Traffic Noise Assessment Results (Existing NSRs outside the Project area)

NSR ID	Description	Existing / Planned Use	NAP ID	Criteria, dB(A)	Predicted Mitigated Overall Noise Level, dB(A)	Major Noise Source causing Exceedance
E1-OA	Tin Wah Estate	Existing Residential	E1-OA_R01	70	65-67	-
E1-OB	Tin Shing Court	Existing Residential	E1-IB_R01	70	62-66	-
E1-OB_E0	Q.E.S. Old Students' Association Branch Primary School	Existing Educational	E1-OB_E01 to E1-OB_E02	65	56-74	Ping Ha Road (East of Hung Tin Road)
E1-OB_E1	YLPMSAA Tang Siu Tong Secondary School	Existing Educational	E1-OB_E11 to E1-OB_E12	65	61-74	Ping Ha Road (East of Hung Tin Road). Hung Tin Road
E1-OB_E2	Lions Clubs International Ho Tak Sum Primary School	Existing Educational	E1-OB_E21	65	54-65	-
E2-OA	Village House at Lau Fau Shan	Existing Village	E2-OA_R01	70	66-67	-
E2-OB	San Hing Tsuen	Existing Village	E2-OB_R01 to E2-OB_R03	70	71-72	Lau Fau Shan Road
E2-OC	Sha Kong Wai	Existing Village	E2-OC_R01 to E2-OC_R04	70	64-70	-
E2-OD	Tin Yan Estate	Existing Residential	E2-OD_R01	70	66-67	-
E4-OA	Tsing Cheun Wai	Existing Village	E4-OA_R01 to E4-OA_R04	70	68-70	-
E4-OB	Nai Wai	Existing Village	E4-OB_R01	70	76-77	Castle Peak Road
E4-OC	Sun Fung Wai	Existing Village	E4-OC_R01 to E4-OC_R03	70	72-78	Castle Peak Road
E4-OD	Chung Uk Tsuen	Existing Village	E4-OD_R01 to E4-OD_R04	70	74-78	Castle Peak Road

NSR ID	Description	Existing / Planned Use	NAP ID	Criteria, dB(A)	Predicted Mitigated Overall Noise Level, dB(A)	Major Noise Source causing Exceedance
E5-OA	Tong Fong Tsuen	Existing Village	E5-OA-R01 & E5-OA-R02	70	66-79	Ping Ha Road (East of Hung Tin Road)
E5-OB	Hang Mei Tsuen	Existing Village	E5-OB-R01	70	75	Ping Ha Road (East of Hung Tin Road)
E5-OC	Tai Tao Tsuen	Existing Village	E5-OC-R01	70	71	Castle Peak Road, Hung Tin Road
E5-OD	Uptown	Existing Residential	E5-OD_R01	70	72-76	Castle Peak Road, Hung Tin Road
E5-OE	Shung Tak Catholic English College	Existing Educational	E5-OE_E01	65	73-74	Castle Peak Road, Hung Tin Road
E5-OF	Treasure Court	Existing Residential	E5-OF_R01	70	73	Castle Peak Road
E5-OG	Lai Hung Garden	Existing Residential	E5-OG_R01	70	74	Castle Peak Road
E5-OH	Bauhinia Garden	Existing Residential	E5-OH_R01	70	74-75	Castle Peak Road

Table 4.31 Summary of Mitigated Road Traffic Noise Assessment Results (Existing NSRs within the Project area)

NSR ID	Description	Use	NAP ID	Criteria	Predicted Overall Noise Level, dB(A)	Major Noise Source causing Exceedance
E1-IA	Tseung Kong Wai	Existing Village	E1-IA_R01	70	69-70	-
E1-IB	Hong Mei Tsuen	Existing Village	E1-IB_R01	70	54-55	-
E1-IC	Tung Tau Tsuen	Existing Village	E1-IC_R01 to E1-IC_R03	70	61-67	-
E1-ID	Lo Uk Tsuen	Existing Village	E1-ID_R01	70	56	-
E1-IE	Sik Kong Wai	Existing Village	E1-IE_R01	70	66-68	-
E1-IF	Ha Tsuen Shi	Existing Village	E1-IF_R01 to E1-IF_R02	70	62-67	-
E1-IG	San Uk Tsuen	Existing Village	E1-IG_R01 to E1-IG_R03	70	61-66	-
E1-IH	Sha Chau Lei	Existing Village	E1-IH_R01 to E1-IH_R02	70	66-70	-
E1-IJ	San Sang Tsuen	Existing Village	E1-IJ_R01 to E1-IJ_R05	70	68-73	Tin Ha Road
E1-IK	Shek Po Tsuen	Existing Village	E1-IK_R01 to E1-IK_R03	70	64-72	Hung Tin Road, Hung Chi Road
E2-IA	Fung Kong Tsuen	Existing Village	E2-IA_R01 to E2-IA_R02	70	67-68	-
E4-IA	Tin Sum Tsuen	Existing Village	E4-IA_R01 to E4-IA_R03	70	72-73	Tin Ha Road

NSR ID	Description	Use	NAP ID	Criteria	Predicted Overall Noise Level, dB(A)	Major Noise Source causing Exceedance
E4-IB	Tsing Chuen Wai	Existing Village	E4-IB_R01	70	67-70	-
E5-IA	Kiu Tau Wai	Existing Village	E5-IA_R01	70	61-66	-
E5-IB_R	Hung Uk Tsuen	Existing Village	E5-IB_R01 to E5-IB_R02	70	61-69	-
E5-IB_C	Chinese Mission Seminary	Existing Church	E5-IB_C01	65	73-74	Castle Peak Road
E5-IC	San Lee Uk Tsuen	Existing Village	E5-IC_R01	70	73	Tin Ha Road
E1-26	Existing Homes for the Elderly (Yeung Chun Pui and Ching Chung)	Homes for the Elderly	E1-26_H01 to E1-26_H02	70	58-60	-
E4-30	Ling Liang Worldwide Evangelistic Mission Hung Shui Kiu Ling Liang Church	Place of Worship	E4-30_C01	65	72	Castle Peak Road
E5-23	Sheffield Villas	Residential	E5-23_R01 to E5-23_R12	70	64-66	-
E5-25	Hung Fuk Estate	Residential	E5-25_R01 to E5-25_R41	70	61-74	Hung Tin Road, Hung Chi Road
E5-31	Existing Residential Development	Residential	E5-31_R01 to E5-31_R22	70	64-73	Castle Peak Road
E5-35	Bellevue Court	Residential	E5-35_R01	70	67-71	Castle Peak Road
E5-39	Existing Residential Development	Residential	E5-39_R01 to E5-39_R05	70	59-75	Castle Peak Road
E5-44	The Church of Jesus Christ of Latter-Day Saints (Hung Shui Kiu)	Residential	E5-44_C01 to E5-44_C02	65	72-73	Castle Peak Road, Tin Ha Road
O5-06	Committed Subsidised Sales Flat	Residential	O5-06_R01 to O5-06_R24	70	58-72	Ping Ha Road (East of Hung Tin Road)
O5-33	Committed Residential Development	Residential	O5-33_R01 to O5-33_R31	70	68-77	Hung Shui Kiu Tin Sum Road

Table 4.32 Summary of Mitigated Road Traffic Noise Assessment Results (Planned NSRs within the Project area)

NSR ID	Use	Description	NAP ID	Criteria	Mitigated Overall Noise Level, dB(A)	Require Additional Mitigation Measures
P1-02	Residential	Site 1-02	P1-02_R01 to P1-02_R31	70	54-72	Yes
P1-03	Educational	Site 1-03	P1-03_E01 to P1-03_E11	65	51-62	No
P1-04	Residential	Site 1-04	P1-04_R01	70	44-58	No
P1-05	Residential	Site 1-05	P1-05_R01 to P1-05_R11	70	56-66	No
P1-06	Residential	Site 1-06	P1-06_R01 to P1-06_R11	70	64-70	No
P1-08	Residential	Site 1-08	P1-08_R01 to P1-08_R22	70	63-70	No

NSR ID	Use	Description	NAP ID	Criteria	Mitigated Overall Noise Level, dB(A)	Require Additional Mitigation Measures
P1-09	Residential	Site 1-09	P1-09_R01 to P1-09_R12	70	58-69	No
P1-12	Residential	Site 1-12	P1-12_R01	70	66-70	No
P1-15	Educational	Site 1-15	P1-15_E01 to P1-15_E05	65	60-66	Yes
P1-17	Residential	Site 1-17	P1-17_R01	70	54-60	No
P1-18	Educational	Site 1-18	P1-18_E01 to P1-18_E03	65	53-65	No
P1-19	Residential	Site 1-19	P1-19_R01 to P1-19_R21	70	60-70	No
P1-21	Residential	Site 1-21	P1-21_R01 to P1-21_R21	70	52-67	No
P1-22	Educational	Site 1-22	P1-22_E01 to P1-22_E11	65	49-69	Yes
P2-02	Residential	Site 2-02	P2-02_R01 to P2-02_R32	70	61-67	No
P2-03	Residential	Site 2-03	P2-03_R01 to P2-03_R21	70	52-65	No
P2-07	Residential	Site 2-07	P2-07_R01 to P2-07_R21	70	62-66	No
P2-09	Educational	Site 2-09	P2-09_E01 to P2-09_E02	65	58-65	No
P2-10	Residential	Site 2-10	P2-10_R01 to P2-10_R31	70	61-69	No
P2-11	Residential	Site 2-11	P2-11_R01 to P2-11_R31	70	65-70	No
P2-14	Residential	Site 2-14	P2-14_R01 to P2-14_R12	70	62-67	No
P2-15	Educational	Site 2-15	P2-15_R01	70	62-64	No
P2-16	Educational	Site 2-16	P2-16_E01 to P2-16_R15	65	60-63	No
P2-18	Residential	Site 2-18	P2-18_R01 to P2-18_R05	70	68-70	No
P2-19	Residential	Site 2-19	P2-19_R01	70	65-67	No
P2-24	Educational	Site 2-24	P2-24_E01 to P2-24_E11	65	54-65	No
P2-25	Residential Home for the Elderly	Site 2-25	P2-25_H01	70	67-69	No
P2-26	Educational	Site 2-26	P2-26_E01 to P2-26_E11	65	59-66	Yes
P2-28	Educational	Site 2-28	P2-28_E01 to P2-28_E02	65	55-63	No
P2-30	Residential	Site 2-30	P2-30_R01 to P2-30_R21	70	47-70	No
P2-31	Residential	Site 2-31	P2-31_R01 to P2-31_R21	70	60-67	No
P4-01	Staff Quarters	Site 4-01	P4-01_R01 to P4-01_R13	70	62-72	Yes
P4-03	Residential	Site 4-03	P4-03_R01 to P4-03_R11	70	58-70	No
P4-04	Residential	Site 4-04	P4-04_R01 to P4-04_R24	70	54-70	No

NSR ID	Use	Description	NAP ID	Criteria	Mitigated Overall Noise Level, dB(A)	Require Additional Mitigation Measures
P4-05	Residential	Site 4-05	P4-05_R01 to P4-05_R14	70	56-72	Yes
P4-06	Residential	Site 4-06	P4-06_R01 to P4-06_R12	70	61-70	No
P4-08	Educational	Site 4-08	P4-08_E01 to P4-08_E11	65	61-64	No
P4-10	Residential	Site 4-10	P4-10_R01	70	61-68	No
P4-20	Residential	Site 4-20	P4-20_R01 to P4-20_R04	70	62-70	No
P4-22	Residential	Site 4-22	P4-22_R01 to P4-22_R42	70	52-70	No
P4-24	Residential	Site 4-24	P4-24_R01 to P4-24_R46	70	63-75	Yes
P4-25a	Other Use (Commercial and Residential)	Site 4-25a	P4-25a_R01	70	63-67	No
P4-25b	Other Use (Commercial and Residential)	Site 4-25b	P4-25b_R01 to P4-25b_R11	70	60-69	No
P4-25c	Other Use (Commercial and Residential)	Site 4-25c	P4-25c_R01 to P4-25c_R11	70	52-60	No
P4-26	Residential	Site 4-26	P4-26_R01 to P4-26_R21	70	61-71	Yes
P4-28	Residential	Site 4-28	P4-28_R01 to P4-28_R44	70	54-71	Yes
P4-29	Other Use (Commercial and Residential)	Site 4-29	P4-29_R01 to P4-29_R31	70	50-69	No
P4-32	Staff Quarters	Site 4-32	P4-32_R01 to P4-32_R14	70	61-70	No
P4-33	Educational	Site 4-33	P4-33_E01 to P4-33_E03	65	59-73	Yes
P4-36	Educational	Site 4-36	P4-36_E02 to P4-36_R01	65	58-65	No
P5-07a	Other Use (Commercial and Residential)	Site 5-07a	P5-07a_R01 - P5-07a_R21	70	53-68	No
P5-07b	Other Use (Commercial and Residential)	Site 5-07b	P5-07b_R01 to P5-07b_R21	70	57-73	Yes
P5-08	Other Use (Commercial and Residential)	Site 5-08	P5-08_R01 to P5-08_R11	70	66-70	No
P5-09	Other Use (Commercial and Residential)	Site 5-09	P5-09_R01 to P5-09_R31	70	57-69	No
P5-14	Educational	Site 5-14	P5-14_E01 to P5-14_E11	65	66-72	Yes
P5-16	Residential	Site 5-16	P5-16_R01 to P5-16_R11	70	55-74	Yes
P5-17	Residential	Site 5-17	P5-17_E04 to P5-17_R12	65	53-73	Yes

NSR ID	Use	Description	NAP ID	Criteria	Mitigated Overall Noise Level, dB(A)	Require Additional Mitigation Measures
P5-21	Educational	Site 5-21	P5-21_E01 to P5-21_E02	65	61-66	Yes
O5-22	Committed Residential Development (Site 5-22)	Residential	O5-22_R01 to O5-22_R03	70	65-77	Yes
P5-24	Residential	Site 5-24	P5-24_R01 to P5-24_R23	70	43-68	No
P5-26	Residential	Site 5-26	P5-26_R01 to P5-26_R04	70	56-70	No
P5-28	Residential Home for the Elderly	Site 5-28	P5-28_H01	70	64-72	Yes
P5-32	Other Use (Commercial and Residential)	Site 5-32	P5-32_R01 to P5-32_R82	70	63-77	Yes
P5-34	Educational	Site 5-34	P5-34_E01 to P5-34_E11	65	48-71	Yes
P5-37	Educational	Site 5-37	P5-37_E01 to P5-37_E02	65	70-74	Yes

4.7.18 As shown in **Table 4.30** and **Table 4.31,** the exceedance at the representative existing NSRs would be due to Ping Ha Road (East of Hung Tin Road), Lau Fau Shan Road, Castle Peak Road, Tin Ha Road, Hung Tin Road, Hung Chi Road and Hung Shui Kiu Tin Sam Road. A review is conducted to determine whether the traffic noise impact on these existing roads due to the Project would be considered significant. Findings as presented in **Appendix 4.7.6** indicate that significant traffic noise impact from the Project would only be predicted on Hung Chi Road and Hung Shui Kiu Tin Sam Road. Implementation of mitigation measures on these existing roads under this Project are proposed to alleviate the traffic noise impact as described in **Sections 4.7.19** to **4.7.21**.

Hung Chi Road

4.7.19 The facades of Hung Fuk Estate (E1-IK_R02 and EI-IK_R03) and Shek Po Tsuen (E5-25_R01) facing Hung Chi Road would be exposed to the traffic noise level exceeding the noise criterion of 70 dB(A) though existing 2 m high noise barrier at the Shek Po Tsuen side and 5 m high noise barrier in the vicinity of Hung Fuk Estate have been in place. To alleviate the adverse traffic noise impact on these facades, LNRS on the concerned road sections is proposed and the results indicate that the above affected facades would comply with the noise criterion of 70 dB(A). Different materials for LNRS are being reviewed and tested by the Government for application in Hong Kong. Environmental reviews will be conducted at the later design stage to review the proposed noise mitigation measures taking into account the latest design standard at that time for the application of the LNRS materials.

Hung Shui Kiu Tin Sam Road (East Section)

4.7.20 The planned NSRs at Sites 5-28, 5-32 and 5-33 facing Hung Shui Kiu Tin Sam Road (East Section) would be exposed to traffic noise levels exceeding 70 dB(A) though existing 3 m high noise barrier in the vicinity of Site 5-28 has been in place. Provision of noise barrier along the concerned road section are studied. It is noted that a number of existing run-ins will be required for Site 5-32, which will break off the alignment of the noise barrier. In addition, the existing space of planting strips allowing installation of noise barriers are limited. Provision of noise barrier for Sites 5-32 and 5-33 is not feasible. LNRS on the concerned road sections is proposed to alleviate the traffic noise

impact and the results indicate that the affected facades at Site 5-28 would comply with the noise criterion of 70 dB(A). Different materials for LNRS are being reviewed and tested by the Government for application in Hong Kong. Environmental reviews will be conducted at the later design stage to review the proposed noise mitigation measures taking into account the latest design standard at that time for the application of the LNRS materials.

Hung Shui Kiu Tin Sam Road (South of San Lee Uk Tsuen)

4.7.21 The planned residential NSRs at Site 5-33 and educational institute at Site 5-37 facing Hung Shui Kiu Tin Sam Road (South of San Lee Uk Tsuen) would be exposed to traffic noise levels exceeding 70dB(A) and 65 dB(A) respectively. Mitigation measure on this road section is studied. Due to the existing bus stops, the existing space of planting strips allowing installation of noise barriers are limited. Provision of noise barrier is not feasible. LNRS on the concerned road sections is proposed to alleviate the traffic noise impact. Different materials for LNRS are being reviewed and tested by the Government for application in Hong Kong. Environmental reviews will be conducted at the later design stage to review the proposed noise mitigation measures taking into account the latest design standard at that time for the application of the LNRS materials.

Castle Peak Road

4.7.22 With the enhancement of the existing noise environment with LNRS described in **Section 4.7.8,** the Project would reduce the noise emissions at Castle Peak Road.

4.7.23 It is noted that a retrofitting programme for Castle Peak Road (Hung Shui Kiu) has also been studied by EPD and an approximately 106 m long 5 m high vertical barrier along westbound of Castle Peak Road (Hung Shui Kiu Section) near Parkview Garden (with absorptive panels) would be implemented. However, for other sections of Castle Peak road (Hung Shui Kiu), there are following site constraints that provision of noise barriers/enclosures are not feasible or ineffectiveness.

4.7.24 Space Availability: The central reserve at Castle Peak Road (Hung Shui Kiu) is less than 2 m in width. Therefore, space is not adequate for horizontal clearance at both sides of the carriageways. In addition, there is not enough space to widen the width of the central reserve as there are bus stop and existing subway on the westbound of the carriageway of Castle Peak Road (Hung Shui Kiu Section) and Light Railway Transit (LRT) on the opposite side of the carriageway. This has imposed constraints for erection of noise enclosure and barrier.

4.7.25 Existing Structure: For the section fronting Treasure Court, there is an existing subway across Castle Peak Road and a ramp structure on both sides of the carriageway. The amenity area between the carriageway and the existing ramp structure is less than 2 m width. If the noise barriers are proposed, its foundations and columns would conflict with the existing subway structure.

4.7.26 Road Safety: A few existing bus bay located along Castle Peak Road. It could be another constraint since sightline of passengers and the bus drivers at bus bay would be much degraded after erection of a barrier structure so that the barrier should be set back from the carriageway. However, it would reduce the effectiveness of noise barrier.

4.7.27 Pedestrian/Vehicle Access: There are also many pedestrian access/vehicle access along Castle Peak Road. This imposed a constraint on the extent of the noise barrier and the short length of barriers (e.g. less than 30 m) is considered not acoustically effective in noise reduction due to the diffracted noise at the two ends of the noise barrier.

4.7.28 For the remaining roads including Ping Ha Road, Lau Fau Shan Road, Tin Ha Road and Hung Tin Road, as reviewed, the Project would not change the nature and the alignment of these roads, and would not induce significant change in the traffic capacity, traffic composition and traffic pattern of these roads. Therefore, the noise impact on these existing roads due to the Project is considered insignificant and should therefore be addressed under the policy endorsed by the Executive Council in November 2000 to address the noise impacts from existing roads if necessary and feasible. The engineering solutions under the policy include retrofitting of barriers and enclosures; and providing LNRS. The feasibility of the installation of noise barriers/enclosures on the concerned existing roads is studied and the outcome is summarised in the following paragraphs.

Ping Ha Road

4.7.29 With the existing 5 m high vertical barrier at Ping Ha Road in the vicinity of YLPMSAA Tang Siu Tong Secondary School and Q.E.S. Old Students' Association Branch Primary School, the predicted noise levels at these two schools would still exceed the noise criterion of 65 dB(A). Upgrading the existing noise barrier to cantilever noise barriers is not feasible as the existing foundation cannot allow extra loading.

Lau Fau Shan Road

4.7.30 Construction of noise barriers along Lau Fau Shan Road is considered not feasible as there is no sufficient space on footpath outside private land and there are a number of entrance accesses.

Tin Ha Road

4.7.31 Vertical noise barriers have been proposed at private land under “Hung Shui Kiu Development, Stage 2 – Widening of Tin Ha Road and Tan Kwai Tsuen Road”. However, no barrier is proposed in the final layout due to address public objections on land resumption for noise barrier. Therefore, it is not feasible to install noise barrier. Resurfacing with LNRS materials may be considered to be implemented where practicable by the Government.

Hung Tin Road

4.7.32 The existing Hung Tin Road outside the Shek Po Tsuen is constructed with retaining structures between the local roads and Hung Tin Road, therefore, there will be no space for installing noise barrier next to Hung Tin Road. In addition, construction of noise barriers along local road of Shek Po Tsuen is also considered not feasible as there is no sufficient space on footpath outside private land.

4.7.33 Except the above existing roads, exceedance at some planned NSRs would be due to KSWH (the section in the vicinity of Site 4-1 and Site 4-5) and some planned local roads. However, this section of the existing KSWH may not have sufficient structural capacity for installing more extensive noise barriers. Construction of a large extent of noise barrier along local roads is not feasible due to insufficient space. Therefore, additional mitigation measures including blank façade/acoustic windows would be considered for these planned residential NSRs. Provision of air conditioning and noise insulated windows for planned educational institutes under Class Assessment Document is proposed to alleviate the adverse traffic noise impact. With these mitigation measures in place, the noise levels at these affected planned NSRs would be within the respective noise criteria. Details of the additional mitigation measures proposed at the affected NSRs are summarised in **Table 4.33** below and its locations are shown in **Figures 4.7.17 – Figure 4.7.27**.

Table 4.33 Proposed Additional Mitigation Measures at Planned Sites

Site Ref. No.	Use	Proposed Additional Mitigation Measures	Applied to	Figure Reference
1-2	Residential	Acoustic Windows	1 - 17/F	Figure 4.7.19
1-15	Educational	Blank Façade	1 edge of 1 building	Figure 4.7.21
1-22	Educational	Noise Insulated Windows and Air Conditioning	1 school	Figure 4.7.21
2-26	Educational	Blank Façade	1 edge of 1 building	Figure 4.7.20
4-1	Staff Quarters	Acoustic Windows	1 - 25/F	Figure 4.7.25
		Blank Façade	1 edge of 2 buildings	
4-5	Residential	Acoustic Windows	20 - 42/F	Figure 4.7.25
4-24	Residential	Acoustic Windows	1 - 20/F	Figure 4.7.26
4-26	Residential	Acoustic Windows	3 - 11/F	Figure 4.7.26
4-28	Residential	Acoustic Windows	1 - 2/F	Figure 4.7.26
4-33	Educational	Noise Insulated Windows and Air Conditioning	1 school	Figure 4.7.27
5-7b	Residential/ Commercial	Acoustic Windows	1 – 9/F	Figure 4.7.22
5-14	Educational	Noise Insulated Windows and Air Conditioning	2 schools	Figure 4.7.23
5-16	Residential	Acoustic Windows	3 – 22/F	Figure 4.7.21
5-17	Residential	Acoustic Windows	4 – 29/F	Figure 4.7.22
5-21	Educational	Noise Insulated Windows and Air Conditioning	1 school	Figure 4.7.23
5-22	Residential	Acoustic Balcony	1 - 21/F	Figure 4.7.24
5-28	Residential Home for the Elderly	Noise Insulated Windows and Air Conditioning[(a)]	1 Residential Home for the Elderly	Figure 4.7.24
5-32	Residential/ Commercial	Acoustic Windows	1 - 28/F	Figure 4.7.24
		Acoustic Balcony	1 – 9/F	
5-34	Educational	Blank Façade	1 edge of 1 building	Figure 4.7.24
		Noise Insulated Windows and Air Conditioning	1 school	
5-37	Educational	Noise Insulated Windows and Air Conditioning	1 school	Figure 4.7.24

Note:
(a) For Site 5-28, the need of additional mitigation measures would subject to whether the LNRS can apply on Hung Shui Kiu Tin Sam Road (East Section).

4.7.34 **Appendix 4.7.4** summarises the predicted traffic noise levels at various NSRs under mitigated scenario. It has been demonstrated that the noise impact can be mitigated to an acceptable level by employing a package of mitigation measures including: (i) noise barriers at some road sections of Project roads; (ii) application of LNRS on some Project road sections; (iii) alternative building layout and special building layout design such as blank façade design and provision of acoustic windows/acoustic balcony for planned residential NSRs; and (iv) boundary wall and provision of noise insulated windows and air conditioning for some educational institutes under Class Assessment Document. However, there is scope for adoption of alternative noise mitigation measures, for example, by means of alternative layout and design of individual developments at detailed design stage. Noise impact assessment at the planned residential sites is proposed to be conducted by future developers at the detailed design stage to study whether the future development layout would avoid exposing excessive traffic noise

levels so as to minimise the scale/extent of the proposed noise mitigation measures such as noise barriers and special building layout design. The requirement of noise impact assessment would be included in the lease condition or planning briefs of the residential sites.

4.7.35 With the implementation of mitigation measures, the total number of dwellings, classrooms and other noise sensitive element that will be benefited and protected has been calculated. It is estimated that a total of about 786 existing dwellings, 4352 planned dwellings, 120 existing classrooms, 455 planned classrooms and 1 existing and 3 planned buildings of home for the elderly would be benefited. A total of about 189 existing dwellings, 4352 planned dwellings, 30 existing classrooms, 455 planned classrooms and 1 existing and 3 planned buildings of home for the elderly would be protected.

Indirect Mitigation Measures for Existing NSRs

4.7.36 Review of the further mitigation measures have been conducted in consideration of the constraints and project nature. It is considered that all practicable direct mitigation measures, including: i) noise barriers at some road sections of Project roads; ii) application of LNRS on some Project Road sections; and iii) alternative building layout and special building design including blank façade design and provision of acoustic windows/acoustic balcony for planned residential NSRs and provision of noise insulated windows and air conditioning for planned educational institutes have been exhausted and no adverse residual impact is anticipated.

4.7.37 **Appendix 4.7.5** presents the eligibility test against the above three testing criteria stated in **Section 4.7.7** for indirect technical noise mitigation measures. All representative NSRs do not fall within the above three testing criteria, indirect mitigation measure is therefore not required under this Project.

Evaluation of Residual Environmental Impacts

4.7.38 The traffic noise impact from the Project can be properly mitigated by implementing the proposed noise control measures such as noise barriers on Project roads, LNRS and special building design on the planned developments. Residual road traffic noise impact is not anticipated.

4.8 Rail Noise Assessment

Assessment Approach and Methodology

Train Noise Source Term for Rail Noise Assessment

4.8.1 The key noise impact from the railway would be the train induced airborne noise at viaduct and at grade section of WRL and LRT respectively across the Project area. Three noise sources have been identified from train operation are listed below:

- Rolling noise from train wheel and rail interaction;
- Noise from air conditioning unit on top of the each train cars; and
- Structure re-radiated noise from the viaduct structure during train passing.

4.8.2 The WRL viaduct structure including the multi plenum system and the above noise source propagation path are further illustrated with diagram in **Appendix 4.8.1**.

WRL Source Terms Reference Review

4.8.3 The rail noise source terms of WRL were referred from the "West Rail Operation Noise Assessment Report" prepared by MTR Corporation Limited in July 2015. The report aims to assess the noise impact of WRL due to the change from 7 cars to 8 cars in accordance with the Environmental Permit (FEP-24/004/1998/J). Noise measurement has been conducted to obtain the in-situ noise source terms for trains running on typical viaduct sections with the installation of multi plenum system. The air conditioning units on top of the WRL train and the structure re-radiated noise from the viaduct were also measured during the study. All three train noise source terms are considered as the latest information and adopted in the WRL noise assessment for this Project.

LRT Source Terms Reference Review

4.8.4 The source term of LRT was adopted from the previous EIA Report No. EIA-027/1999 entitled "Light Rail Transit (LRT) works for Tin Shui Wai Reserve Zone and Grade Separation of the LRT with Pui To Road and Tsing Lun Road in Tuen Mun" under a EIA Study Brief No: ESB-027/1999. The Sound Exposure Level (SEL) of the LRT was evaluated from the Lmax referenced from the previous EIA report. Air conditioning noise and re-radiated noise are considered insignificant contribution compared with the rolling noise of a LRT in this Project.

Rail Noise Assessment Procedures

4.8.5 The rail noise assessment was carried out to assess the worst case scenario under normal, abnormal, transient and emergency operation. The rail noise from the noise source to the NSRs will be corrected by distance, angle of view, barrier, speed, frequency, façade etc. The railway track was divided into segments and the respective noise level of each segment was calculated. The assessment procedure is listed in **Table 4.34** summarised the steps and assumption of the rail noise calculation. Major assessment parameters are including train frequency and speed are provided by MTRC and summarised in **Table 4.35**.

Table 4.34 Rail Noise Assessment Procedures

Description	Assumptions / Remarks	Ref.
Locate the NAPs which would most likely be affected by air-borne noise from the rail operation	The first layer of existing/planned NSRs located in the vicinity of WRL and LRT, have been selected as representative NAPs for rail noise prediction and evaluation	Section 4.4
Identify the train noise sources origination	The rolling noise source is located from the gap between the train and the walkway on the viaduct. Part of the train wheel rail interaction noise had been absorbed by a multi-plenum system and the effect of the system has been included in the source measurement in previous study A/C noise located on the roof of each car and source level was measured in previous study Structural re-radiated noise generated from the vibration of the viaduct during event was measured in previous study	**Appendix 4.8.1**
Identify the train frequency and make appropriate frequency correction	+10 log10 (N) where N = Train frequency per 30 min per Direction	A
Identify the train speed for correction	+20 log10 (V/Vref) where V = Train speed, Vref = Reference train speed	A
Identify the distance from the NSR to the train and make distance correction	-10 log10 (d'/dref) where d' = Slant distance from track to NSR and dref = Reference distance	A

Description	Assumptions / Remarks	Ref.
Identify the train type and the source term of the train	Relevant noise level referred to previous study	B & C
Identify the angle of view and make appropriate angle of view correction	+10 log10 ($\pi\theta/180 - \cos 2\alpha \sin\theta$) -5 where θ= Angle of View and α= Acute angle between a line drawn through the NSR, parallel to the track, and the line bisecting the angle of view, θ	A
Incorporate Façade Correction	+2.5 dB(A)	A
Incorporate Track Wear Correction	+3 dB(A)	A
Incorporate Joint/Turnout Correction	With conservative assumption, +7.0dB(A) for a representative 20 m segment	A
Incorporate Barrier Correction	Shadow Zone: -21dB(A) for δ > 2.5 m where δ is the Path Difference in metre -7.75 log10 (5.2 + 203 δ) dB(A) for 0≤ δ <2.5 m Illuminated Zone: 0dB(A) for δ > 0.4 m 0.88 + 2.14 log10 (10-3 + δ) dB(A) for 0≤ δ <0.4 m	Chart 6(a) of A
Calculate the overall noise level from all rail segments including rolling noise, A/C noise and structural re-radiated noise	Standard acoustic principal	-
Calculate the Lmax	SEL = Lmax + 10log(L/V) +10.5 – 10log[(4D/(4D2+1)) + 2tan-1(1/(2D))]	Equation 15.21 of D
Calculate the Leq, 24hr	Standard acoustic principal	-
Model validation	On site measurement was carried out to record the in-situ noise level and compared against the modelled results.	**Appendix 4.8.2**

Reference:
A - Calculation of Railway Noise (1995).
B - "West Rail Operation Noise Assessment Report" prepared by MTR Corporation Limited in July 2015.
C - "Light Rail Transit (LRT) works for Tin Shui Wai Reserve Zone and Grade Separation of the LRT with Pui To Road and Tsing Lun Road in Tuen Mun" under a EIA Study Brief No: ESB-027/1999.
D - Transportation Noise Reference Book.

Table 4.35 Rail Noise Assessment Input Summary

Parameters	WRL	LRT
Train Type and No. of Car	SP1900, 9 cars total 225 m in accordance with the VEP-45/2001	LRV, 2 cars of total 40 m as conservative scenario
Rolling Noise	SEL Northbound = 81.4 dB(A) [a] SEL Southbound = 80.7 dB(A) [a] (8 cars running 130kph at 25 m)	Lmax = 65 dB(A) [b] (2 cars running 50kph at 25 m)
Air Conditioning Noise	Lmax at Viaduct = 48.8 dB(A) [a] Lmax at Station = 54.8 dB(A) [a] (8 cars at 25 m)	Not applicable as insignificant contribution
Structure Re-radiated Noise	$L_{eq\ 30mins}$ = 40.6 dB(A) Viaduct [a]	Not applicable as LRT in HSK is at grade
Train Frequency per 30 min per Direction	14 number during peak daytime; 10 number during peak night-time	Between Nai Wai LRT Station to HSK LRT Station: Southbound: 20 number during peak daytime; 13 number during peak night-time Northbound: 20 number during peak daytime; 24 number during peak night-time From Tin Shui Wai LRT Station to Hang Mei Tsuen LRT Station: Southbound: 14 number during peak daytime; 13 number during peak night-time Northbound: 14 number during peak daytime; 18 number during peak night-time
Train Frequency per 24hrs both directions	680 trains	Between Nai Wai LRT Station to HSK LRT Station: 917 trains From Tin Shui Wai LRT Station to Hang Mei Tsuen LRT Station: 717 trains
Train Speed	Refer to speed profile in Appendix 2.1 in [a] • Station speed, 50 kph • Approach and leaving Station, 80 kph • Other viaduct section, 100 kph	LRT speed was measured on site. • Section near planned NSR, 60 kph • Section approaching and leaving junctions and stations, 40 kph • Section near NAP LR-E1, 50 kph • Section near NAP LR-E2, 36 kph

Notes:
[a] "West Rail Operation Noise Assessment Report" prepared by MTR Corporation Limited in July 2015.
[b] "Light Rail Transit (LRT) works for Tin Shui Wai Reserve Zone and Grade Separation of the LRT with Pui To Road and Tsing Lun Road in Tuen Mun , 1999".

Proposed HSK Station

4.8.6 A new HSK railway station (HSK Station) is proposed within the Project area. No detail layout is available at this EIA stage. The new station should match the existing West Rail alignment such that station platforms would be constructed on both sides of the tracks (i.e. sided platform design) and enclosed by 4 m screen doors, side wall and top cover. The track section of the station is remain uncovered in the centre. The assumed layout of proposed HSK Station is shown in **Figure 4.8.1**.

4.8.7 As mentioned in **Section 2.4.44**, the proposed HSK Station is a separate DP under Item A.2 of Schedule 2 of EIAO-TM, a separate EIA study would be conducted by the rail operator to fulfil all the statutory requirements and procedures under the Ordinance.

Phasing of HSK Development

4.8.8 The site layouts for rail noise assessment have been confirmed with Planning Department. Commercial developments Site 5-18 of 125 mPD are strategically planned to protect planned residential Site 5-17 of 115 mPD. Therefore Site 5-18 should be developed prior to Site 5-17 in purpose. This phasing approach also applied to the HSK WRL Station which should be developed before or concurrent with the residential development Sites 4-25 and 4-29 nearby. A review would be conducted for the environmental noise performance of the sites if the above development phasing programme cannot meet the specific arrangement.

Prediction and Evaluation of Noise Impacts

WRL Noise

4.8.9 The predicted air-borne noise levels from WRL with respect to the representative site layouts confirmed with relevant departments are presented in **Table 4.36**. The predicted noise level with full mPD level of all NSRs are shown in **Appendix 4.8.3.** Exceedances were predicted at representative NSRs WR-P5d, P5e, P5f, P7b & P7c at night-time period. While exceedance of day time criteria was predicted at WR-P10. It is predicted that a total of about 155 planned dwellings will be exposed to WR noise impact under unmitigated scenario. Mitigation measures are therefore required to reduce the rail noise impact for above NSRs during WRL operation.

Table 4.36 WRL Noise Calculation Summary – Unmitigated

NAP	Description	Area Sensitivity Rating	Noise Criteria, dB(A)		mPD	Leq, 30min, dB(A)		L_{max}, dB(A)	Leq, 24 hrs, dB(A)
			0700-2300 hrs	2300-0700 hrs		0700-2300 hrs	2300-0700 hrs		
Existing WRL NSRs									
WR-E1	Premises to the southwest of Yick Yuen Tsuen	A	60	50	8.7	50	48	57	49
WR-E2	Premises to the east of Oakland Court	A	60	50	8.5	48	47	54	48
WR-E3	Premises to the southwest of Tin Sum Tsuen	A	60	50	7.7	43	41	48	43
WR-E4	Premises to the West of San Lee Uk Tsuen	A	60	50	6.5	47	46	51	47

NAP	Description	Area Sensitivity Rating	Noise Criteria, dB(A)		mPD	Leq, 30min, dB(A)		L_{max}, dB(A)	Leq, 24 hrs, dB(A)
			0700-2300 hrs	2300-0700 hrs		0700-2300 hrs	2300-0700 hrs		
WR-E5	Premises to the West of San Lee Uk Tsuen Block 3,Galore Garden	A	60	50	6.9-9.9	47-48	46-47	52-54	47-48
WR-E6	Shek Po Tsuen Village House Premises to the southwest of Yick Yuen Tsuen Premises to the east of Oakland Court	A	60	50	6.4-15.4	47-49	45-47	51-54	47-49
WR-E7	Premises to the southwest of Tin Sum Tsuen	A	60	50	5.2-8.2	45	44	46-47	45
Project Planned WRL NSRs									
WR-P1	Residential Development in Site 4-6	B	65	55	10.0-141.0	42-52	41-51	42-53	42-52
WR-P2	School in Site 4-8	B	65	[a]	11.0-47.0	42-47	[a]	41-48	42-48
WR-P3	Residential Development in Site 4-10	B	65	55	11.0-166.0	43-54	41-52	43-55	43-54
WR-P4	Residential Development in Site 4-20	B	65	55	11.0-19.0	50-54	48-53	56-62	49-54
WR-P5a	Residential Development in Site 4-22	B	65	55	11.0-127.0	42-53	41-52	48-62	40-53
WR-P5b	Residential Development in Site 4-22	B	65	55	11.0-127.0	41-53	40-51	49-64	40-53
WR-P5c	Residential Development in Site 4-2	B	65	55	11.0-127.0	39-49	39-48	45-60	36-49
WR-P5d	Residential Development in Site 4-22	B	65	55	11.0-127.0	45-58	44-**56**	51-67	44-58
WR-P5e	Residential Development in Site 4-22	B	65	55	11.0-127.0	46-58	45-**57**	51-65	45-58
WR-P5f	Residential Development in Site 4-22	B	65	55	11.0-127.0	47-59	46-**58**	52-66	47-59
WR-P6a	Residential Development in Site 4-25	B	65	55	12.0-178.0	39-52	38-51	40-57	37-52
WR-P6b	Residential Development in Site 4-25	B	65	55	12.0-178.0	39-52	37-51	39-57	37-52
WR-P6c	Residential Development in Site 4-25	B	65	55	12.0-178.0	38-51	37-49	37-55	37-51

NAP	Description	Area Sensitivity Rating	Noise Criteria, dB(A)		mPD	Leq, 30min, dB(A)		L_{max}, dB(A)	Leq, 24 hrs, dB(A)
			0700-2300 hrs	2300-0700 hrs		0700-2300 hrs	2300-0700 hrs		
WR-P7a	Residential Development in Site 4-29	B	65	55	13.0-171.0	44-56	43-55	49-63	44-56
WR-P7b	Residential Development in Site 4-29	B	65	55	13.0-171.0	47-59	45-**57**	50-64	47-59
WR-P7c	Residential Development in Site 4-29	B	65	55	13.0-171.0	45-57	44-**56**	50-64	45-57
WR-P8	School in Site 4-36	B	65	[a]	12.0-48.0	43-52	[a]	46-56	43-52
WR-P9a	Residential Development in Site 5-17	B	65	55	7.0-111.0	42-54	41-53	48-61	42-54
WR-P9b	Residential Development in Site 5-17	B	65	55	7.0-111.0	44-55	42-54	49-62	44-55
WR-P10	School in Site 5-21	B	65	[a]	6.0-43.0	49-**67**	[a]	56-74	48-**67**
WR-P11	School in Site 5-34	B	65	[a]	7.0-43.0	42-46	[a]	42-47	42-46

Notes:
[a] Educational use is not considered as a night-time NSR.
Numbers in bold denote exceedances of the noise criteria.

LRT Noise

4.8.10 The predicted air-borne noise levels from LRT with respect to the representative site layouts confirmed with relevant departments are presented in **Table 4.37**. The full list of predicted levels with respect to mPD is shown in **Appendix 4.8.4.** It is noticed that low level of the NSRs LR-P5a & P6 would exceed the night-time criteria. It is predicted that a total of about 30 planned dwellings will be exposed to LRT noise impact under unmitigated scenario. Mitigation measures are required to reduce the rail noise impact during LRT operation.

Table 4.37 LRT Noise Calculation Summary – Unmitigated

NAP	Site No.	Area Sensitivity Rating	Noise Criteria, dB(A)		mPD	Leq, 30min, dB(A)		Lmax, dB(A)	Leq, 24 hrs, dB(A)
			0700-2300 Hrs	2300-0700 hrs		0700-2300 hrs	2300-0700 hrs		
Existing LRT NSRs									
LR-E1	Parkview Garden Block 24 - 27	C	70	60	12.5-21.5	60-61	60	71-72	59
LR-E2	Tak Cheung Building	C	70	60	11.6-29.6	57-60	57-60	66-70	56-59
LR-E3	Beauty Court Block 1	C	70	60	10.4-46.4	54-57	54-57	61-66	53-56
LR-E4	Coronet Court	C	70	60	10.1-46.1	55-58	55-58	62-67	54-57
LR-E5	Park Nara Tower 1	C	70	60	10.6-46.6	55-58	54-58	62-68	53-57
LR-E6	Village House in Hung Uk Tsuen facing Kiu Hung Road	C	70	60	5.5-14.5	58	58	64	57
LR-E7	Ling Liang Church	C	70	[a]	13.5-16.5	58	[a]	66	56-57
Project Planned LRT NSRs									
LR-P1	Residential Development in Site 4-22	C	70	60	11.0-127.0	48-51	48-51	49-55	46-50
LR-P2	Residential Development in Site 4-26	C	70	60	12.0-152.0	49-54	48-54	48-58	47-52
LR-P3a	Residential Development in Site 4-28	C	70	60	13.0-144.0	51-57	51-57	55-66	50-55
LR-P3b	Residential Development in Site 4-28	C	70	60	13.0-144.0	50-58	50-57	55-68	49-56
LR-P4	School in Site 4-33	C	70	[a]	12.0-48.0	59-64	[a]	69-78	58-63
LR-P5a	Residential Development in Site 5-7b	B	65	55	5.0-142.0	50-55	50-**56**	49-61	50-55
LR-P5b	Residential Development in Site 5-7b	B	65	55	5.0-142.0	49-55	49-55	49-60	49-55
LR-P5c	Residential Development in Site 5-7b	B	65	55	5.0-142.0	48-54	49-55	50-62	49-54
LR-P6	Residential Development in Site 5-9	B	65	55	5.0-152.0	50-56	51-**56**	52-63	50-56

Notes:
[a] Educational use and church is not considered as a night-time NSR.
Numbers in bold denote exceedances of the noise criteria.

Rail Noise Mitigation Measures Options

4.8.11 Based on the assessment results, those rail noise level of exceeding the criteria are less than 3 dB(A), the acoustic fins, non-sensitive use or fixed glazing and layout set back at the NSR are considered effective to reduce the noise impact. Moreover, planning strategy including commercial buildings screening and residential buildings orientation are considered against rail noise during the development of the revised RODP. Predicted noise levels of mitigated NSRs are enclosed in **Appendix 4.8.5**. None of the dwellings will exceed the noise criteria due to rail noise under mitigated scenario.

Architectural Fins

4.8.12 Architectural fins are typical noise mitigation measure for buildings units impacted by traffic or rail noise. For NSR WR-P5d and LR-P5a, 2 m fins has been proposed for the dwellings to narrow the angle of view to the railway track. The noise level of the NSR screened by architectural fins were assessed and the mitigated noise level satisfy the noise criteria as shown in **Table 4.38** and **Figure 4.8.2 – 4.8.4**.

Non-Sensitive Use or Fixed Glazing

4.8.13 For those façade NAP with exceedance, the NAP would be relocated to another positions with smaller angle of view facing the railway and the façade NAP with exceeded the criteria are designed as non-sensitive use or fixed glazing. The noise level of relocated NAPs at WR-P5e, P5f and LR-P6a are satisfied the noise criteria and shown in **Table 4.38** and **Figure 4.8.2 & 4.8.4**.

Layout Setback

4.8.14 Some of the proposed development sites has a potential of shifting the layout as well as set-back away from the existing railway line. The setback provides a sufficient buffer distance and attenuation in terms of noise propagation. It was proposed to setback the layout of Site 4-29 and Site 5-21 from the WRL to a distance of 70 m and 25 m as shown in **Figure 4.8.5 and Figure 4.8.6 respectively**, the assessed noise level at NAP WR-7b, WR-P7c and WR-P10 would satisfy the noise criteria as shown in **Table 4.38**.

Table 4.38 Rail Noise Calculation Summary – Mitigated

NAP	Proposed Mitigation Measures	Noise Criteria, dB(A)		mPD	Leq, 30min, dB(A)		Lmax, dB(A)	Leq, 24 hrs, dB(A)
		Daytime /evening	Night-time		0700-2300 hrs	2300-0700 hrs		
WR-P5d	2 m Architectural fin	65	55	11.0-127.0	42-51	41-50	45-60	37-51
WR-P5e.1	1.5 m Architectural fin + Relocation to non-sensitive use or fixed glazing	65	55	11.0-127.0	43-55	42-54	49-63	43-55
WR-P5f.1	Relocation to non-sensitive use or fixed glazing	65	55	11.0-127.0	45-57	43-55	51-65	44-57
WR-P7b	Layout Set Back to 70 m from WRL	65	55	13.0-171.0	45-57	44-55	48-61	45-57
WR-P7c	Layout Set Back to 70 m from WRL	65	55	13.0-171.0	44-56	43-54	48-61	44-56
WR-P10	Layout Set Back to 25 m from WRL	65	[a]	6.0-43.0	48-63	[a]	54-70	47-63
LR-P5a	2 m Architectural fin	65	55	5.0-142.0	49-54	49-55	49-61	49-54
LR-P6.1	Relocation to non-sensitive use or fixed glazing	65	55	5.0-152.0	48-53	49-54	53-63	48-53

NAP	Proposed Mitigation Measures	Noise Criteria, dB(A)		mPD	Leq, 30min, dB(A)		Lmax, dB(A)	Leq, 24 hrs, dB(A)
		Daytime /evening	Night-time		0700-2300 hrs	2300-0700 hrs		
LR-P6.2	1.3 m Architectural fin + Relocation to non-sensitive use or fixed glazing	65	55	5.0-152.0	47-53	48-54	51-62	47-53
LR-P6.3	Relocation to non-sensitive use or fixed glazing	65	55	5.0-152.0	49-55	49-55	52-63	49-55

Notes:
[a] Educational use is not considered as a night-time NSR.

Rail Noise Impact Assessment for Development Sites

4.8.15 For the new developments in the OZP, the future development proponents should conduct a stand-alone Railway Noise Impact Assessment (RNIA) to assess the potential noise impacts from rail operations on future occupants, and implement the aforesaid or other specific designed noise mitigation measures at their respective development sites to the satisfaction of EPD, to ensure full compliance with the statutory noise limits. The above development requirements would be imposed through relevant Planning Briefs or Land Lease conditions to be fulfilled by the future development proponents. Sites require RNIA include: 4-6, 4-8, 4-10, 4-22, 4-25c, 4-26, 4-28, 4-29, 4-31, 4-32, 4-33, 4-36, 5-7b, 5-9, 5-17, 5-21, 5-32 and 5-34.

Environmentally Friendly Transport Services

4.8.16 An EFTS is proposed to be introduced to operate within the Project. The alignment of EFTS is shown in **Figure 4.8.7.**

4.8.17 Several modes of EFTS, including LRT, trolley bus, automatic people mover (APM), monorails, modern tramway and electric / LPG bus, will be investigated in later stage. No updated information was available at the time of preparing this report. For conservative noise assessment, the rail based EFTS was assumed for rail noise impact assessment. As shown in **Figure 4.8.7**, all of the NSRs are planned with at least 10 m buffer distance from the proposed alignment for EFTS. A reference scheme was developed with the assumptions and assessed level shown in **Table 4.39**.

Table 4.39 Rail based EFTS Reference Scheme

Items	Description
Type of source	Automated guideway transit, steel wheel, welded rail
Noise source level	SEL 80 dB(A) at 50 feet and 50 mph Ref: Table 5-1 Transit Noise and Vibration Impact Assessment FTA-VA-90-1003-06 May 2006 by Office of Planning and Environment Federal Transit Administration
NSR Distance from track	10 m
Operating speed	Assumed 40 kph
Frequency	Assumed 5 trains per headway
Calculated Noise level at NSR boundary	57.2 dB(A) > night-time noise criteria of 55 dB(A)
Track enhancement	• Embedded rail of 3 dB(A) reduction • Green track with vegetation 2 dB(A) reduction Reference material as shown in **Appendix 4.8.6**
Noise level at NSR after track enhancement	< 55 dB(A) complied with night-time noise criteria of HSKNDA

4.8.18 In case a rail cross or joint is designed for the EFTS, such cross joint location should be located at the section with largest distance between the track and the NSR. Alternatively, erect a noise barrier or enclosure at the cross joint section to control the noise impact to the nearby NSRs.

4.8.19 There is no design and operation detail of the EFTS during this Schedule 3 EIA. In any case there are changes to the assumption in **Table 4.39** and causing an exceed of the noise level compared with the criteria, the EFTS project proponent should implement further noise mitigation measures such as noise barrier and enclosure to control the noise impact by the EFTS to satisfy the noise criteria. Noise barrier and enclosure are effective and traditional mitigation measures to reduce rail noise by 10 dB(A) and 15 dB(A) respectively.

4.8.20 As discussed in **Chapter 2** of this report, the proposed EFTS is a Schedule 2 DP under the EIAO. The proposed EFTS will be designed, constructed and operated by the others. The associated environmental impacts will be adequately addressed in a further detailed EIA study to be prepared and submitted under the EIAO by the respective project proponent. Adverse noise impacts from EFTS are therefore not expected.

Cumulative Impact

4.8.21 The existing WRL is running at the centre of the Project area and the LRT is running along Castle Peak Road close to the southern Project area. The cumulative impact is not anticipated by both existing rail to the planned and existing NSRs. However, the proposed EFTS would run from the southern Project area near Nai Wai LRT station to the northern of the Project area and end at TSW Station. As shown in the EFTS alignment, there is potential cumulative rail noise impact to those NSRs by both EFTS, LRT and WRL for Site 4-22 at the southern Project area. If the cumulative noise level to Site 4-22 evaluated by the EFTS project proponent exceeds the noise criteria, the noise level of the EFTS should be further mitigated by means of measure mentioned in **Section 4.8.17** to a level of ANL-10 dB(A) at the Site 4-22 NSR.

Evaluation of Residual Environmental Impacts

4.8.22 WRL would be the major mass transit railway for the Project area. A new HSK Station would be built to serve the community to and from urban area. The operation of WRL will generate noise impact to some of the planned development. Noise control measure should be considered during the planning of the land use and the design of building. Different noise mitigation measures such as layout set back, relocation of noise sensitive uses or implementation of fixed glazing and architectural fins are considered effective mitigation measures. With the mitigation measures in place, the predicted noise level of all dwellings would satisfy the noise criteria in the TM.

4.8.23 LRT will continue running in Castle Peak Road along the southern Project area. Planned developments may expose to the LRT noise without any screening obstacles and result in noise impact. Mitigation measures such as architectural fin and relocation of habitable windows are suggested to abate the rail noise impact.

4.8.24 In conclusion, the future development proponents should conduct a stand-alone Noise Impact Assessment (NIA) to assess the potential noise impacts from rail operations on future occupants, and implement the aforesaid or other specific designed noise mitigation measures at their respective development sites to the satisfaction of EPD, to ensure full compliance with the statutory noise limits. The above development requirements would be imposed through relevant Planning Briefs or Land Lease conditions to be fulfilled by the future development proponents.

4.9 Fixed Plant Noise Assessment

Assessment Approach and Methodology

4.9.1 The assessment for fixed plant noise impact is conducted in accordance with the IND-TM and the criteria set in Annex 5 of the EIAO-TM. The worst operation mode with 100% operation is assumed in the prediction. The noise levels from operation of existing fixed noise sources such as Sik Kong Tsuen Stormwater Pumping Station, Sik Kong Wai Stormwater Pumping Station, Lo Uk Tsuen Stormwater Pumping Station, Kiu Tau Wai Stormwater Pumping Station, Sheung Cheung Wai Floodwater Pumping Station and Ha Tsuen Sewage Pumping Station are determined by noise measurement. As there is no design information on planned fixed noise sources, maximum allowable sound power level for the planned fixed plant noise source is predicted based on backward calculation of separation distance between the noise source and the nearest NSRs regardless on the percentage usage. Cumulative fixed noise impact arising from fixed noise sources located within 300 m assessment area at each identified NSR is assessed. Validity of the operation modes have been confirmed with the Project Engineer. Project Proponent would further liaise and agree with the relevant departments on the responsibility of implementation and maintenance of the predicted maximum allowable sound power levels for the planned fixed plant noise sources in the detailed design stage.

4.9.2 Fixed plant noise impact assessment is conducted based on the following procedures:

- Determine the assessment area;
- Identify and locate representative NSRs that may be affected by the noise sources;
- Determine the noise criteria for both daytime and night-time;
- Use standard acoustic principle for attenuation and directivity;
- Adopt correction of tonality, impulsiveness and intermittency as stipulated in TM-places;
- Calculate the noise impacts using assumed plant inventories and utilisation schedule, if available; and
- Calculation repeats for various levels from the fixed plant noise sources

4.9.3 Locations of the potential fixed noise sources during operation phase are shown in **Figure 4.4.37** – **Figure 4.4.51**.

Prediction and Evaluation of Environmental Impacts

Upgraded San Wai STW

4.9.4 The existing San Wai STW is a preliminary treatment plant located within the Project area. Under the approved EIA study for "Upgrading and expansion of San Wai STW and expansion of Ha Tsuen Pumping Station"[1] (San Wai STW EIA) completed in 2003, it was proposed to expand the design flow of San Wan STW and upgrade its treatment level.

[1] Upgrading and expansion of San Wai STW and expansion of Ha Tsuen Pumping Station (Register No.: AEIAR-072/2003) was approved on 12 May 2013

4.9.5 The existing San Wai STW is located at Site 3-17. According to the San Wai STW EIA, the predicted fixed plant noise levels at San Wai village house were 41-42 dB(A) under mitigated scenario. Referring to the Revised RODP, the San Wan village will be removed, and the nearest NSR is at Tseung Kong Wai village house, which is about 130 m far away from the original San Wai village house. Hence, the fixed plant noise levels predicted at Tseung Kong Wai village house would be less than 42 dB(A). As refer to **Table 4.6**, the night-time background noise levels measured at Tseung Kong Wai village house is 46 dB(A). Thus, the fixed plant noise levels due to the operation of San Wai STW at the nearest NSR, i.e. Tseung Kong Wai village house, would still comply with the noise criteria of 46 dB(A). The nearest planned NSRs under the Project would be located at least 300 m from the upgraded San Wai STW. In addition, the upgraded Sai Wai STW would be totally screened by the planned logistics facilities and there is no direct line-of-sight between the STW and the nearest planned NSRs. Thus, the Sai Wai STW would not induce noise impact to the nearby planned NSRs under the Project.

Existing Pumping Stations

4.9.6 There are four existing stormwater pumping stations, a floodwater pumping station and a SPS located within/in vicinity of the Project area. **Table 4.40** presents the measured noise levels of the existing pumping stations within the 300 m area from the boundary of the Project. Corrections for tonality, impulsiveness and intermittency have been made to the corrected noise levels. The measured noise levels have included the effects of all noise sources in the vicinity such as road traffic noise and community noise.

Table 4.40 Corrected Noise Levels of the Existing Pumping Stations

Source ID	Pumping Stations	Maximum Corrected SPL, LAeq (30-min), dB(A)		
		Daytime (07:00 – 19:00)	Evening-time (19:00-23:00)	Night-time (23:00 – 07:00)
EPS1	Sik Kong Tsuen Stormwater Pumping Station	65	64	64
EPS2	Sik Kong Wai Stormwater Pumping Station	66	63	63
EPS3	Lo Uk Tsuen Stormwater Pumping Station	57	56	59
EPS4	Kiu Tau Wai Stormwater Pumping Station	62	63	65
EPS5	Sheung Cheung Wai Floodwater Pumping Station	69	61	67
ESP6	Ha Tsuen SPS	82	85	84

Remarks:
- Corrections for tonality, impulsiveness and intermittency have been considered.
- Free-field measurements were made at 1 m from louvre of the SPSs.

4.9.7 As a conservative approach, the maximum corrected SPL showing in **Table 4.40**, i.e. the SPLs measured in daytime, are adopted in the calculation for all time modes. The predicted SPL at the representative planned NSRs due to the operation of above existing pumping stations and other cumulative fixed plant noise sources (if any) are summarised in **Table 4.41**. Detailed calculation is shown in **Appendix 4.9.1**.

Table 4.41 Predicted Sound Pressure Level at Planned NSRs for Existing Pumping Stations

Source ID	Nearest Planned NSR	AP	ASR	Shortest Distance from the Source (m)	Max Predicted SPL, $L_{eq30mins}$ dB(A) [Criteria, dB(A)]		
					Day time	Evening time	Night-time
EPS1	1-19	P1-19_R11	B	210	24(1) [60]	22(1) [60]	22(1) [50]
EPS2	1-19	P1-19_R11	B	310	24(1) [60]	22(1) [60]	22(1) [50]
EPS3	1-12	P1-12_R01	B	75	18 [60]	17 [60]	20 [50]
EPS4	5-9	P5-9_R32	B	115	32(1) [60]	32(1) [60]	28(1) [50]
EPS5	5-6	O5-06_R25	B	150	54(1) [60]	54(1) [60]	44(1) [50]
EPS6	5-16	P5-16_R01	B	40	55(1) [60]	55(1) [60]	48(1) [50]

Remarks:
(1) Cumulative impact from other fixed noise sources has been considered.

Planned Sewage Pumping Stations (DP9)

4.9.8 Four new SPS at Sites 2-34, 3-41, 3-48 and 4-35 with adequate capacities will be provided under this Project to handle the additional sewage flow from this Project and other concurrent projects. The major noise sources of the planned SPSs are from the operation of pumps, mechanically racked screens, transformer and ventilation fan of deodorisation unit. The separation distances for the NSRs nearest to the planned SPSs are listed as below:

- SPS (Site 2-34): Nearest NSR at about 135 m;
- SPS (Site 3-41): Nearest NSR at about 90 m;
- SPS (Site 3-48): Nearest NSR at about 100 m; and
- SPS (Site 4-35): Nearest NSR at about 10 m.

4.9.9 As confirmed with the Project Proponent, the design for the proposed SPSs have not been finalised, the fixed plant noise assessment is therefore conducted to determine the maximum allowable SWL based on the respective separation distances. There are other fixed noise sources located within 300 m, hence, cumulative impacts to the nearest NSRs are considered. **Table 4.42** below shows the maximum allowable SWL. Detailed calculation is shown in **Appendix 4.9.1**.

Table 4.42 Summary of Maximum Allowable SWL for Planned Sewage Pumping Stations

Plant Item	Maximum Allowable SWL, dB(A)		
	Day time	Evening time	Night-time
SPS (Site 2-34)(1)	95	95	90
SPS (Site 3-41)(1)	95	95	85
SPS (Site 3-48)(1)	103	103	93
SPS (Site 4-35)(1)	100	100	90

Note:
(1) Cumulative impacts arising from other fixed noise sources are included.

4.9.10 It is proposed to enclose the noisy sewage facilities and install acoustic silencers for vent shaft of the planned pumping station as appropriate to ensure the specified maximum SWLs as shown in **Table 4.42** will not be exceeded. The future design and selection of the equipment shall also aim to reduce the effect of tonality at the NSRs as much as practicable. However, the Contractor shall also take into account the latest available information at time of detailed design to review and update the maximum allowable SWL as appropriate.

Planned Sewage Treatment Works

4.9.11 In order to handle the sewage effluent generated from the Project area, a new STW will be built to handle the sewage flow from Project. The environmental impact arising from the construction and operation of the STWs will be addressed in a separate Schedule 2 EIA in later stage.

4.9.12 The new HSK STW at Site 3-26 and the part of the land of existing San Wai STW will be provided to handle the sewage flow from the Project. The design capacity of this new STW is 85,500 m^3 per day. With reference to the San Wai STW EIA, the design capacity is 246,000 m^3 per day. Based on this larger design capacity, the predicted fixed plant noise levels at San Wai village house were 41-42 dB(A) under mitigated scenario. Predicted fixed plant noise impact due to the new STW is expected to be less than the prediction for San Wai STW. However, as the worst assumption, same fixed plant noise levels of 41-42 dB(A) are used for a reference. Referring to the Revised RODP, the San Wai village will be removed, and the nearest NSR is at Tseung Kong Wai village house, which is about 100 m far away from the original San Wai village house. Hence, the fixed plant noise levels predicted at Tseung Kong Wai village house would be expected less than 42 dB(A). As shown in **Table 4.6**, the night-time background noise levels measured at Tseung Kong Wai village house is 46 dB(A). Thus, the fixed plant noise levels due to the operation of San Wai STW at the nearest NSR, i.e. Tseung Kong Wai village house, would still comply with the noise criteria of 46 dB(A). In addition, this STW will be totally screened by the planned logistics facilities and there is no direct line-of-sight between the STW and the NSRs. Thus, the STWs would not induce noise impact to the nearby NSRs.

4.9.13 As the proposed STW is a DP under Item F.1 of Part 1 Schedule 2 of EIAO-TM, a separate EIA study will be conducted by the future operator to fulfil all the statutory requirements and procedures under the EIAO.

Planned Port Back-up, Storage and Workshop Uses

4.9.14 Approximately 24 ha of land at the northern fringe of the Project area is reserved for port back-up, storage and workshop uses. This area may accommodate some of the existing brownfield operations through the possible development of multi-storey industrial compounds or other land efficient means. The provision of this dedicated area will help to alleviate existing industrial/residential interface issues resulting from existing container storage and port back-up areas. In addition, new roads will be provided to connect this area to KSWH directly thereby eliminating the current movement of heavy good vehicles through built-up areas including residential areas.

4.9.15 The separation distance between the nearest NSRs and planned port back-up, storage and workshop uses at Site 3-5, Site 3-6, Site 3-7, Site 3-8, Site 3-11, Site 3-13 and Site 3-14 are greater than 300 m. Besides, the planned port back-up, storage and workshop uses at Site 3-7, Site 3-8, Site 3-11, Site 3-13 and Site 3-14 would be screened by the surrounding planned 10 to 11-storeys high logistics buildings at Site 3-15, Site 3-16 and Site 3-18, and therefore there is no line-of-sight at the NSRs. Hence, the planned port back-up, storage and workshop uses at these Sites would not induce noise impact to the nearby NSRs.

4.9.16 For those planned port back-up, storage and workshop uses at Site 3-1 and Site 3-4, there is no design information, the fixed plant noise assessment is conducted to determine the maximum allowable SWL based on the respective separation distances. A planned SPS at Site 2-34 is located opposite to Site 3-4. Cumulative fixed plant noise impacts arising from these planned port back-up, storage and workshop uses and planned SPS are included. **Table 4.43** below shows the maximum allowable SWL. Detailed calculation is shown in **Appendix 4.9.1**.

Table 4.43 Summary of Maximum Allowable SWL for Planned Port Back-up, Storage and Workshop Uses

Plant Item	Maximum Allowable SWL, dB(A)		
	Day time	Evening time	Night-time
PBU+SWU (Site 3-1) (1)	85	80	73
PBU+SWU (Site 3-4) (1)	106	101	94

Note:
(1) Cumulative impact arising from other fixed noise sources are included.

4.9.17 The Contractor shall install noise barrier or acoustic enclosure as appropriate to ensure the specified maximum SWLs as shown in **Table 4.43** will not be exceeded. The future design and selection of the equipment shall also aim to reduce the effect of tonality at the NSRs as much as practicable. However, the Contractor shall also take into account the latest available information at time of detail design to review and update the maximum allowable SWL as appropriate.

4.9.18 As the proposed port back-up, storage and workshop uses is considered as a DP under Item B.5 of Part 1 Schedule 2 of EIAO-TM, a separate EIA study will be conducted by the future operator to fulfil all the statutory requirements and procedures under the EIAO.

Planned District Cooling System

4.9.19 Two DCS plants at Site 1-24 and Site 4-18 are proposed under this Project. The major noise sources come from the pump system. The separation distances for the NSRs nearest to the planned DCS are list below:

- DCS (Site 1-24): Nearest NSR at about 80 m; and
- DCS (Site 4-18): Nearest NSR at about 150 m.

4.9.20 Based on the latest information, there is no design information on the proposed DCSs, the fixed plant noise assessment is conducted to determine the maximum allowable SWL based on the respective separation distances.

4.9.21 For planned DCS at Site 1-24, it is located adjacent to the existing Ha Tsuen SPS at Site 1-23. For planned DCS at Site 4-18, it is located adjacent to the planned covered vehicle depot at Site 4-21, planned ESS at Site 3-37, planned SPS at Site 4-35 and planned industry at Site 3-52. Hence, cumulative impacts to the nearest NSRs are considered in the assessment. **Table 4.44** below shows the maximum allowable SWL. Detailed calculation is shown in **Appendix 4.9.1**.

Table 4.44 Summary of Maximum Allowable SWL for Planned Cooling System

Plant Item	Maximum Allowable SWL, dB(A)		
	Day time	Evening time	Night-time
DCS (Site 1-24)(1)	102	102	92
DCS (Site 4-18)(1)	106	106	96

Note:
(1) Cumulative impact arising from other fixed noise sources are included.

4.9.22 The Contractor shall install acoustic silencers, noise barrier or acoustic enclosure as appropriate to ensure the specified maximum SWLs as shown in **Table 4.44** will not be exceeded. The future design and selection of the equipment shall also aim to reduce the effect of tonality at the NSRs as much as practicable. However, the Contractor shall also take into account the latest available information at time of detail design to review and update the maximum allowable SWL as appropriate.

Planned Electricity Substation

4.9.23 Four ESS at Sites 3-19, 4-23, 4-37 and 5-19 are proposed under this Project. The major noise sources of ESS are generated from the operation of transformers and ventilation systems.

4.9.24 Based on the latest information, there is no design information on the proposed ESS, the fixed plant noise assessment is conducted to determine the maximum allowable SWL based on the respective separation distances.

4.9.25 For planned ESS at Site 3-19, the surrounding logistics facility would block the line-of-sight between this planned ESS and the NSRs. Hence, planned ESS at Site 3-19 would not induce noise impact to the nearby NSRs. There are other fixed noise sources located within 300 m, hence, cumulative impacts to the nearest NSRs are considered in the assessment. **Table 4.45** below shows the maximum allowable SWL. Detailed calculation is shown in **Appendix 4.9.1**.

Table 4.45 Summary of Maximum Allowable SWL for Planned Electricity Substation

Plant Item	Maximum Allowable SWL, dB(A)		
	Day time	Evening time	Night-time
ESS (Site 4-23) (1)	89	89	79
ESS (Site 4-37)	94	94	84
ESS (Site 5-19)(1)	94	94	84

Note:
(1) Cumulative impact arising from other fixed noise sources are included.

4.9.26 The Contractor shall install acoustic silencers, noise barrier or acoustic enclosure as appropriate to ensure the specified maximum SWLs as shown in **Table 4.45** will not be exceeded. The future design and selection of the equipment shall also aim to reduce the effect of tonality at the NSRs as much as practicable. However, the Contractor shall also take into account the latest available information at time of detail design to review and update the maximum allowable SWL as appropriate.

Planned Public Transport Interchange and Vehicle Depot

4.9.27 Four PTIs at Site 2-23, Site 4-13, Site 4-29 and Site 5-5 are provided under this Project. One vehicle depot at Site 4-21 is provided under this Project. The PTIs and vehicle depot would be decked under proposed building structure and designed with no line-of-sight at the noise sensitive use. The major fixed plant noise sources of PTIs and vehicle depot include the operation of ventilation fans.

4.9.28 Since there is no design information on the proposed PTIs and vehicle depot, the fixed plant noise assessment is conducted to determine the maximum allowable SWL based on the respective separation distances. The separation distances for the NSRs nearest to the planned PTIs and vehicle depot are list below:

- PTI (Site 2-23): Nearest NSR at about 10 m;

- PTI (Site 4-13a): Nearest NSR at about 170 m;
- PTI (Site 4-29): Nearest NSR at about 35 m;
- PTI (Site 5-5): Nearest NSR at about 35 m; and
- Vehicle depot (Site 4-21): Nearest NSR at about 5 m.

4.9.29 There are other fixed noise sources located within 300 m, hence, cumulative impacts to the nearest NSRs are considered in the assessment. **Table 4.46** below shows the maximum allowable SWL. Detailed calculation is shown in **Appendix 4.9.1**.

Table 4.46 Summary of Maximum Allowable SWL for Planned Public Transport Interchange

Plant Item	Maximum Allowable SWL, dB(A)		
	Day time	Evening time	Night-time
PTI (Site 2-23)(1)	85	85	75
PTI (Site 4-13a)(1)	99	99	89
PTI (Site 4-29)(1)	96	96	86
PTI (Site 5-5)(1)	96	96	86
Vehicle Depot (Site 4-21)(1)	87	87	77

Note:
(1) Cumulative impact arising from other fixed noise sources are included.

4.9.30 Since the PTI and vehicle depot are covered and the possible noise source would be the ventilation shafts. It is proposed to install acoustic silencers, as appropriate to ensure the specified maximum SWLs as shown in **Table 4.46** will not be exceeded. The future design and selection of the equipment shall also aim to reduce the effect of tonality at the NSRs as much as practicable. However, the Contractor shall also take into account the latest available information at time of detail design to review and update the maximum allowable SWL as appropriate.

Planned Sports Ground

4.9.31 A sports ground is proposed at Site 1-29. Based on the latest information, there is no design information on the planned sports ground, the fixed plant noise assessment is conducted to determine the maximum allowable SWL based on the respective separation distances. The maximum allowable SWLs for the planned sports ground are 93 dB(A), 93 dB(A) and 83 dB(A) during Day-time, Evening-time and Night-time, respectively. Detailed calculation is shown in **Appendix 4.9.1**.

4.9.32 It is recommended to incorporate the following measures for the public announcement system and/or the sound amplification system, if used during the outdoor noise activities:

- To use a cluster of small power loudspeakers instead of a few large power loudspeakers;
- To use directional loudspeakers and orientate them to point towards the audience and away from the nearby NSRs; and
- To include a “Limiter” device in the system to set the upper bound of the output sound level.

4.9.33 With proper event management, the noise generated form the outdoor activities from the planned sports ground would be under control and adverse noise impact on the nearby NSRs would be minimised.

Planned Fire Station

4.9.34 A divisional fire station is proposed at Site 4-1. The major noise sources of fire station include condenser, transformer, etc. The nearest NSRs with facades facing the fire station would be the planned residential premises at Site 4-3 which is approximately 50 m away from the station.

4.9.35 Since there is no design information on the proposed fire station, the fixed plant noise assessment is conducted to determine the maximum allowable SWL based on the respective separation distances. There are some planned industries at Site 3-43 to 3-50 located opposite to Site 4-1 and hence cumulative impact is assessed. **Table 4.47** below shows the maximum allowable SWL. Detailed calculation is shown in **Appendix 4.9.1**.

Table 4.47 Summary of Maximum Allowable SWL for Planned Fire Station

Plant Item	Maximum Allowable SWL, dB(A)		
	Day time	Evening time	Night-time
Fire Station (Site 4-1)(1)	98	98	88

Note:
(1) Cumulative impact arising from other fixed noise sources are included.

4.9.36 The Contractor shall install acoustic silencers, noise barrier or acoustic enclosure as appropriate to ensure the specified maximum SWLs as shown in **Table 4.47** will not be exceeded. The future design and selection of the equipment shall also aim to reduce the effect of tonality at the NSRs as much as practicable. However, the Contractor shall also take into account the latest available information at time of detail design to review and update the maximum allowable SWL as appropriate.

4.9.37 Other noise sources associated with this fire station would be related to the emergency response of the station, i.e. the station loudspeakers, siren and the vehicle sirens. Noise from fire station may sometimes be disturbing particularly during emergency duties. Yet such occurrences are of short duration and infrequent, and could not be avoided when emergencies occur. The sirens are intended to warn surrounding emergency, so that priority may be given to them. Emergency vehicles could also be assisted in their response by installation of “hurry call” or other signalisations at intersection along the route.

Planned Hospital

4.9.38 A hospital is proposed at Site 5-13. The major noise sources of hospital include air-cooled chillers, cooling tower, ventilation louvers, etc. The nearest NSRs with facades facing the planned hospital would be the planned commercial & residential premises at Site 5-8, planned education institution at Site 5-21 and existing village at Shek Po Tsuen. For NSR at Site 5-8, existing EPS4 and a planned PTI at Site 5-5 are located within 300 m, and hence cumulative impact is considered in the assessment. For NSR at planned education institution at Site 5-21 and Shek Po Tsuen, cumulative impact from planned hospital and ESS at Site 5-19 are also assessed.

4.9.39 Based on the latest information, there is no design information on the planned hospital, the fixed plant noise assessment is conducted to determine the maximum allowable SWL based on the respective separation distances. The maximum allowable SWL for the planned hospital is 98, 98 and 88 dB(A) during Day-, Evening- and Night-time, respectively. Detailed calculation is shown in **Appendix 4.9.1**.

4.9.40 The Contractor shall install acoustic silencers or acoustic enclosure as appropriate to ensure the specified maximum SWLs as mentioned in above section will not be exceeded. The future design and selection of the equipment shall also aim to reduce the effect of tonality at the NSRs as much as practicable. However, the Contractor shall also take into account the latest available information at time of detail design to review and update the maximum allowable SWL as appropriate.

Planned EFTS Depot

4.9.41 A EFTS Depot is planned at Site 3-18. It will be decked by 9-storey logistics facility. The major fixed plant noise sources of EFTS Depot include the operation of ventilation fans. A planned SPS at Site 2-34 is located less than 300 m separation distance, and hence, cumulative impact at the nearest NSRs are assessed.

4.9.42 Based on the latest information, there is no design information on the planned EFTS Depot, the fixed plant noise assessment is conducted to determine the maximum allowable SWL based on the respective separation distances. The maximum allowable SWL for the planned EFTS Depot is 106, 106 and 96 dB(A) during Day-, Evening- and Night-time, respectively. Detailed calculation is shown in **Appendix 4.9.1**.

4.9.43 Since the PTI and vehicle depot are covered and the possible noise source would be the ventilation shafts. It is proposed to install acoustic silencers, as appropriate to ensure the specified maximum SWL as mentioned in above section will not be exceeded. The future design and selection of the equipment shall also aim to reduce the effect of tonality at the NSRs as much as practicable. However, the Contractor shall also take into account the latest available information at time of detail design to review and update the maximum allowable SWL as appropriate.

4.9.44 As the planned EFTS and its associated facilities are considered as a DP under Item A.3 of Part 1 Schedule 2 of EIAO-TM, a separate EIA study will be conducted by the future operator to fulfil all the statutory requirements and procedures under the EIAO.

Planned Parking and Operational Facilities for EFTS

4.9.45 The planned parking and operational facilities for EFTS proposed at Site 2-29 will be enclosed inside a building structure provision of ventilation fans. The fixed plant noise sources of planned parking and operational facilities for EFTS include the operation of ventilation fans. The proposed ventilation fans will be facing to the west and designed with no line-of-sight at the noise sensitive use. Thus, the planned EFTS Depot would not induce noise impact to the nearby NSRs.

4.9.46 As the planned EFTS and its associated facilities are considered as a DP under Item A.3 of Part 1 Schedule 2 of EIAO-TM, a separate EIA study will be conducted by the future operator to fulfil all the statutory requirements and procedures under the EIAO.

Planned West Rail HSK Station

4.9.47 A West Rail HSK station is planned at Site 4-34. The major fixed plant noise sources of the station include the operation of ventilation fans. A planned fire station, ESS at Site 4-23 and 4-29, PTI at Site 4-13a are located less than 300 m separation distance, and hence, cumulative impact at the nearest NSRs are assessed.

4.9.48 Based on the latest information, there is no design information on the station, the fixed plant noise assessment is conducted to determine the maximum allowable SWL based on the respective separation distances. The maximum allowable SWL for the planned station is 93, 93 and 83 dB(A) during Day-, Evening- and Night-time, respectively. Detailed calculation is shown in **Appendix 4.9.1**.

4.9.49 It is proposed to install acoustic silencers, as appropriate to ensure the specified maximum SWL as mentioned in above section will not be exceeded. The future design and selection of the equipment shall also aim to reduce the effect of tonality at the NSRs as much as practicable. However, the Contractor shall also take into account the latest available information at time of detail design to review and update the maximum allowable SWL as appropriate.

Planned Industrial Area

4.9.50 The sites at the west of KSWH including Sites 3-43, 3-44, 3-45, 3-46, 3-47, 3-50, 3-51 and 3-52 are planned as industrial sites. It may have potential industrial noise impacts arising from chillers, cooling towers and/or ventilation fans on the nearby NSRs. As there is no design information at this stage, the fixed plant noise assessment is conducted to determine the maximum allowable SWL based on the respective separation distances. The maximum allowable SWL for these planned industrial sites are summarised in **Table 4.48.** Detail calculation is shown in **Appendix 4.9.1**.

4.9.51 It is proposed to enclose the noisy equipment and install acoustic silencers, as appropriate to ensure the specified maximum SWL as summarised in **Table 4.48** will not be exceeded. The future design and selection of the equipment shall also aim to reduce the effect of tonality at the NSRs as much as practicable. However, the operators shall also take into account the latest available information at time of detail design to review and update the maximum allowable SWL as appropriate.

Table 4.48 Summary of Maximum Allowable SWL for Industrial Area

Plant Item	Maximum Allowable SWL, dB(A)		
	Day time	Evening time	Night-time
Site 3-43	106	106	96
Site 3-44	106	106	96
Site 3-45	104	104	94
Site 3-46	104	104	94
Site 3-47	105	105	95
Site 3-50	104	104	94
Site 3-51	No NSR within 300 m assessment area		
Site 3-52	104	104	94

Mitigation Measures

4.9.52 In accordance with the above predicted results, the fixed plant noise impacts predicted at the existing and planned NSRs would comply with the respective noise criteria. As there is no design information on those proposed fixed plants, the following tentative noise mitigation measures are recommended for the planned fixed plant noise sources which are located near to the existing and planned NSRs:

- All the pumps and noisy plants should be enclosed inside the building structure;
- Proper selection of quiet plant aiming to reduce the tonality at NSRs;
- Installation of silencer/acoustic enclosure/acoustic louvre for the exhaust of ventilation system; and
- Openings of ventilation systems should be located away from NSRs as far as practicable.

4.9.53 The feasibility, practicability, programming and effectiveness of the above mitigation measures have been reviewed by the Project Engineer.

Evaluation of Residual Environmental Impacts

4.9.54 Residual fixed plant noise impacts are not anticipated.

4.10 Helicopter Noise Impact Assessment

Identification of Helicopter Noise Source

Helicopter Fleet

4.10.1 The current helicopter models operating at the existing helipad near Ha Tsuen Interchange are Super Puma and EC155. With reference to Government Flying Service (GFS) "Procurement of seven helicopters and associated mission equipment of the Government Flying Service" submitted to Finance Committee (FC) of Legislative Council in June 2013, these helicopters models will reach their end of service lifespan after Year 2017 and will be replaced by a new fleet of helicopter. With the approval from FC on 29 October 2013, the old helicopter models would be replaced by a new helicopter model after 2017. The new helicopter model would be Airbus H175 (known as EC175 previously) as advised by GFS. All above references are documented in **Appendix 4.10.1**.

Noise Performance of H175 Helicopter

4.10.2 This assessment is conducted based on the new H175 helicopter model, which will operate at the existing helipad. The noise performance of H175 helicopter during operation modes of take-off, overflight and approach are made reference to ICAO standard. There are other operation modes, which do not involve lateral movements, such as "hovering", "touchdown", "idling" and "lift-off". However, the ICAO's standard does not have these data. The noise performance for these movements is made reference to the Helipad at Yung Shue Wan, Lamma Island (YSW) EIA Report (Report No. 094/2004). These assumptions and noise performance of H175 helicopter are detailed in **Appendix 4.10.1**.

Emergency Night-time Operation

4.10.3 Night-time operation of the helipad is not expected, with exception of emergency flying services, such as air ambulance, search and rescue and emergency law enforcement operation support. According to GFS statistics, there was only one emergency use of the helipad since 2011. Operation details as advised by GFS are presented below:

- Air ambulance - GFS provides a 24 hours air ambulance service. After receiving emergency callouts from clinics throughout the territory, GFS helicopters can arrive on scene within 20 minutes for locations within Island Zone such as Hong Kong Island, Cheung Chau, Hei Ling Chau, Lantau, Peng Chau and Soko Islands, and 30 minutes for elsewhere within the Hong Kong territory.

- Search and rescue – The GFS has the duty to perform offshore search and rescue operations up to 700 nm south of Hong Kong, including dawn searches for lost or injured hikers and climbers. Such operations sometimes require the winching of people from inaccessible places in the rocky hills of the New Territories or from the many outlying islands, often under bad weather conditions.

- Law enforcement operation support - GFS provides 24/7 emergency response to law enforcement operations, and such operations are often short notice, time critical and situation dictated. This involves providing flying support to various police operations for law enforcement purposes and for maintaining law and order.

4.10.4 Given the emergency nature of these night-time flying services, no night-time helicopter noise impact to the environment is anticipated.

Assessment Approach and Methodology

4.10.5 Noise associated with the existing helipad will be generated during helicopter overflight, i.e. manoeuvring over the helipad, during approach and during take-off flight.

4.10.6 According to Table 1A of Annex 5 "A Summary of Noise Criteria" in the EIAO-TM, helicopter noise impacts shall be assessed in terms of the L_{max} level, which is the maximum instantaneous sound pressure level at the noise sensitive receiver. Since all the identified NSRs are located at considerable distances (over 1km) from the helipad, helicopter noise can be considered as a "point" source. Therefore, the sound pressure level at NSRs can be evaluated based on standard acoustic principle of a 'point' source, i.e., the sound pressure level in any direction (open air) will decrease at a rate of 6 dB per doubling of distance away from the source. The difference in noise levels at two different distances, d_1 and d_2, can be calculated using the following formula:

$$\text{Noise Level Difference (dB)} = 20 \log_{10} (d_1/d_2)$$

4.10.7 As a worst-case assumption, zero vertical distance between the NSRs and the helicopter is assumed, the minimum separation would then be the minimum horizontal separation distance.

Prediction and Evaluation of Environmental Impacts

4.10.8 **Appendix 4.10.1** presents the detailed calculation results of helicopter noise impact at the identified NSRs. With reference to **Appendix 4.10.1**, the helicopter noise levels predicted at the representative NSRs, including an existing NSR at Tseung Kong Wai (NSR ID: E1-1A_R01) and a planned NSR at Site 2-28 (NSR ID: P2-28_E01) which are existing and planned NSRs closest to the existing helipad, would comply with the Lmax 85dB(A) limit.

Mitigation Measures

4.10.9 With reference to **Appendix 4.10.1**, the predicted noise levels at the representative NSRs would comply with the criteria, no mitigation measure is required.

Evaluation of Residual Environmental Impacts

4.10.10 Residual helicopter noise impacts are not anticipated.

4.11 Environmental Acceptability of Schedule 2 Designated Projects

4.11.1 An application for an EP would be submitted under this EIA for DP1, DP2, DP5, DP6, DP9 and DP12 (refer to **Figure 1.3** and **Section 4.1.2**).

New Roads (DP1, DP2, DP5, DP6 and DP12)

4.11.2 With proper implementation of the recommended mitigation measures for construction activities (as detailed in **Sections 4.6.5** to **4.6.13**, and **4.6.17** to **4.6.18** above), as well as the mitigation measures to reduce traffic noise impact during the operational phase (as described in **Sections 4.7.14** and **4.7.33** above), no unacceptable noise impact would be resulted from the proposed roads (DP1, DP2, DP5, DP6 and DP12).

Sewage Pumping Station (DP9)

4.11.3 With proper implementation of the recommended mitigation measures for construction activities (as detailed in **Sections 4.6.5 to 4.6.13** above), and the design of SPSs follow the maximum allowable sound power level as described in **Table 4.42** to minimise potential fixed plant noise impact (as described in **Sections 4.9.8** to **4.9.10** above), no unacceptable noise impact would be resulted from the proposed SPSs (DP9).

Other DPs

4.11.4 There will be separate EIA studies to assess the following Schedule 2 DPs (refer to **Figure 1.2**):

- DP3 - Construction of new West Rail HSK Station (Site 4-34)
- DP4 - Construction of EFTS – Potential DP subject to further review
- DP7 - Construction of a new container back-up and storage area (Site 3-1, 3-4, 3-5, 3-13 and 3-14) – Potential DP subject to further review
- DP8 - Construction of new HSK STW (Site 3-26 and part of the existing San Wai STW)
- DP10 - Construction of flushing water service reservoirs for reuse of reclaimed water in Fung Kong Tsuen and Tan Kwai Tsuen (Sites 3-3 and 5-40)
- DP11 - Construction of one RTS (Site 3-12)

4.11.5 The noise impact of these Schedule 2 DPs during construction phase and operation phase will be further investigated in separate EIA studies under the EIAO. The EM&A requirements for these Schedule 2 DPs will also be provided under separate EIA studies.

4.12 Conclusion

Construction Phase

4.12.1 Assessment on the construction noise associated with the use of PME for different phases of construction works has been conducted. With the implementation of practical mitigation measures including good site management practices, use of movable noise barrier, noise barrier and full enclosure, use of “quiet” plant, proper workfront management, proper grouping of PMEs for some construction activities at critical work areas and provision of minimum separations between the affected schools and the critical works areas during the examination period, no unacceptable impact arising from the construction of the Project will be anticipated.

Operation Phase

4.12.2 Operational road traffic noise impact on existing and planned noise sensitive uses within and in the vicinity of the Project have been predicted for the worst case year. Results indicate that the noise impacts from the Project roads can be mitigated by a combination of noise mitigation measures including: (i) application of LNRS materials on some roads; (ii) vertical noise barriers/cantilever noise barriers along some Project road sections; (iii) alternative building layout and special building design such as provision of blank façade/acoustic windows/acoustic balcony in some buildings; and (iv) provision of boundary wall, air conditioning and noise insulated windows for affected planned educational institutes to within the respective noise criteria. These mitigation measures would also ensure that the noise levels at the planned NSRs under this Project are within the respective noise criteria. Regarding the adverse noise impact from the existing road sections, LNRS would be considered to be implemented under the policy endorsed by the Executive Council to address the existing noise impacts from existing roads or by the Project if significant noise impact from the existing roads is due to the Project. Different materials for LNRS are being reviewed and tested by the Government for application in Hong Kong. Environmental reviews will be conducted at the later design stage to review and ascertain the proposed provisional noise mitigation measures taking into account the latest design standard at that time for the suitability and application of the LNRS materials.

4.12.3 However, there is scope for adoption of alternative noise mitigation measures, for example, by means of alternative layout and design of individual developments at detailed design stage. Noise impact assessment at the planned residential sites is proposed to be conducted by future developers at the detailed design stage to study whether the future development layout would avoid exposing excessive traffic noise levels so as to minimise the scale/extent of the proposed noise mitigation measures.

4.12.4 Fixed noise assessment has been conducted. Noise impact from planned fixed plant could be effectively mitigated by implementing noise control measure at source during the detailed design stage. With the adoption of the proposed maximum permissible SWLs for the proposed fixed plant, the impact noise levels at representative NSRs would comply with the relevant noise criteria. The predicted noise levels at the planned NSRs due to existing fixed plant noise sources would also comply with the relevant noise criteria. Therefore, adverse fixed noise sources impact to the NSRs is not anticipated.

4.12.5 Rail noise assessment has been conducted based on rail noise source measurement and operational information from the railway operator. Results indicate that the noise impacts from WRL on NSRs would comply with the statutory requirement with the provision of architectural fins and acoustic windows on the affected NSRs, and consideration of layout setback. The noise impacts from LRT on NSRs would be mitigated with the provision of architectural fins and acoustic windows on the affected NSRs.

4.12.6 An EFTS is proposed to be introduced to operate within the Project, which may be in the form of rail based or road based mode of transport. For conservative noise assessment, the rail based EFTS was assumed for rail noise impact assessment. Results indicate that the noise impacts on NSRs would comply with the statutory requirement after considering some track enhancement measures such as embedded rail, green track with vegetation, etc. With the provision of 10 m buffer distance, the operation of EFTS would not pose adverse noise impact on the nearby NSRs.

4.12.7 The existing helipad near Ha Tsuen Interchange of KSWH is for emergency use. There is no routine flight expected. Helicopter noise assessment at the existing helipad during occasional take-off, overflight and approach has been conducted. Results indicate that the operation of the helipad would not pose environmental impact to the planned NSRs in the Project area and the existing NSRs.

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